

# ROUTE ANYTHING: A THEORY OF INFERENCE ROUTERS

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## ABSTRACT

An inference router chooses which model, tool, or combination should handle a task using the current state and records from earlier outcomes. We organize routing around three mechanisms: what capabilities to combine, when to spend more computation, and what information to retain. We develop a reference-relative accounting framework that compares delivered quality with total resource costs, including routing, verification, and information costs. On CodeRouterBench and RouterBench, cost-aware residual coverage selects useful candidates more reliably than standalone success rates; on RouterBench it selects the best held-out candidate in 67% of test cases, versus 13% for the cost-unaware baseline. After the failure of candidate model, our proposed conditional planning selects shorter chains, using online feedback to repair 41–54% of failures compared with 5–18% for blind retries. Verifier errors and charges can reduce cascade gains, and routing signals value most when task-type priors are not explicitly specified. Together, these findings suggest that effective routing depends on evaluating complementary capabilities, additional computation, and retained information by their marginal quality gains and full resource costs.

## 1 INTRODUCTION

An inference router chooses the next computation from a heterogeneous capability pool. For example, a smaller model may solve an easy query, a specialist may resolve a particular uncertainty, and a tool may certify an answer that would otherwise require another model call. A router can select one model, combine complementary proposers with an aggregator, or assign different roles and context views. Its decision may occur once per task or repeatedly within an agent execution. The central problem is how to choose sufficient capability at a resource cost justified by the value it delivers.

Model routing and cascades formalize parts of this problem. FrugalGPT learns cheaper cascades (Chen et al., 2024b), RouteLLM learns routing from preferences (Ong et al., 2025), and unified formulations connect routing and cascading under a budget model (Dekoninck et al., 2025). These settings distinguish routers from its execution: an ex-ante router dispatches a selected path, whereas a verified cascade attempts a cheap path and escalates after observing insufficiency. A portfolio router additionally decides which capabilities should work together. The value of each routing decision depends on joint failures, routing overhead, and what the router has learned from earlier outcomes.

We study *inference routers* as stateful decision policies, with a common comparison of delivered value and total resource cost. We verify the proposed theory through *model routing* to facilitate the evaluation: selecting and combining models, deciding whether to continue after failure, and retaining useful outcomes. We aim to answer these central questions:

What should a router combine, when should it spend more, and what should it remember?

These are coupled as a routed portfolio must cover failures left by the current selection. A later attempt must be evaluated on the cases that remain after earlier failures. A verifier or routing signal must improve a decision enough to pay for its own cost. A stored record must improve the policy it replaces, after exploration, mistaken writes, and retention are charged.

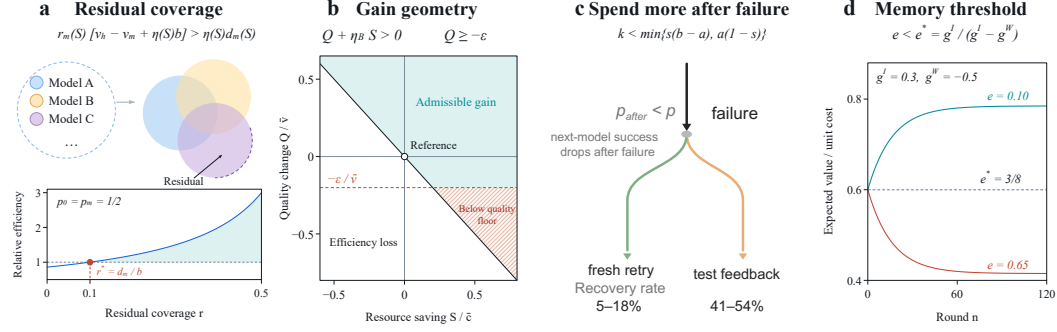


Figure 1: **A decision map for additional inference.** (a) Admissible policies combine an efficiency gain with a quality floor; axes use the reference means  $\bar{v}$  and  $\bar{c}$ . (b) Newly covered failures must justify the added model’s cost (Section 3.1). (c) Under quality-preserving, exact revelation with  $0 < a < b$ , information is worth buying only inside the shaded band; this is an upper bound on the value of a learned signal (Section 4.3). (d) In the illustrated record model, memory helps below an erroneous-write fraction  $e$  set by record gains  $g^I, g^W$  (Appendix F; empirical study in Appendix Q). Curves and parameter values are theoretical illustrations, not fitted experimental results.

Figure 1 previews the common accounting and its implications for complementarity, information, and memory. For a named reference policy, we measure a router’s change in delivered quality and its resource saving on the same tasks. Their combination gives an exact efficiency test, while a separate quality floor states how much degradation is permitted. The main evidence comes from CodeRouterBench’s 9,999 coding tasks and RouterBench’s 36,497 prompts, supplemented by live retries, verifier calls, and prompt-only forecasts.

**Decision rules and headline evidence.** Three model-routing calculations make the framework concrete. For portfolio  $S$  and candidate  $m$ , residual coverage  $r_m(S)$ , added cost  $d_m$ , and current efficiency  $\eta(S)$  give the gain score

$$N_S = r_m(S) [v_h - v_m + \eta(S)b] - \eta(S)d_m. \quad (1)$$

Here accepted value  $v_h$ , fallback value  $v_m$ , and fallback cost  $b$  are shared across the compared portfolios. For failure state  $k$ , conditional success  $p_k$ , attempt cost  $a_k$ , resource price  $\lambda$ , and continuation value  $\Pi_{k+1}$ , backward planning gives

$$\Pi_k = \max\{J^{\text{direct}}, p_k v_h - \lambda a_k + (1 - p_k)\Pi_{k+1}\}. \quad (2)$$

In a quality-preserving attempt–fallback decision, let  $s$  be primary success probability,  $a$  attempt cost,  $b$  fallback cost, and  $s_Z$  the posterior after a signal priced at  $k$ . Its net value is

$$\min\{a + (1 - s)b, b\} - \mathbb{E} \min\{a + (1 - s_Z)b, b\} - k. \quad (3)$$

The corresponding studies show that cost-aware residual coverage selects the best held-out RouterBench partner in 67% of 566 portfolios, versus 13% for marginal success; conditional planning improves gain-direction agreement from 77.2% to 93.2%; and feedback repairs 41–54% of failures versus 5–18% for blind retries. Charging the verifier reduces the judge cascade from  $2.63\times$  to  $1.09\times$  efficiency, while signal capture falls from 55–56% pooled to 6–8% within task dimensions. Memory results remain in the appendix because their record process requires a separate stream and retention analysis.

**Study design and scope.** The empirical sections instantiate these equations in order: residual coverage for combination, conditional continuation value for failure states, total-cost operating points for verification, and net value of information for signals. We use fitting and held-out splits wherever the data permit, paired task-bootstrap intervals, and explicit resource prices. The appendices provide proofs and supplementary studies of policy optimization, capacity allocation, agent chains, Jev dispatch, sensitivity, and memory.

**Relation to existing work.** Our questions connect model cascades and cost-aware routing (Chen et al., 2024b; Aggarwal et al., 2024; Dekoninck et al., 2025), complementary algorithm portfolios (Rice, 1976; Huberman et al., 1997; Gomes & Selman, 2001; Kotthoff, 2014; Xu et al., 2012), and the value of information in sequential decisions (Blackwell, 1953; Howard, 1966; Russell & Wefald, 1991). They also relate to reject-option verification (Chow, 1970; Gupta et al., 2024), agent and harness routing (Zhou et al., 2026a; Liu et al., 2026), and persistent records and caches (Fagin, 1977; Bang, 2023; Chen et al., 2024c). We use fractional programming and state-wise optimization (Dinkelbach, 1967); the contribution is their connection to measurable routing decisions and the accompanying evidence. Appendix O gives the broader mapping across routing granularity.

## 2 A UNIFIED MODEL OF INFERENCE ROUTERS

**Router policies and actions.** At decision round  $n$ , the observable state  $x_n$  contains the task and available execution history, while  $I_n$  contains records from previous rounds. A stateful router chooses an execution plan and updates its information from feedback:

$$u_n = g(x_n, I_n), \quad I_{n+1} = \text{Upd}(I_n, x_n, u_n, f_n). \quad (4)$$

An action may select a model, assign a portfolio, invoke an agent tool, or allocate submodels, steps, or tokens within a model. Execution feedback  $f_n$  updates the state and stored information. The common abstraction is the choice of computation together with its resulting value and complete cost; the available actions and feedback depend on the system. Appendix P.2 maps representative systems to these primitives.

**Reference-relative comparison.** Let  $V$  be delivered task value and  $D$  the complete resource cost, including model calls, routing, aggregation, verification, and information updates. Let  $Y_B, C_B$  be the outcomes of a named reference policy on the same task population. We report aggregate efficiency  $\eta = \mathbb{E}[V]/\mathbb{E}[D]$ , reference efficiency  $\eta_B = \mathbb{E}[Y_B]/\mathbb{E}[C_B]$ , quality change  $Q = \mathbb{E}[V - Y_B]$ , and saving  $S = \mathbb{E}[C_B - D]$ . Then

$$\eta - \eta_B = \frac{Q + \eta_B S}{\mathbb{E}[D]}, \quad \text{quality requirement: } Q \geq -\epsilon. \quad (5)$$

Thus an efficiency improvement requires the quality change plus reference-valued saving to be positive. The quality requirement is separate. If the objective has an external resource price  $\lambda$ , we use  $J_\lambda = \mathbb{E}[V - \lambda D]$ ; dollars, tokens, seconds, and latency are therefore reported as distinct resources.

**Specialization to model routing.** We now take the capability pool to be callable language models. The state contains the prompt, task metadata, and any preceding outputs or failures; an action selects a model or portfolio and an execution rule; feedback supplies scores or test results; records retain information for later choices. This preserves the policy and objective above while making their quantities observable on matched tasks. Direct dispatch commits to the selected output. A verified cascade instead accepts a primary output  $Y_P$  on event  $H \in \{0, 1\}$  and calls a fallback on rejection. With primary-plus-verification cost  $A$ , its outcome is

$$V = HY_P + (1 - H)Y_B, \quad D = A + (1 - H)C_B. \quad (6)$$

A failed primary is still paid for. Verifier and signal calls enter  $A$  when measured or explicitly priced; free oracle checks are identified as such. Because  $H$  can select easier tasks, the reference outcome and cost need not have the same conditional means on accepted and rejected branches. This is why the experiments use matched held-out comparisons and why verifier and signal costs are treated as part of the decision.

**Can this identify the best router type?** For a fixed finite action menu and state representation, Theorem 5 chooses the action maximizing value minus cost subject to a quality floor. The policy comparison in Appendix A instantiates this result: the unconstrained dollar objective selects direct non-Opus dispatch, while cascades buy additional quality at higher cost. Model routing thus tests the framework’s decision rules within a specified menu and task population.

### 3 WHAT SHOULD A ROUTER COMBINE?

**Data and evaluation.** CodeRouterBench (Zhou et al., 2026a) records scores, token usage, and latency for eight models on 9,999 coding tasks. We fit primitives and choose policies on 7,080 probing tasks, then evaluate on 2,919 `id_test` tasks; states are nine task dimensions or 21 dimension $\times$ source cells. Opus is the reference. RouterBench (Hu et al., 2024) supplies a second outcome matrix with 36,497 prompts and eleven models, split into 25,542 fitting and 10,955 held-out prompts for the portfolio study. Its reference is gpt-4-1106-preview. These are random splits of the same task pools.

Unless stated otherwise, a recorded primary is accepted if its released score meets or exceeds the threshold  $\theta \in \{1, 0.5\}$ , with the reference used on rejection. This oracle-verifier replay excludes verifier and router fees. The verifier and signal studies explicitly replace or price these components. Live retry runs generate new answers but retain historical Opus outcomes. Intervals are paired 95% task-bootstrap intervals; a gain has a *decided sign* when its interval excludes zero. Repeated configurations share tasks. Appendix B.1 provides the sample and accounting details.

#### 3.1 WHICH MODEL ADDS COST-AWARE COVERAGE

**Question.** If a router can call a second model, should it choose the model with the highest standalone success rate? Such a model may reproduce the existing portfolio’s successes while missing the same difficult tasks. Our proposed metric is *residual coverage*: the probability that the new member succeeds when every current member fails.

**Decision rule and test.** For portfolio  $S$  and candidate  $m$ , write  $r_m(S)$  for residual coverage,  $d_m$  for the additional calling cost, and  $\eta(S)$  for current efficiency. Proposition 2 gives the gain score in Eq. (1): its first term values newly covered tasks and its second prices the added model. Positive  $N_S$  predicts an efficiency improvement. Proposition 3 shows why standalone success probabilities cannot identify the required overlap.

We estimate the rule on the fitting split and predict held-out additions. CodeRouterBench uses seven candidates, portfolios of up to three models, and nine dimensions. RouterBench considers ten candidate additions to 21 initial portfolios within source and task contexts. The selecting verifier picks a passing member, otherwise the reference; costs sum the member calls. The main control replaces measured residual coverage by the portfolio failure probability times the candidate’s marginal success, discarding their dependence. Appendix B.2 gives the filtering and ranking protocol.

Table 1: Does adding a model improve efficiency? Agreement with the held-out gain direction. “Decided” selects comparisons whose gain interval excludes zero; these percentages are not task success rates.

| Dataset and comparison set      | Comparisons | Cost-aware coverage | Independence proxy |
|---------------------------------|-------------|---------------------|--------------------|
| CodeRouterBench, $\theta = 1$   | 674         | <b>91.4%</b>        | 71.5%              |
| CodeRouterBench, $\theta = 0.5$ | 958         | <b>93.1%</b>        | 71.2%              |
| RouterBench, all                | 3,602       | <b>93.2%</b>        | 85.9%              |
| RouterBench, decided            | 2,559       | <b>98.2%</b>        | 91.8%              |

**Finding: complementary coverage must also be affordable.** The cost-aware rule outperforms the independence proxy on both datasets (Table 1). More directly, it selects the held-out best partner in **67% of 566 RouterBench portfolios**, versus **13%** when ranking by marginal success. Its median rank correlation with realized gain is 0.928, compared with 0.080 for residual coverage alone and  $-0.412$  for marginal success. Thus measuring complementarity without pricing it is also insufficient.

The common-conditional-value assumption is approximate: additions change the tasks left for the fallback. CodeRouterBench miss-conditional value decreases in 97% of comparisons, and some predicted gain magnitudes or directions reverse as fallback cost changes. The evidence supports useful prediction and partner selection on these splits, rather than an exact empirical model of every addition.

**Takeaway.** Choose a partner for the failures it resolves at an affordable cost. Measure joint outcomes on the same tasks; standalone model rankings do not reveal this information.

## 4 WHEN SHOULD A ROUTER SPEND MORE?

### 4.1 WHAT SHOULD HAPPEN AFTER A FAILURE

**Question.** Suppose the first model has already failed. The next model’s success rate over all tasks is now the wrong starting point: the router is facing the first model’s failure subset. Should it try again, switch models, or escalate immediately?

**Decision rule and test.** We plan backwards from a terminal Opus call. At each failure state, the router compares escalating now with paying for another attempt and retaining the option to continue after another rejection. Proposition 7 formalizes this comparison; its shared-value recursion is Eq. (2). The critical input is  $p_k$  after the previous failures. Fully matching values and reference costs to each state gives the general recursion. The single-attempt threshold is the final-step case; its supplementary evaluation is in Appendix B.3.

The chain-planning study evaluates all 259 ordered chains of one to three distinct non-Opus models, followed by Opus. We fit conditional outcomes on probing and evaluate the fixed plans on held-out tasks, at two acceptance thresholds and five resource prices. The main control uses marginal success rates; a fully state-matched recursion additionally updates the conditional values and costs.

**Finding: a failure often makes further attempts less attractive.** At  $\theta = 1$ , all 42 ordered model pairs have lower second-model success after the first model fails. The median conditional-to-marginal ratio is **0.49**. Failure-conditioned planning selects shorter chains in **36 of 105** state–price settings and equally long chains in the others. Where plans differ, the held-out gain difference favors conditional planning with an interval above zero in 27 settings, and favors the marginal plan in two.

Across 57,405 decided chain-segment comparisons, gain-direction agreement is 77.2% for marginal planning, 88.5% for the simplified conditional rule, and 93.2% for full state matching. These comparisons reuse tasks and overlapping chains. Excluding degenerate comparisons leaves 34,746, with 69.6% versus 81.2% agreement for marginal versus simplified conditional planning (Appendix B.3).

**Live supplement: feedback changes what a retry can do.** The recorded matrix has one output per model and task, so it cannot measure another sample from the same model. We therefore run five models on 283 executable test tasks, then compare a fresh retry with a retry receiving failed-test output on the same first-failure subsets. Table 2 shows that **feedback repairs 41–54% of failures, versus 5–18% for a fresh retry**. This is information acquired from execution, not the original marginal capability being sampled again.

Table 2: Live retries supplement chain planning. Recovery rates use the same initial failures within each model; brackets are paired 95% intervals. The sandbox tests supply both feedback and the final score.

| Model           | Failed tasks | First success | Fresh recovery    | Feedback recovery        |
|-----------------|--------------|---------------|-------------------|--------------------------|
| Qwen3.5-Plus    | 37           | 0.87          | 0.11 [0.02, 0.21] | <b>0.54</b> [0.39, 0.69] |
| Qwen3-Max       | 38           | 0.87          | 0.05 [0.00, 0.12] | <b>0.47</b> [0.31, 0.64] |
| Kimi-K2.5       | 46           | 0.84          | 0.15 [0.05, 0.26] | <b>0.50</b> [0.36, 0.64] |
| DeepSeek-v4-pro | 47           | 0.83          | 0.13 [0.04, 0.24] | <b>0.51</b> [0.37, 0.67] |
| DeepSeek-flash  | 61           | 0.78          | 0.18 [0.09, 0.27] | <b>0.41</b> [0.29, 0.54] |

Feedback recovery is measured against the same tests used to guide the retry, rather than an untouched test suite. Three-attempt feedback chains cost 0.56–3.05 m\$ per resolved task versus 5.93 m\$ for recorded Opus, under the evaluated prices, but all have seconds-based efficiency ratios at most 0.83. Appendix B.3 reports model switching, budgets, and visible-test acceptance. Separately, the released OOD176 agent chain resolves 0.733 of tasks versus Opus’s 0.636; its cost-sensitive comparison is confined to Appendix B.6.

**Takeaway. Plan the next attempt from what remains after failure, and value feedback separately from blind resampling.** Conditional planning can save unproductive calls; feedback can change the prospects of the calls that remain.

#### 4.2 WHICH VERIFIER MAKES A CASCADE WORTHWHILE?

**Question.** An ideal verifier can reject incorrect cheap answers at no cost. A deployed verifier can accept errors, reject useful answers, and consume resources itself. Which accuracy–threshold–cost combination actually improves the cascade?

**Decision rule and test.** We treat each verifier and threshold as an action with its own final score and total cost. Equation (5) evaluates its efficiency gain, subject to the separate quality floor. This comparison uses accepted and rejected task outcomes; AUROC alone does not determine either quantity. Appendix H.5 derives the general operating-point criterion.

The verifier study uses recorded outputs from three primaries, 500 source-stratified probing tasks, and 334 runnable held-out tasks. It compares visible examples, locally executed hidden tests, a released-score oracle, and a Grok judge on two primaries. Verifier thresholds are evaluated for both efficiency and quality. The hidden tests are close to the benchmark’s scoring mechanism, so they provide an oracle-like comparison. The judge’s tokens are charged at the primary model’s blended token price, a proxy rather than the judge’s actual invoice. Table 3 reports Qwen3-Max; Appendix B.4 gives the complete protocol.

Table 3: Verifier choice changes the cascade’s efficiency and quality.  $r = \eta/\eta_B$ ;  $Q$  is mean score change. Judge rows include the pricing proxy. Test verifiers have no dollar charge; local verification time enters seconds.

| Verifier              | AUROC | $r(\theta = 0.5)$ [95% CI] | $r(\theta = 1)$ [95% CI] | $Q(\theta = 0.5)$ |
|-----------------------|-------|----------------------------|--------------------------|-------------------|
| Visible examples      | 0.537 | 1.98 [1.72, 2.35]          | 1.06 [1.03, 1.10]        | −0.036            |
| Hidden tests          | 0.994 | 2.87 [2.38, 3.53]          | 2.39 [2.05, 2.85]        | +0.024            |
| LLM judge             | 0.824 | 1.09 [1.00, 1.21]          | 0.64 [0.61, 0.66]        | −0.012            |
| Released-score oracle | 1.000 | 2.46 [2.09, 2.93]          | 2.46 [2.09, 2.93]        | +0.045            |

**Finding: verification gains can disappear after charging for them.** At  $\theta = 1$ , visible tests produce  $1.06\times$  efficiency, versus  $2.39\times$  for hidden tests. At  $\theta = 0.5$ , a free judge would yield  $2.63\times$ , but the cost proxy reduces this to  $1.09\times$ ; raising its threshold to 1 lowers the ratio to 0.64. Better discrimination than the visible tests therefore does not ensure better end-to-end efficiency.

Quality also changes the comparison. Accepting every Qwen3-Max output reaches  $19.65\times$  efficiency but loses 0.063 mean score, with interval  $[-0.108, -0.024]$ . A high ratio alone cannot identify a suitable verifier. The operating-point rule predicts 97.6% of 416 decided gain directions; predicted minimum quality-feasible thresholds fall within realized intervals in 95.0% of 360 comparisons. These are comparisons across shared outputs and thresholds.

**Takeaway. Choose a verifier by the quality and total cost of the resulting policy at its acceptance threshold.** Verification accuracy is an input to this decision, not its final objective.

#### 4.3 IS ROUTING INFORMATION WORTH BUYING?

**Question.** Before calling a model, a router may ask a predictor whether a cheap attempt is likely to work. Does the predictor reveal useful information beyond task metadata, and can the resulting savings pay for the prediction itself?

**Decision rule.** In the quality-preserving two-action model, attempting costs  $C_{\text{attempt}}(s) = a + (1 - s)b$ , where  $s$  is the primary’s success probability,  $a$  its cost, and  $b$  the fallback cost; direct fallback costs  $b$ . Let  $s_Z$  be the posterior after a signal costing  $k$ . Equation (3) gives the signal’s net value. Proposition 6 bounds the gross savings by exact information and identifies when even a perfect signal cannot recover its price. We report *capture*, gross saving divided by this perfect-information bound, and subtract the actual or stated signal charge separately.

**Test 1: what information do released routers supply?** The nine released CodeRouterBench router files expose model choices, not calibrated probabilities. We turn those choices into binary signals, such as “selected Qwen3-Max,” and estimate their posterior success rates by ten half-split cross-fits within the held-out population. We then compare the best signal-conditioned attempt decision with the best uninformed choice. This is a diagnostic analysis of information, with  $k = 0$ , rather than a probing-only trained deployment policy.

Table 4: Cross-fitted capture for the signal “selected Qwen3-Max,” at  $\theta = 1$  and no signal charge. Dimension-conditioned capture divides aggregated gross saving by the aggregated perfect-information bound; the baseline already knows the dimension.

| Signal source       | Pooled capture [95% CI] | Within-dimension capture |
|---------------------|-------------------------|--------------------------|
| Fine-tuned Qwen 27B | 0.555 [0.490, 0.614]    | 0.062                    |
| Fine-tuned Qwen 9B  | 0.546 [0.475, 0.608]    | 0.078                    |

**Finding: much of the apparent signal value is task-type information.** The two fine-tuned signals capture **55–56%** of perfect-information value when tasks are pooled, falling to **6–8%** after conditioning on dimension (Table 4). Both the uninformed decision and the available information bound change after conditioning. This comparison isolates incremental decision value beyond task type. Signal encoding and use also matter: some classical routers have negative native capture under “selected the primary,” but nonnegative capture under posterior-threshold use or a different binary encoding (Appendix B.5).

**Test 2: does a paid LLM forecast earn its cost?** The paid-forecast study makes 1,200 prompt-only DeepSeek-v4-pro calls, predicting whether Qwen3-Max and Kimi-K2.5 will pass hidden tests. It uses 600 probing and 600 held-out tasks, fits posterior calibration on probing, and compares with TF-IDF logistic regression on the same prompts. Held-out AUROC is 0.541/0.618 for the LLM versus 0.895/0.768 for TF-IDF. The LLM’s task-weighted capture across twelve non-degenerate source–primary settings is  $-0.037$   $[-0.225, 0.077]$ , indicating little gross decision saving.

At the evaluated peak list price, the forecast averages \$0.000944 and 1.89 seconds per call. Its dollar charge equals **2.4 Qwen3-Max attempts** as a ratio of task means; the mean net value is  $-\$0.000994$  **per task**. Appendix B.5 reports alternative price bases. An additional direct-choice Jev replay is retained as an appendix baseline (Appendix B.5).

**Takeaway. Buy a routing signal when it changes a profitable decision and the savings exceed its acquisition cost.** Prediction accuracy and a sophisticated signal source do not establish this value; the measured forecast’s negative result is specific to its decisions, population, and price.

## 5 WHAT SHOULD A ROUTER REMEMBER?

**Empirical bridge.** The main memory evidence appears in Appendix Q, where the same record policy is compared with both an always-Opus reference and an already efficient prior. This study connects the unified framework to persistent routing decisions and reports writer quality, retention rules, an ideal-write control, and the semi-synthetic construction in full.

## 6 DISCUSSION AND TAKEAWAYS

The three router decisions share one design rule: evaluate additional computation where it changes the decision and charge for the entire plan. A partner should cover failures the current portfolio leaves behind at an affordable cost. A later attempt should be assessed on the state left after earlier failures, with feedback separated from blind resampling. A verifier or routing forecast should be selected by the resulting quality and total cost, not by discrimination or accuracy alone. A record should be retained only when its expected improvement over the current prior pays for exploration, erroneous writes, and retention.

Table 5: A measurement checklist for additional routing work. Each decision needs the quantity that changes the reference-relative comparison, not a proxy reported in isolation.

| Decision        | Measure before acting                                                | Failure when omitted                                       |
|-----------------|----------------------------------------------------------------------|------------------------------------------------------------|
| Combine         | Joint failures, residual coverage, and added call cost               | A high-scoring partner repeats the same errors             |
| Spend more      | Conditional success after failure, feedback value, and verifier cost | A retry or judge looks useful only on the marginal average |
| Buy information | Decision improvement net of signal price and task-type baseline      | An accurate signal costs more than the choice it improves  |
| Remember        | Record-conditioned gain, write error, and retention cost             | Coverage grows while the deployed policy loses value       |

**A practical reading.** The equations give a compact audit trail for a routing change. First fix the quality floor and resource being optimized; then estimate the state-specific quantity in the relevant row of Table 5; finally charge every call used to obtain or act on that estimate. This ordering prevents a favorable success rate, AUROC, or cache hit rate from standing in for the end-to-end objective.

**Three implications for deployment.** Equation (1) makes portfolio construction a joint-outcome problem: a weaker specialist can be preferable when it resolves failures that the current portfolio leaves uncovered. Equation (2) makes escalation stateful: after a failure, the relevant population changes, so a router should re-estimate continuation value instead of replaying a marginal ranking. Equation (3) makes information conditional on an action: a predictor matters only when it changes which computation is purchased, and its own acquisition cost belongs in that comparison. These implications explain why the measured gains are strongest for residual coverage and feedback, while the paid prompt-only forecast is negative.

**What the experiments do and do not establish.** The evidence does not provide a universal ordering among dispatch, cascades, live agents, and intra-model routing. Outcome matrices support paired replay but use an acceptance oracle unless a verifier is specified. Live comparisons retain historical Opus outcomes, and retry feedback and final scoring use the same sandbox tests. The signal diagnostics and memory streams have their own estimation protocols, stated with their results. Prices, quality requirements, and task populations are part of every conclusion. The contribution is therefore a set of auditable decision rules and matched measurements, rather than a single leaderboard.

**Connections to emerging routing systems.** Conductor, Trinity, Fugu, and Fusion expand the router’s action menu through heterogeneous model coordination (Nielsen et al., 2026; Xu et al., 2026; Sakana AI, 2026; Zhou et al., 2026b). OpenSquilla’s harness-native routing makes execution history and feedback part of the state (Liu et al., 2026); SCOPE-Router, VIP-Router, and R2R specialize decisions to visual tasks, token pruning, and token generation (Yu et al., 2026; Liang et al., 2026; Fu et al., 2025). Cache-to-Cache, CacheBridge, and latent hand-off change switching cost and retained task value (Fu et al., 2026; Qu et al., 2026; Zou et al., 2026; Mostik AI, 2026). These mechanisms enter the same value–cost comparison. Appendix P.2 develops the mappings, while our costed Jev dispatch study (Appendix B.5) evaluates one structured decision model on matched backend outcomes. The main experiments establish the model-routing case of this broader framework.

## AI USE STATEMENT

Generative AI tools assisted with manuscript drafting and revision, literature lookup, mathematical derivation checks, and preparation of code and figures.

## ETHICS STATEMENT

Lower inference cost may improve access to capable systems, while extra calls and verification consume resources. We report quality separately from efficiency and identify cases where routing or erroneous records reduce quality. Applications must evaluate their own quality requirements and measured costs.

## REPRODUCIBILITY STATEMENT

The appendices contain proofs, analytical illustrations, and experimental protocols and results. The source README and REPRODUCIBILITY.md provide data versions, scripts, seeds, and reproduction instructions.

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## A WHICH ROUTING POLICY DOES THE OBJECTIVE SELECT?

**Question.** When direct dispatch and verified escalation are both available, which actions maximize score per dollar, and how does a quality requirement change the answer? This experiment applies Theorem 5 to a fixed menu rather than ranking router architectures in general.

**Design.** We use the released CodeRouterBench outcomes for eight models, fitting all action values, costs, and state frequencies on 7,080 probing tasks and evaluating the resulting policies on 2,919 `id_test` tasks. States are the 21 non-empty dimension $\times$ source cells, with nine dimensions as a coarser alternative. Every state has 22 actions: direct dispatch to each of eight models, or one of seven non-Opus primaries followed by Opus when its released score is below  $\theta \in \{1, 0.5\}$ . Cascade acceptance is an oracle replay; router and verifier overhead are excluded. Dollar costs use the released token-price estimates. Always-Opus has test mean score 0.4383 and cost \$34.02.

The unconstrained policy maximizes aggregate score per dollar. Constrained policies additionally impose  $\mathbb{E}[V] \geq \bar{v} - \epsilon$  on probing for  $\epsilon \in \{0, 0.005, 0.01, 0.02, 0.05\}$ . Best fixed dispatch and cascade baselines are selected on probing. ACRouter and DimensionBest use their released model selections and pay only the selected backend’s recorded cost. The per-task oracle is optimized on the test outcomes and supplies an in-sample ceiling. Intervals use 1,000 paired task-bootstrap resamples, with fitted policies held fixed.

**Findings.** **The dollar-optimal policies select non-Opus direct dispatch on every task, even though cascades are available.** This holds for both state partitions and all five fitted quality floors. The 21-cell unconstrained policy reaches 28.187-fold Opus efficiency but loses 0.0201 mean score (Table 6). Fitting a zero-loss floor yields 25.516-fold efficiency and a test score difference of  $-0.0044 [-0.0175, +0.0097]$ . Only 26.6% of paired bootstrap samples satisfy the zero-loss floor; this resampling fraction is not a calibrated probability of future compliance. The fitted constraint therefore does not establish quality preservation on new tasks. By comparison, the best fixed cascade improves quality by 0.0558 while reaching 2.004-fold efficiency.

Table 6: Dollar efficiency, cost, and quality on `id_test`.  $\Delta V = \mathbb{E}[V - Y_B]$ ; costs are m\$ per task. Brackets are paired 95% intervals. Quality floors are fitted on probing.

| Policy                      | $\eta/\eta_B$ [CI]      | Cost   | $\Delta V$ [CI]            |
|-----------------------------|-------------------------|--------|----------------------------|
| 21 cells, unconstrained     | 28.187 [26.840, 29.659] | 0.395  | -0.0201 [-0.0343, -0.0056] |
| 21 cells, $\epsilon = 0$    | 25.516 [24.278, 26.903] | 0.452  | -0.0044 [-0.0175, +0.0097] |
| 21 cells, $\epsilon = 0.01$ | 27.612 [26.311, 29.072] | 0.409  | -0.0140 [-0.0272, -0.0004] |
| Best single dispatch        | 20.097 [18.965, 21.297] | 0.525  | -0.0416 [-0.0549, -0.0275] |
| Best fixed cascade          | 2.004 [1.881, 2.140]    | 6.556  | +0.0558 [+0.0461, +0.0660] |
| ACRouter decisions          | 1.745 [1.666, 1.826]    | 7.643  | +0.0631 [+0.0528, +0.0731] |
| DimensionBest               | 3.389 [3.166, 3.638]    | 3.912  | +0.0603 [+0.0484, +0.0722] |
| Per-task oracle             | 33.510 [32.065, 35.206] | 0.368  | +0.0248 [+0.0117, +0.0376] |
| Always-Opus                 | 1.000                   | 11.655 | 0                          |

**Takeaway.** **The preferred routing mechanism depends on the objective and required quality.** On this menu and dollar basis, optimizing the efficiency ratio chooses direct dispatch; cascades buy additional quality at a substantially higher cost. A fitted quality floor must be evaluated separately on held-out tasks. These results do not establish a universal ordering among dispatch, cascades, live agents, and intra-model routing.

## B EXPERIMENTAL DETAILS

### B.1 DATA, ACCOUNTING, AND ESTIMATION

The source `README` and `REPRODUCIBILITY.md` provide data snapshots, scripts, seeds, and execution instructions.

We use the released CodeRouterBench outcome matrix `id_results_long.csv` and its accompanying `models.json` price list (Zhou et al., 2026a): 9,999 coding tasks, eight models, nine dimensions, and 21 non-empty dimension $\times$ source cells. The release splits these tasks into probing (7,080; train and validation combined) and `id_test` (2,919). The models are claude-opus-4-6, claude-sonnet-4-6, gpt-5.4, Qwen3-Max, Kimi-K2.5, Qwen3.5-Plus, glm-5, and MiniMax-M2.7. The reference, always-Opus, has mean `id_test` score 0.4383 and total estimated cost \$34.02. RouterBench uses the released 0-shot matrix of 36,497 prompts and 11 models (Hu et al., 2024); OOD176 uses ACRouter’s released 176-task agentic traces.

**Replay and live measurements.** Unless stated otherwise, a replay accepts the primary when its released score is at least  $\theta \in \{1, 0.5\}$ , then uses the recorded Opus outcome on a miss:  $V = HY_P + (1 - H)Y_B$  and  $D = C_P + (1 - H)C_B$ . This is an oracle-verifier replay with no verifier charge. Dollar costs are token-price estimates; router overhead is excluded unless explicitly priced. Sequential seconds add recorded primary and fallback latencies. The live retry, verifier, paid-forecast, Jev dispatch, and memory studies include fresh calls through one relay, but all Opus and Sonnet outcomes remain recorded; those comparisons combine live calls with replayed reference outcomes. No end-to-end concurrent system is timed.

**Samples and intervals.** Primitives and policy choices are fitted on probing and evaluated on `id_test`, except the explicitly cross-fitted signal analysis below. Intervals are 95% percentile task-bootstrap intervals, paired for policy comparisons, with 500 or 1,000 resamples; record simulations also resample streams. A sign is decided when its realized interval excludes zero. These random splits measure stability within the same task pools. Configuration counts reuse tasks and are not counts of independent experiments. Efficiency is the ratio of aggregate value to aggregate resource, not the average of task-wise ratios.

The released matrix includes 148 missing Qwen3-Max runs recorded as score 0 and cost 0. They remain in the dollar analyses; their removal is a sensitivity. Latency analyses exclude zero-latency rows unless noted; the fixed-menu optimization study retains the release’s latency entries. Machine-precision identity checks and simulations that impose a theorem’s assumptions are implementation checks, not empirical evidence for those assumptions.

### B.2 RESIDUAL COVERAGE ON TWO DATASETS

An oracle selecting verifier commits the best passing portfolio member and uses the reference on a miss; costs sum member calls, with no aggregation fee. CodeRouterBench uses the seven non-Opus candidates, portfolios of up to three members, and nine dimension contexts. A candidate needs at least 20 passes and 20 failures, and a non-trivial comparison at least ten tasks in each conditional cell. RouterBench uses gpt-4-1106-preview as reference, ten candidate additions to 21 one- or two-member portfolios, and contexts from eight source families and 57 MMLU subsets. Its split is 25,542 fit and 10,955 held-out prompts, stratified by source and reference success; filtering and ranking use the fit split.

The score  $N_S = r_m(S)[v_h - v_m + \eta(S)b] - \eta(S)d_m$  uses the candidate’s residual coverage and incremental cost. The independence baseline replaces  $r_m(S)$  with  $(1 - p(S))\mathbb{P}(E_m)$ ; other baselines rank by marginal success or residual coverage alone. In addition to the main-text sign and best-partner results, RouterBench gives median Spearman correlation with realized gain 0.928 for  $N_S$ , 0.080 for residual coverage alone, and  $-0.412$  for marginal success.

The common-conditional-value assumption is imperfect on both datasets. Adding a CodeRouterBench model lowers miss-conditional value in 97% of the comparisons, with median changes of  $-13.7\%$  and  $-17.5\%$  at the two thresholds. In the pooled Qwen3-Max portfolio, adding Qwen3.5-Plus is predicted to lose 1.4 score per dollar but gains 5.0 [3.5, 6.6], because the new failure set lowers

fallback cost by 14%. RouterBench shows conditional-value violations in 56% of 6,392 comparisons and median absolute relative magnitude error of 7.4%. These results support useful sign prediction on these splits, not exact magnitude prediction under changing conditional values.

### B.3 FAILURE-CONDITIONED CHAINS AND LIVE RETRIES

**Recorded chains.** The chain-planning study enumerates 259 ordered chains of one to three distinct non-Opus models, followed by Opus, at  $\theta \in \{1, 0.5\}$ . A post-failure state contains the identities of all models already tried and their failures. Probing estimates determine whether to attempt another model or stop and call Opus; the selected chain is then replayed on test tasks. Five resource prices yield 57,405 interval-decided 21-cell segments, shared across tasks and chains. Sign agreement is 88.5% for the simplified attempt rule, 77.2% for marginal planning, and 93.2% for the fully state-matched recursion. Restricting to 34,746 non-degenerate segments gives 81.2% and 69.6% for the first two rules.

**Single-step control.** The corresponding one-step decision compares an attempted model followed by Opus against calling Opus directly. Primitives are fitted on probing, and the gain sign is evaluated on test tasks. Across the 9-dimension and 21-cell partitions, the attempt rule agrees with 92.5% and 93.3% of interval-decided signs (148/160 and 305/327 decided comparisons, from 180 and 420 total), versus 76.3% and 63.9% for always attempting. This supplies the one-step control for the chain-planning study; the non-positive-margin branch contains only one physical cell.

**Live retries with a recorded terminal.** The live retry study makes 1,415 first-attempt chains on 283 executable `id.test` tasks, excluding tasks scored zero by all eight recorded models. Five models use one relay and completion budgets of 2,048 tokens, or 6,144 for DeepSeek-flash; exhausted attempts count as failures. After failure, we compare a fresh retry, a retry receiving failing-test output, and a model switch receiving the same context. Opus is the recorded terminal. The hidden-test sandbox both supplies retry feedback and scores the output, so improvement measures repair against those tests, not a separate untouched test set.

Three-attempt feedback chains cost 0.56–3.05 m\$ per resolved task against 5.93 m\$ for recorded Opus; DeepSeek dollar prices are proxies. Every live chain has seconds-based  $\eta/\eta_B \leq 0.83$ . Replacing hidden tests with visible tests at  $\theta = 0.5$  yields erroneous acceptance rates 0.169–0.233 and chain quality differences of  $-0.085$  to  $-0.141$ . These additional measurements separate recovery with test feedback from the quality loss caused by imperfect acceptance.

### B.4 VERIFIER OPERATING POINTS

The verifier study evaluates recorded responses from three primaries on 500 source-stratified probing and 334 runnable `id.test` tasks. Verifiers are visible examples extracted by ACRouter’s tool, hidden tests run locally, the released score oracle, and a grok-4.6 judge (thinking off; Qwen3-Max and Kimi-K2.5 only; 1,668 live calls). The judge was selected using a 60-item, five-configuration pilot. Its token cost is charged at the primary’s blended token price: this is a pricing proxy, not the judge’s own invoice or listed price.

Accepting every Qwen3-Max output reaches 19.65-fold efficiency but loses  $Q = -0.063$   $[-0.108, -0.024]$ , violating floors with  $\epsilon \leq 0.02$ . The judge reaches 2.63-fold efficiency if its calls are free, versus 1.09 under the proxy charge. The operating-point floor predicts 97.6% of 416 decided signs among 450 operating points, and the predicted minimum quality-feasible threshold lies in the realized interval in 95.0% of 360 cells. These are threshold comparisons on shared outputs; false acceptance still matters for quality even where it cancels from a particular efficiency-sign condition.

### B.5 RELEASED AND LIVE ROUTING SIGNALS

**Released signals.** The nine CodeRouterBench router files expose the chosen model, not a full forecast. The released-signal analysis therefore forms binary signals: chose Qwen3-Max or chose any non-Opus model. Primitive costs and base rates come from probing, while posterior lifts are cross-fitted over ten half-splits of the `id.test` state. This is a diagnostic use of labeled test data,

not a router trained exclusively on probing. Native use attempts on  $Z = 1$ ; posterior-threshold use attempts when  $s_Z > a/b$ . Capture is gross cost saving divided by the perfect-information value, before the router’s own charge.

For  $Z = \text{chase}$  Qwen3-Max, classical routers have negative native capture because their  $Z = 0$  tasks still satisfy  $\mathbb{P}(H = 1 \mid Z = 0) > a/b = 0.043$ : withholding the attempt loses profitable opportunities. Posterior-threshold use of those signals has zero capture. Defining  $Z$  as choosing any non-Opus model changes the question and gives positive capture for some of the same routers.

**A fresh LLM forecast.** The paid-forecast study samples 600 probing and 600 `id_test` tasks by source and cell from 3,601 executable, non-all-zero tasks. DeepSeek-v4-pro reads the prompt and predicts whether Qwen3-Max and Kimi-K2.5 pass all hidden tests (1,200 calls). Its posterior is fitted by pooled probing isotonic regression and anchored to the state’s probing base rate; the decision threshold is  $a/b$ . Held-out AUROC is 0.541 and 0.618, versus 0.895 and 0.768 for TF-IDF logistic routing fitted on the same prompts. The forecast’s task-weighted capture across 12 non-degenerate source $\times$ primary rows is  $-0.037 [-0.225, 0.077]$ .

At the selected peak list price, a call averages 635 tokens, 1.89 seconds, and \$0.000944. It costs 2.4 Qwen3-Max attempts as a ratio of task means, or 2.87 when averaging state-wise ratios. The task-weighted net value is  $-\$0.000994$ , approximately the signal cost with a negative sign. This finding is specific to the forecast and price basis:  $k \geq a$  holds in 22 of 24 Qwen3-Max states at peak price, 19 at off-peak price, and none under the self-hosted proxy; measured seconds also give none. High predictive discrimination alone need not create savings when the profitable action seldom changes.

**Jev direct-choice routing.** The Jev dispatch study tests a different use of a structured decision model: ex-ante dispatch rather than post-output verification. For each task, Jev receives only the prompt, source and dimension metadata, and short descriptions of the eight released backends; its typed Choice interface follows TypeSafe AI (2026c). It returns one `Choice`; no candidate output, score, label, or price is sent to the model. The selected backend is then scored offline against the released outcome matrix, with the Jev input charge included in the policy cost. The primary run covers the complete 2,919-task `id_test` split and the complete OOD176 set (3,095 calls; zero errors; 3.085M input tokens; \$0.12956 estimated at \$0.042 per million input tokens). The request model was `jev-latest`, resolved as `jev-1.13.0`.

Table 7: Direct Jev Choice dispatch scored on released outcomes. Policy cost includes the Jev input charge; baseline costs are backend costs.

| Split                | Policy          | Mean score | Pass@1 | Cost/task (\$) |
|----------------------|-----------------|------------|--------|----------------|
| <code>id_test</code> | Jev, all eight  | 0.388      | 21.6%  | 0.000662       |
|                      | Fixed Qwen3-Max | 0.397      | 22.0%  | 0.000525       |
|                      | Fixed Opus      | 0.438      | 24.8%  | 0.011655       |
| OOD176               | Jev, all eight  | 0.267      | 26.7%  | 0.075656       |
|                      | Fixed Qwen3-Max | 0.330      | 33.0%  | 0.023903       |
|                      | Fixed Opus      | 0.636      | 63.6%  | 1.107433       |

On `id_test`, Jev’s mean score is 0.388 versus 0.397 for fixed Qwen3-Max, while its total cost is 26% higher after charging the query. On OOD176, it scores 0.267 versus 0.330 and costs 3.17 times as much as fixed Qwen3-Max. Jev selected Qwen3-Max, Qwen3.5-Plus, and claude-sonnet-4-6 only (2,724/193/2 on `id_test`; 83/81/12 on OOD176), and never selected GPT-5.4 or Opus. Thus this direct-choice run supplies an end-to-end costed routing baseline, not evidence that Jev can verify a generated answer or preserve a quality floor. It is an offline replay of recorded backend outcomes; no backend answers were regenerated.

## B.6 RELEASED AGENTIC CHAIN

On OOD176, ACRouter’s released chain resolves 0.733  $[0.670, 0.795]$  of tasks versus 0.636 for Opus. At released costs it reaches  $\eta/\eta_B = 2.59 [1.97, 3.54]$ , versus 1.91 for the best oracle cascade. However, 168 of 469 chain attempts have no recorded run and are billed at zero; imputing their cost

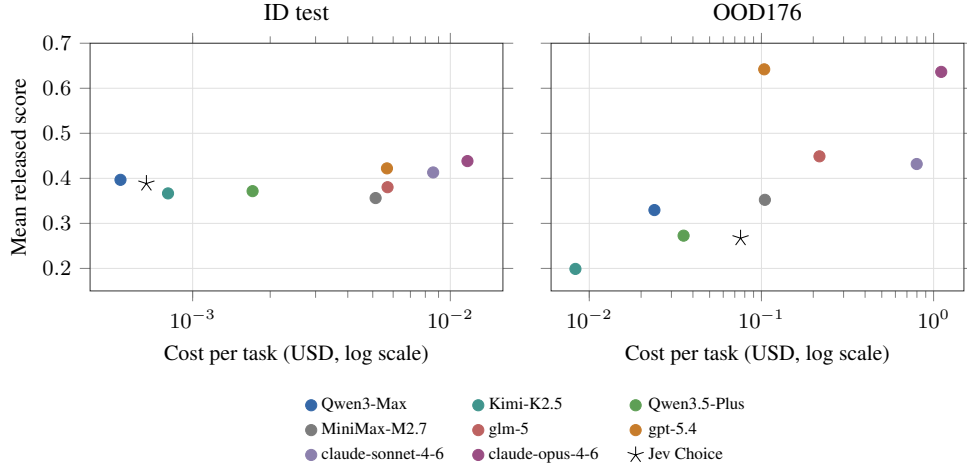


Figure 2: Cost–quality replay for Jev dispatch. Each panel shows fixed-backend outcomes and the Jev Choice policy; Jev input cost is included in its starred point. The OOD panel uses the 176-task released resolved indicator.

from the model’s mean recorded old112 cost reduces the ratio to 1.75 [1.41, 2.23]. The new64 subset also uses provider-imputed costs and source-model substitutions. The task-resolution comparison is therefore more interpretable than the released cost ranking.

Charging \$0.01 or \$0.05 per sandbox attempt reduces the released-cost ratio to 2.46 or 2.06. The most efficient ex-ante OOD176 baseline reaches 24.29-fold efficiency but loses 0.307 mean quality and fails the  $\epsilon = 0.01$  floor. These descriptive replay comparisons illustrate why cost rankings must be read together with quality.

#### B.7 ALLOCATION UNDER OBSERVED COVERAGE CURVES

The allocation study asks whether the closed-form allocation law remains useful on empirical coverage curves. A capacity unit adds one prepared model, up to seven per task dimension, in an order fixed on probing. All allocation rules, including the dynamic-programming optimum of the empirical probing curves, are fitted on probing and then evaluated on `id_test`; intervals use 1,000 paired task resamples. Price-ordered marginal coverage gains increase by more than 0.02 in three of nine dimensions at each threshold, and fitted exponents differ across dimensions. Thus the strict-concavity and common-exponent assumptions do not generally hold. At  $\theta = 1$  with unit preparation costs, the literal  $d_k = 1$  closed form achieves 11.4% less held-out coverage than the probing-selected dynamic-programming allocation; the free-intercept variant loses 27%. Equalizing separately fitted per-dimension marginals is 3.1% below that same baseline and 10.2% above uniform, but is a heuristic outside the stated hypotheses. These gaps average over the budget sweep; the baseline is not optimized on test outcomes. The nearly uniform literal dimension-level solution provides little discrimination against uniform allocation.

#### B.8 SUPPLEMENTARY REFERENCE AND RESOURCE SENSITIVITY

This appendix-only analysis varies the reference (Opus, DimensionBest, a per-task oracle, and 20 random draws), resource price, dollars/tokens/seconds, simulated verifier noise, and a common shift of score zero. Record streams reuse the oracle-write control’s constructed arrivals. On latency-valid tasks Opus has mean latency 7.8 seconds, compared with 8.2–44.9 for the other models; none of the 14 fixed primary-then-Opus cascades improves seconds-based efficiency. These are comparisons of recorded latencies, not controlled concurrency measurements.

Records gain +28.62 [26.22, 30.90] score per dollar against Opus but lose −0.0153 [−0.0166, −0.0142] score per second. Against DimensionBest the dollar difference is −23.83 [−27.47, −20.17]. The probing record criterion tracks these reversals. Simulated false-accept and false-reject probabilities up to 0.1 do not reverse the reported qualitative statements; redrawing

verifier randomness widens record-stream intervals by a median factor of 1.21. Translating every score changes ratio comparisons through the saved-resource term, so the chosen value zero remains part of the interpretation.

## B.9 STATISTICAL SCOPE

The analyses are exploratory. Configuration grids share tasks, thresholds, chains, and replay streams; binary-score dimensions duplicate their two threshold settings. Some noisy-verifier intervals condition on a single draw of verifier errors. Probing and test are random splits of the same pools, and recorded Opus cost and latency differ by about 6% between them. Chain-sign intervals hold fitted predictions and the interval-decided comparison set fixed. These intervals describe stability on the stated task populations.

## C FORMAL ROUTER MODEL AND COMPARISON

**Router policies and actions.** A decision round is one decision with a measurable outcome; its observable state  $x_n$  includes the task and the available history, and an information state  $I_n$  holds records from earlier rounds. A router  $g$  selects an action and updates its information from feedback  $f_n$ :

$$u_n = g(x_n, I_n), \quad I_{n+1} = \text{Upd}(I_n, x_n, u_n, f_n). \quad (7)$$

An action is an execution plan: which capabilities run (models, tools, skills, or submodels), how (one path, a parallel portfolio, or a sequence of attempts), and a commit rule that ends the task (Table 8). An ex-ante router commits before execution, possibly after paying for a routing signal. A verify-then-escalate router runs and checks a cheap primary, then escalates on rejection. An agentic router may dispatch one model per task and retain its verified outcome (Zhou et al., 2026a), or re-route and retry within the agent loop (Liu et al., 2026). An intra-model router selects submodels, steps, or token budgets, with full computation as a possible reference (Liang et al., 2026; Tang et al., 2026; Zhao et al., 2026; Fu et al., 2025). All four consume resources and deliver value; the cascade is the running example because its costs are explicit. Theorem 5 requires a policy-independent state distribution. Within-task retries after failure are instead covered by the escalation recursion of Proposition 7.

**Comparing routing policies.** On a common task population, let  $(V, D)$  be the value delivered and resource consumed under the router and  $(Y_B, C_B)$  those of a reference policy, all nonnegative and integrable, with  $\bar{c} = \mathbb{E}[C_B] > 0$ ,  $\mathbb{E}[D] > 0$ ,  $\bar{v} = \mathbb{E}[Y_B]$ , and  $\eta_B = \bar{v}/\bar{c}$ . For a stationary ergodic sequence of rounds, the long-run value per resource is

$$\eta = \lim_{N \rightarrow \infty} \frac{\sum_{n < N} V_n}{\sum_{n < N} D_n} = \frac{\mathbb{E}[V]}{\mathbb{E}[D]} \quad \text{almost surely}, \quad (8)$$

a ratio of aggregates, not the average of  $V_n/D_n$ .

**Lemma 1** (Reference-relative gain). *Under the model above, define the quality change  $Q$  and the resource saving  $S$  by*

$$Q = \mathbb{E}[V - Y_B], \quad S = \mathbb{E}[C_B - D]. \quad (9)$$

*Then  $\mathbb{E}[V] = \bar{v} + Q$ ,  $\mathbb{E}[D] = \bar{c} - S$ , and*

$$\eta - \eta_B = \frac{Q + \eta_B S}{\bar{c} - S}. \quad (10)$$

*Equivalently,*

$$Q + \eta_B S = \mathbb{E}[V - \eta_B D], \quad \mathbb{E}[V - \lambda D] = \mathbb{E}[D] (\eta - \lambda) \quad (\lambda > 0). \quad (11)$$

*Consequently, efficiency improves if and only if the net gain  $Q + \eta_B S$  is positive; a quality constraint  $\mathbb{E}[V] \geq \bar{v} - \epsilon$  is the separate condition  $Q \geq -\epsilon$ . (Proof in Appendix H.1.)*

| Regime               | Commits                         | Evidence                                             | Fallback or reference                                                                   | Examples                                               |
|----------------------|---------------------------------|------------------------------------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------|
| Ex-ante dispatch     | before execution                | none, or a paid routing signal $Z$                   | none                                                                                    | RouteLLM, SCOPE-Router                                 |
| Verify-then-escalate | after a cheap attempt           | explicit verifier, $H$                               | a fixed stronger path                                                                   | FrugalGPT, AutoMix, LLM cascades                       |
| In-loop agentic      | per task, or at each agent step | implicit and execution-grounded: sandbox runs, tests | none within the task (the record improves later tasks), or any re-route/retry at a step | ACRouter (per task), harness-native routing (per step) |
| Intra-model          | per submodel, step, or token    | task loss or reward in training                      | full computation as reference                                                           | VIP-Router, RDS, RAPID <sup>3</sup>                    |

Table 8: Four routing regimes under one accounting. An executed fallback and a comparison reference play different roles.

**Scope of the ratio objective.** The ratio  $\eta$  is the right objective when saved resources can be redeployed at the reference’s marginal value, so that  $\eta_B$  prices a unit of resource; under a fixed workload or an external quality requirement, the linear objective  $J_\lambda = \mathbb{E}[V - \lambda D]$  with an exogenous price  $\lambda$  and the floor  $\mathbb{E}[V] \geq \bar{v} - \epsilon$  is the right form. Values need a meaningful zero: adding a constant to every value can change the sign of any criterion below with nonzero resource saving. Every criterion below is (11) against a named reference (Appendix H.2) and compares efficiencies only; Theorem 5 adds the floor (Appendix H.3 shows the ratio alone accepting a lossy router).

**Verified execution with escalation.** The cascade rule (choose a primary and a fallback plan, run and verify the primary, commit it on acceptance  $H = 1$ , otherwise ( $H = 0$ ) run the fallback, then record the outcomes) specializes the router. An output is *sufficient* when the verifier accepts it as adequate for the task; a *hit* is an accepted primary output, and a *miss* invokes the fallback. Let  $Y_P$  be the primary value,  $(Y_F, C_F)$  the fallback value and *additional* resource if invoked, and  $A$  the primary cost including routing, aggregation, verification, and update resources. The verifier may be a model judge or an execution check such as a sandbox run or a test suite; the accounting is the same, its cost enters  $A$ , and its false accepts enter  $Q$ . Then

$$V = HY_P + (1 - H)Y_F, \quad D = A + (1 - H)C_F, \quad (12)$$

$$Q = \mathbb{E}[H(Y_P - Y_B) + (1 - H)(Y_F - Y_B)], \quad S = \mathbb{E}[HC_B - A - (1 - H)(C_F - C_B)]. \quad (13)$$

Here  $\mathbb{E}[HC_B]$  is the avoided reference work and  $A$  includes gate calls, routing signals, and lost cache discounts (Appendix H.8); no independence between  $H$  and the reference outcomes is assumed.

**The homogeneous cascade.** With hit probability  $p$ , hit and miss values  $v_h, v_m$ , primary cost  $a > 0$ , and fallback cost  $b > 0$ ,

$$\eta(p) = \frac{v_m + p(v_h - v_m)}{a + (1 - p)b}. \quad (14)$$

If the fallback restores the reference on every miss and the reference’s conditional means do not depend on  $H$  (a reference-restoring fallback), then  $v_m = \bar{v}$ ,  $b = \bar{c}$ , and for  $v_h, \bar{v} > 0$  the cascade is at least as efficient as the reference if and only if

$$p \geq \frac{\bar{v}}{v_h} \frac{a}{b}. \quad (15)$$

With  $v_h = \bar{v}$  and  $b = \bar{c}$ , a tenfold price advantage needs 10% acceptance. The floor holds only without selection: if accepted tasks are those the reference also solves cheaply, so that  $\bar{c}_h = \mathbb{E}[C_B \mid H = 1] < \bar{c}$  and  $\bar{v}_h = \mathbb{E}[Y_B \mid H = 1]$ , the condition is  $p(v_h - \bar{v}_h) + \eta_B(p\bar{c}_h - a) \geq 0$  (Appendix H.6); when the accepted output matches the reference on the same tasks ( $v_h = \bar{v}_h$ ), the floor is  $a/\bar{c}_h$ , or 40% at  $\bar{c}_h = \bar{c}/4$ . The normalized efficiency

$$\frac{\eta}{\eta_B} = \frac{pG + 1 - p}{1/P + 1 - p}, \quad G = \frac{v_h}{\bar{v}}, \quad P = \frac{b}{a}, \quad (16)$$

separates quality retention from cost ratio; an ex-ante router, lacking the wasted primary term, is a different specialization (Appendix H.4).

**The verifier’s operating point.** Let  $H = 1\{\Sigma \geq \theta\}$ , let  $\pi_0$  be the primary’s correctness probability, and let  $\text{TPR}(\theta), \text{FPR}(\theta)$  be acceptance probabilities conditional on correctness or error. With binary value,  $p = \pi_0 \text{TPR} + (1 - \pi_0) \text{FPR}$  and  $p v_h = \pi_0 \text{TPR}$ . If reference success is independent of primary correctness, its cost is constant, and  $\Sigma$  is conditionally independent of  $Y_B$  given primary correctness, the homogeneous floor reduces to  $\pi_0 \text{TPR}(\theta) b \geq \bar{v} a$ ; only under these assumptions does the false-accept term cancel. Accepting everything is then efficiency-optimal whenever  $a/b \leq \pi_0/\bar{v}$ , but can violate the separate quality floor. Appendix H.5 gives the general criterion with the false-accept term and the quality-constrained operating point.

## D FORMAL RESULTS FOR MODEL COMBINATIONS

Let  $E_m$  be the event that candidate  $m$  supplies a sufficient output on a shared task, and let an exact selecting verifier pick a sufficient candidate whenever one exists. For a portfolio  $S$ ,

$$p(S) = \mathbb{P}\left(\bigcup_{m \in S} E_m\right), \quad r_m(S) = \mathbb{P}\left(E_m \cap \bigcap_{i \in S} E_i^c\right); \quad (17)$$

the residual coverage  $r_m(S)$  is the extra success probability supplied by  $m$ , under arbitrary correlations.

**Proposition 2** (Efficiency-aware residual coverage). *Suppose the verifier is exact (it selects a sufficient candidate whenever one exists), the compared portfolios have common conditional values  $v_h, v_m$  and fallback cost  $b$ , and they follow (14) with primary cost  $a(S)$  and efficiency  $\eta(S)$ . Let  $d_m(S) = a(S \cup \{m\}) - a(S)$ , including any change in aggregation and verification cost. Adding  $m$  strictly improves efficiency if and only if*

$$r_m(S)[v_h - v_m + \eta(S)b] > \eta(S)d_m(S). \quad (18)$$

(Proof in Appendix L.1.)

This is Lemma 1 with the current portfolio as reference. Imperfect verifiers require comparing realized values and costs through that identity. Under exact selection,  $r_m(S) = p(S \cup \{m\}) - p(S)$  is a joint property that marginal scores cannot identify:

**Proposition 3** (A limitation of marginal-quality selection). *Let a mandatory primary  $m_0$  be paired with one of two candidates  $m_1, m_2$ , all of cost  $c > 0$ , with a perfect fallback of cost  $b > 0$  and unit value for every completed round. Two task distributions with identical individual success probabilities  $(1/2, 1/2, 1/2)$  exist such that the optimal pair costs  $2c$  while every possibly randomized selector fed only these marginals has worst-case expected cost at least  $2c + b/4$ . (Proof in Appendix L.2; Remark 14 gives the identifiable range of  $r_m(S)$ .)*

**Steering and allocation.** A role or context view can steer a candidate toward the portfolio’s failures, improving conditional coverage if and only if the receiving answer region has greater residual sufficiency (Proposition 15, Appendix L.3); since  $p(S)$  is monotone submodular, greedy and knapsack guarantees apply (Nemhauser et al., 1978; Sviridenko, 2004), while adaptive selection needs adaptive submodularity (Golovin & Krause, 2011) (Appendix K.4).

## E FORMAL RESULTS FOR ROUTING DECISIONS AND INFORMATION

A router also decides whether to attempt its primary, dispatch directly to a stronger path, or pay for information that changes this choice. If the primary costs  $a > 0$ , is sufficient with probability  $p$ , and delivers  $v_h$  on a hit, while a miss adds fallback cost  $b$  and yields the direct path’s value  $v_b$ , define for  $\lambda > 0$

$$W_\lambda = \lambda b + v_h - v_b, \quad (19)$$

the value of turning a miss into a hit.

**Proposition 4** (Attempt-then-escalate threshold). *Attempting the primary first is at least as good as direct fallback if and only if  $pW_\lambda \geq \lambda a$ . If  $W_\lambda > 0$ , this is  $p \geq \lambda a/W_\lambda$ ; if  $W_\lambda \leq 0$ , attempting the primary is strictly worse. A routing refinement costing  $k$  that increases  $p$  by  $\Delta p$  while preserving the other quantities improves the attempt-then-escalate objective if and only if  $\Delta p W_\lambda \geq \lambda k$ ; it is worthwhile overall only if the refined cascade also clears the direct-path comparison. (Proof in Appendix I.1.)*

Here  $b$  and  $v_b$  are miss-conditional means. If the reference’s outcomes depend on acceptance, a hit saves the reference’s cost on the accepted tasks,  $\bar{c}_h$ , rather than  $b$ ; then  $W_\lambda = \lambda \bar{c}_h + v_h - \bar{v}_h$  and the band (21) below holds with  $\bar{c}_h$  in place of  $b$  (Appendix H.6).

**Theorem 5** (Efficiency-optimal routing with a quality floor). *Let the observable state  $x$  take finitely many values with router-independent probabilities  $\pi(x) > 0$ , let each state offer a finite menu  $\mathcal{U}(x)$  of actions with expected value  $v(x, u)$  and cost  $c(x, u) > 0$ , and let a router choose  $u$  with probability  $g(u | x)$ . Put  $\Phi(\lambda) = \sum_x \pi(x) \max_{u \in \mathcal{U}(x)} [v(x, u) - \lambda c(x, u)]$ . (i)  $\Phi$  is convex and strictly decreasing with a unique root  $\eta^* = \max_g \eta(g)$ , and  $g$  is efficiency-optimal if and only if in every state it uses only actions maximizing  $v(x, u) - \eta^* c(x, u)$ ; for the menu {direct, attempt-then-escalate} this is the rule  $p(x)W_{\eta^*}(x) \geq \eta^* a(x)$ . (ii) If the reference policy is admissible, the iteration  $\lambda_0 = \eta_B$ ,  $\lambda_{t+1} = \eta(g_t)$  with  $g_t$  maximizing the  $\lambda_t$ -objective reaches  $\eta^*$  in finitely many steps, and its first step is the gain test of Lemma 1:  $\Phi(\eta_B) = \max_g [Q_g + \eta_B S_g]$ . (iii) If the reference is admissible, then under the floor  $\mathbb{E}_g[V] \geq \bar{v} - \epsilon$ , which it satisfies, an optimal router exists that randomizes in at most one state, and a multiplier  $\kappa \geq 0$  exists such that every optimal router uses only actions maximizing  $(1 + \kappa)v(x, u) - \eta_\epsilon^* c(x, u)$ ,  $\eta_\epsilon^*$  being the constrained optimum. For the two-action menu this is Proposition 4 at price  $\eta_\epsilon^*/(1 + \kappa)$ . (Proof in Appendix I.3.)*

The proof combines Dinkelbach’s method (Dinkelbach, 1967) with linear-programming duality for the floor; Lemma 1 is its first step, verifier operating points can enter  $\mathcal{U}(x)$ , and the floor acts only through  $\kappa$ , which raises a hit’s value relative to its cost.

**Paying for a more informed routing decision.** Suppose accepted outputs and the fallback preserve value, the primary is sufficient with probability  $s$ ,  $0 < a < b$ , and exact revelation of the sufficiency event costs  $k \geq 0$ ; the three choices cost

$$C_{\text{cascade}}(s) = a + (1 - s)b, \quad C_{\text{direct}}(s) = b, \quad C_{\text{info}}(s) = k + sa + (1 - s)b, \quad (20)$$

the informed choice avoiding an unsuccessful attempt but paying for a successful one.

**Proposition 6** (The routing-information band). *Exact routing information is strictly cheaper than both uninformed choices if and only if*

$$\frac{k}{b - a} < s < 1 - \frac{k}{a}. \quad (21)$$

*This band is nonempty if and only if  $k < a(b - a)/b$ , and it then contains the uninformed crossover  $s = a/b$ . (Proof in Appendix I.4.)*

For an imperfect experiment  $Z$  with posterior  $s_Z$ , the net value is

$$\min\{C_{\text{cascade}}(s), b\} - \mathbb{E} \min\{C_{\text{cascade}}(s_Z), b\} - k; \quad (22)$$

since  $\mathbb{E}[s_Z] = s$  and exact revelation dominates any experiment about the same event (Blackwell, 1953), the band bounds where a learned router can pay for itself (Figure 3, Appendix N). Exact information pays only if  $k < a$ , and at equal cost it beats raising sufficiency by  $\Delta p$  if and only if  $b \Delta p < a(1 - s)$  (Corollary 12; Appendix I.5); an uncalibrated signal can be worse than none even at  $k = 0$ .

**Escalation inside a loop.** Let a task allow up to  $K$  attempts; attempt  $k$  costs  $a_k$  including its verification, is accepted with probability  $p_k$  given  $k - 1$  failures, and delivers  $v_h$ ; after  $K$  failures a terminal path of cost  $b$  and value  $v_b$  is used. Write  $J^{\text{direct}} = v_b - \lambda b$  and let  $\Pi_k$  be the optimal  $J_\lambda$  after  $k - 1$  failures, with  $\Pi_{K+1} = J^{\text{direct}}$ .

**Proposition 7** (Escalation decisions inside a loop).  $\Pi_k = \max\{J^{\text{direct}}, p_k v_h - \lambda a_k + (1 - p_k)\Pi_{k+1}\}$ , and attempt  $k$  is optimal if and only if  $p_k W_\lambda + (1 - p_k)\Delta_{k+1} \geq \lambda a_k$ , where  $\Delta_{k+1} = \Pi_{k+1} - J^{\text{direct}} \geq 0$ . For the last attempt  $\Delta_{K+1} = 0$  and the rule is Proposition 4; the option value  $\Delta_{k+1}$  of later attempts can only lower the bar. ACRouter’s per-task reward  $\epsilon_1 s_i + \epsilon_2 \kappa_i$  for verified score  $s_i$  and cost  $\kappa_i$  (Zhou et al., 2026a), with  $\epsilon_1 > 0 > \epsilon_2$ , is  $\epsilon_1 J_\lambda$  with  $\lambda = -\epsilon_2/\epsilon_1$ , with  $J_\lambda(x, u)$  the objective of action  $u$  in state  $x$ , and cumulative regret against a per-task oracle is  $\epsilon_1 \sum_i [\max_u J_\lambda(x_i, u) - J_\lambda(x_i, u_i)]$ . (Proof in Appendix I.2.)

A step-level agentic router (Liu et al., 2026) is thus the fixed-price rule of Proposition 4 at each post-failure state plus the option value of later attempts, with  $p_k$  from execution-grounded verification and  $a_k$  including the sandbox run. The cascade is  $K = 1$ . In its released ID dispatch mode, ACRouter (Zhou et al., 2026a) is  $K = 1$  with no terminal fallback ( $b = 0$ ,  $v_b$  the value of the failed output): its verification does not rescue the current task but writes the record that Section F values.

**Parallelism changes the resource being optimized.** Under concurrent execution of primary and fallback, latency is set by completion times rather than summed charges, and cancellation saves only the unbilled suffix; Appendix J.2 gives the latency and dollar criteria.

## F WHAT SHOULD A ROUTER REMEMBER?

A stateful router uses past outcomes through  $I$  in (7). ACRouter stores the chosen model, verified score, cost, and execution trace in a bounded embedding-keyed store and retrieves nearest neighbors (Zhou et al., 2026a). We study an idealized partition into fixed cells that share a record, with retention as in Section F.1; this does not model the full retrieval process. Covering a cell is distinct from learning which arm is best within it (Appendix M.10).

**A record process.** Tasks fall into  $J$  cells arriving independently with probabilities  $\nu_j > 0$ . At the start of a round, each cell is uninformed ( $U$ ) or holds an informative ( $I$ ) or erroneous ( $W$ ) record. After the round’s decision, each record expires independently with probability  $\delta_j$ , and the arriving cell receives an informative write with probability  $\omega_j$  or an erroneous write with probability  $\psi_j$ , replacing its old record. Write types are independent of the past and of the current record (Remarks 16 and 17 and Appendix M.4 relax this). Set

$$r_j = \omega_j + \psi_j > 0, \quad \alpha_j = (1 - \delta_j)(1 - \nu_j r_j) < 1. \quad (23)$$

A record written after round  $n$  first affects round  $n + 1$ ;  $v_j^s, c_j^s$  denote the router’s expected value and cost in cell  $j$  and record state  $s \in \{U, I, W\}$ , including the record update.

**Measuring a record by its gain.** The uninformed reference router, which collects records at the same cost but always routes as if uninformed, has efficiency

$$\eta_U = \frac{\sum_j \nu_j v_j^U}{\sum_j \nu_j c_j^U}. \quad (24)$$

Assume its denominator is positive, let  $\bar{c}_\infty > 0$  and  $\eta_\infty$  be the stationary expected cost and efficiency, and define the record gain and the stationary gain of a cell by

$$g_j^s = (v_j^s - v_j^U) - \eta_U (c_j^s - c_j^U), \quad s \in \{I, W\}, \quad (25)$$

$$G_j = \frac{\nu_j (\omega_j g_j^I + \psi_j g_j^W)}{1 - \alpha_j}. \quad (26)$$

**Proposition 8** (Transient and stationary memory gains). *From an empty store, the probabilities of informative and erroneous records before round  $n \geq 0$  are*

$$i_{j,n} = \frac{\nu_j \omega_j}{1 - \alpha_j} (1 - \alpha_j^n), \quad w_{j,n} = \frac{\nu_j \psi_j}{1 - \alpha_j} (1 - \alpha_j^n). \quad (27)$$

*The stationary efficiency satisfies  $\eta_\infty \geq \eta_U$  if and only if  $\sum_j \nu_j G_j \geq 0$ . At every horizon  $N$ ,*

$$\sum_{n=0}^{N-1} \mathbb{E}[V_n - \eta_U D_n] = \sum_j \nu_j G_j \left[ N - \frac{1 - \alpha_j^N}{1 - \alpha_j} \right]. \quad (28)$$

*If  $g_j^I > 0 > g_j^W$ , cell  $j$  contributes a positive stationary gain if and only if*

$$\frac{\psi_j}{r_j} < \frac{g_j^I}{g_j^I - g_j^W}. \quad (29)$$

(Proof in Appendix M.1.)

The recurrence  $i_{j,n+1} = \nu_j \omega_j + \alpha_j i_{j,n}$  (a good record is freshly written or survives expiration and overwriting) and the summed gains give the proposition. Under  $g_j^I > 0 > g_j^W$ , write quality fixes the sign of a cell’s contribution and write frequency and expiration fix its weight and rate  $\alpha_j$ , so heterogeneous weights can flip the sign of the total and improvement need not be monotone in time; the statement concerns expected aggregates, not a Markov hit sequence.

**Proposition 9** (Records inside a verified cascade). *Suppose that in cell  $j$  the router runs the cascade of Section 2 with record-independent costs  $a_j, b_j$  and values  $v_h, v_b$ , and that a record changes only the hit probability,  $p_j^s$  for  $s \in \{U, I, W\}$ . Then  $v_j^s = p_j^s v_h + (1 - p_j^s) v_b$ ,  $c_j^s = a_j + (1 - p_j^s) b_j$ , and  $g_j^s = (p_j^s - p_j^U) W_{\eta_U, j}$  with  $W_{\eta_U, j} = \eta_U b_j + v_h - v_b$ . If  $W_{\eta_U, j} > 0$ , condition (29) becomes  $\psi_j / r_j < \Delta_j^I / (\Delta_j^I - \Delta_j^W)$  with  $\Delta_j^s = p_j^s - p_j^U$ . If instead a record is served without verification at lookup cost  $c_\ell$ , an erroneous record (of value zero) has  $g_j^W = \eta_U (c_j^U - c_\ell) - v_j^U$ , which is nonnegative whenever the cell’s uninformed efficiency  $v_j^U / c_j^U$  is at most  $\eta_U (1 - c_\ell / c_j^U)$ ; the hypothesis  $g_j^W < 0$  of Proposition 8 then fails. (Proof in Appendix M.2.)*

When a record is written whenever the primary is accepted and is erroneous if and only if the accepted output is worse than the reference,  $\psi_j / r_j = \mathbb{P}(Y_P < Y_B \mid H = 1, \text{cell } j)$ ; calibration at level  $\theta$  relative to the selection information, which for a calibrated score is the acceptance threshold (Appendix J.1), gives  $\psi_j / r_j \leq 1 - \theta$ , tying the noise threshold to the verifier’s operating point. Against a reference without a store, with  $J_U(\lambda) = \sum_j \nu_j (v_j^U - \lambda c_j^U)$  and (25) evaluated at that reference’s efficiency  $\lambda$ , the stationary condition becomes

$$J_U(\lambda) + \sum_j \nu_j G_j(\lambda) \geq 0; \quad (30)$$

if the uninformed-state process matches a no-memory reference of efficiency  $\eta_0$  except for a maintenance cost  $h$  per round, then  $J_U(\eta_0) = -\eta_0 h$ , so the records must deliver at least  $\eta_0 h$  in additional gain (Appendix M.5).

## F.1 CAPACITY AND THE TAIL OF TASK FREQUENCIES

A record expiring after  $E$  global rounds (a time-to-live, TTL) gives cell  $j$  stationary coverage (31); an event FIFO of the last  $M$  write events, duplicates included, with  $\beta = \sum_j \nu_j r_j$  and the newest matching record consulted, gives (32):

$$\chi_j^{\text{TTL}} = 1 - (1 - \nu_j r_j)^E, \quad (31)$$

$$\chi_j^{\text{FIFO}} = 1 - (1 - \nu_j r_j / \beta)^M. \quad (32)$$

Both have informative fraction  $\omega_j / r_j$  given coverage, and substituting the mean span  $M / \beta$  for  $E$  in (31) under-estimates FIFO coverage (Appendix M.6). For clean writes ( $\psi_j = 0$ ) with a common write probability  $\omega$  and a common informed-minus-uninformed hit gain  $\Delta$ , the uncovered hit deficit at TTL horizon  $E$  is  $L_E = \Delta \sum_j \nu_j (1 - \omega \nu_j)^E$ ; from an empty store without expiration the deficit in round  $n$  is the same  $L_n$ , with cumulative version  $R_N = \sum_{n < N} L_n$ .

**Proposition 10** (Heavy-tailed information deficit). *Suppose cells are countably infinite with  $\nu_j = j^{-z} / \zeta(z)$ ,  $z > 1$ . Writes are clean with constant probability  $\omega \in (0, 1]$ , and every cell has the same positive informed-minus-uninformed hit gain  $\Delta$ . Under a TTL of  $E$  rounds, the stationary uncovered hit deficit is  $L_E = \Theta(E^{-(z-1)/z})$ . From an empty store without expiration, the cumulative uncovered hit deficit over  $N$  rounds is  $R_N = \Theta(N^{1/z})$ . (Proof in Appendix M.8.)*

Multiplied by  $W_{\eta_U}$ ,  $L_E$  and  $R_N$  are the shortfalls of (28) under Proposition 9. The exponents and constants are the occupancy asymptotics of Karlin (1967) and Gneden et al. (2007) applied to write events, and the TTL formula is the independent-reference miss ratio of Fagin (1977); the routing content is the contrast with finitely many cells, whose cumulative loss stays bounded (Corollary 18 restates the law in stored records; Remark 19 gives break-even horizons).

## F.2 ALLOCATING ROUTER MEMORY ACROSS TASK FAMILIES

For independent task families with increasing, strictly concave coverage curves  $p_k(F_k)$ , weights  $w_k$ , and unit prices  $\gamma_k$ , a budget is optimally allocated by equalizing weighted marginal coverage per unit price,  $w_k p'_k(F_k) / \gamma_k = \Lambda$  on funded families ( $F_k > 0$ ; Appendix K). For  $J_\lambda$  the weight of a family routed cheap-first is  $w_k = \nu_k W_{\lambda, k}$ ; because its benefit  $\nu_k \max\{0, p_k(F_k) W_{\lambda, k} - \lambda a_k\}$  is not concave in  $F_k$ , the budget problem (46) of Appendix K is the inner problem for a fixed set of families routed cheap-first, whose outer choice follows Theorem 5 (Appendix K.3). The exponent depends on what capacity counts: for a retention horizon or an event FIFO length, Proposition 10 gives

$1 - p_k \approx d_k(1 + F_k)^{-(1-1/z)}$  for a family constant  $d_k$  and capacity proportional to  $(w_k d_k / \gamma_k)^{z/(2z-1)}$  for large funded allocations; for stored records the deficit is  $\Theta(F_k^{-(z-1)})$  (Corollary 18) and capacity scales as  $(w_k d_k / \gamma_k)^{1/z}$ .

## G NOTATION

Tables 9–11 collect the main symbols. A dagger ( $\dagger$ ) marks a letter with section-specific meanings. Probability is  $\mathbb{P}$ , expectation is  $\mathbb{E}$ , the verifier score is  $\Sigma$ , the portfolio is  $S$ , and savings are  $\mathcal{S}$ . The italic and upright labels  $U, I, W$  denote the same record states.

Table 9: Symbols for the router model, model combinations, and routing decisions.

| Symbol                                                   | Meaning                                                                                                                                               | Where            |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| $x_n, I_n, u_n, f_n$                                     | observable state, information state (records), action and feedback of round $n$                                                                       | Eq. (7)          |
| $g$                                                      | router policy $u_n = g(x_n, I_n)$ ; in Thm. 5 the randomized state-wise rule $g(u \mid x)$ ; also $g(s)$ in App. I.5 and $g_k$ in App. K.2 $^\dagger$ | Sec. 2           |
| $V, D$                                                   | value delivered and resource consumed by the router in a round                                                                                        | Sec. 2           |
| $Y_B, C_B; \bar{v}, \bar{c}; \eta_B$                     | reference value and cost, their means, reference efficiency $\bar{v}/\bar{c}$                                                                         | Sec. 2           |
| $\eta$                                                   | efficiency $\mathbb{E}[V]/\mathbb{E}[D]$ , long-run value per resource                                                                                | Eq. (8)          |
| $Q, \mathcal{S}$                                         | quality change $\mathbb{E}[V - Y_B]$ and resource saving $\mathbb{E}[C_B - D]$ ; $Q + \eta_B \mathcal{S}$ is the net gain                             | Lem. 1           |
| $\epsilon$                                               | tolerance of the quality floor $\mathbb{E}[V] \geq \bar{v} - \epsilon$                                                                                | Lem. 1           |
| $Y_P, H, Y_F, C_F, A$                                    | primary value, acceptance (hit) indicator, fallback value and additional resource, primary and routing cost                                           | Eq. (12)         |
| $p, v_h, v_m, a, b$                                      | hit probability, hit- and miss-conditional values, primary cost, miss-conditional fallback cost                                                       | Eq. (14)         |
| $G, P$                                                   | $v_h/\bar{v}$ and $b/a$                                                                                                                               | Eq. (16)         |
| $\bar{v}_h, \bar{v}_m, \bar{c}_h, \bar{c}_m$             | reference value and cost means conditional on $H = 1$ and $H = 0$                                                                                     | App. H.6         |
| $e$                                                      | erroneous-acceptance probability $\mathbb{P}(H = 1, Y_P < Y_B)$ ; in the caption of Fig. 4 the erroneous write fraction $\psi_j/r_j^\dagger$          | App. H.1         |
| $\Sigma, \theta$                                         | verifier score and acceptance threshold, $H = \mathbf{1}\{\Sigma \geq \theta\}$                                                                       | Sec. 2           |
| $\pi_0, \text{TPR}(\theta), \text{FPR}(\theta)$          | probability that the primary is correct; acceptance probabilities given a correct and an incorrect primary                                            | Sec. 2, App. H.5 |
| $\beta_1, \beta_0$                                       | reference success probabilities given a correct and an incorrect primary                                                                              | App. H.5         |
| $E_m, S, p(S), r_m(S)$                                   | success event of candidate $m$ , portfolio, its coverage, residual coverage of $m$ given $S$                                                          | Eq. (17)         |
| $a(S), d_m(S), \eta(S)$                                  | primary cost of portfolio $S$ , added cost of candidate $m$ , efficiency of $S$                                                                       | Prop. 2          |
| $c$                                                      | candidate cost in Prop. 3; $c(x, u)$ is the expected cost of action $u$ in state $x$ in Thm. 5 $^\dagger$                                             | Prop. 3          |
| $\rho, T_0, T_1$                                         | steering parameter and the de-emphasized and retained answer-type regions                                                                             | App. L.3         |
| $\lambda, J_\lambda, J_\lambda(x, u), \eta(g)$           | resource price; linear objective $\mathbb{E}[V - \lambda D]$ , the same objective for action $u$ in state $x$ ; efficiency of router $g$              | Sec. 4.1         |
| $W_\lambda, v_b$                                         | margin $\lambda b + v_h - v_b$ (value of turning a miss into a hit); value of the direct path                                                         | Eq. (19)         |
| $k$                                                      | cost of a routing refinement or routing signal; also the family index in Sec. F.2 and the attempt index in Prop. 7 $^\dagger$                         | Prop. 4          |
| $\Delta p$                                               | increase in hit probability bought by a refinement                                                                                                    | Prop. 4          |
| $s$                                                      | probability of primary sufficiency; also a record state in $\{U, I, W\}^\dagger$                                                                      | Eq. (20)         |
| $C_{\text{cascade}}, C_{\text{direct}}, C_{\text{info}}$ | expected costs of the two uninformed choices and of the informed choice                                                                               | Eq. (20)         |
| $Z, s_Z$                                                 | experiment (routing signal) and the posterior sufficiency probability it induces                                                                      | Eq. (22)         |
| $\pi(x), \mathcal{U}(x), v(x, u), c(x, u)$               | state probabilities, action menu, expected value and cost of action $u$ in state $x$                                                                  | Thm. 5           |
| $\Phi(\lambda), \eta^*, \eta_\epsilon^*, \kappa$         | Dinkelbach function, optimal efficiency, floor-constrained optimum, floor multiplier                                                                  | Thm. 5           |

| Symbol                                                | Meaning                                                                                                                                                    | Where   |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $K, a_k, p_k, \Pi_k, \Delta_{k+1}, J^{\text{direct}}$ | attempt count, cost, conditional acceptance, continuation value, option value, and terminal objective; $a_k$ also denotes family primary cost <sup>†</sup> | Prop. 7 |
| $\epsilon_1, \epsilon_2; s_i, \kappa_i$               | weights of the per-task reward of Zhou et al. (2026a), and the verifier score and cost of task $i$ ; $\epsilon$ is the quality tolerance <sup>†</sup>      | Prop. 7 |

Table 10: Symbols of Section F and its replay experiments.

| Symbol                                      | Meaning                                                                                                                                                                                | Where             |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| $J, \nu_j$                                  | number of cells and arrival probability of cell $j$                                                                                                                                    | Sec. F            |
| $U, I, W$                                   | record states: uninformed, informative, erroneous                                                                                                                                      | Sec. F            |
| $\epsilon$                                  | exploration probability of the semi-synthetic logger in Sec. 3                                                                                                                         | Sec. 3            |
| $\delta_j, \omega_j, \psi_j, r_j, \alpha_j$ | expiration probability, informative and erroneous write probabilities, write rate $\omega_j + \psi_j$ , retention factor                                                               | Eq. (23)          |
| $v_j^s, c_j^s$                              | conditional value and cost in cell $j$ and record state $s$                                                                                                                            | Sec. F            |
| $\eta_U, g_j^s, G_j$                        | efficiency of the uninformed reference, record gain, stationary gain of cell $j$                                                                                                       | Eqs. (24)–(26)    |
| $i_{j,n}, w_{j,n}$                          | probabilities of an informative and of an erroneous record before round $n$                                                                                                            | Eq. (27)          |
| $\eta_\infty, \bar{c}_\infty$               | stationary efficiency and stationary expected cost                                                                                                                                     | Prop. 8           |
| $J_U(\lambda), h$                           | objective of the uninformed state at price $\lambda$ ; maintenance cost per round                                                                                                      | Eq. (30)          |
| $\eta_0, c_0$                               | efficiency and mean cost of a no-memory reference                                                                                                                                      | Eq. (30)          |
| $p_j^s, \Delta_j^s, W_{\eta_U, j}, c_\ell$  | hit probability in record state $s$ , its change $p_j^s - p_j^U$ , margin at price $\eta_U$ , lookup cost of an unverified record                                                      | Prop. 9           |
| $E, M, \beta$                               | TTL horizon in rounds, FIFO length in write events, total write probability $\sum_j \nu_j r_j$ ; $\mathbb{E}$ is expectation <sup>†</sup>                                              | Eqs. (31), (32)   |
| $\chi_j$                                    | stationary coverage of cell $j$                                                                                                                                                        | Eq. (31)          |
| $z, \zeta(z), \kappa_z$                     | Zipf exponent, Riemann zeta function, normalizer $1/\zeta(z)$                                                                                                                          | Prop. 10          |
| $\omega, \Delta$                            | common clean-write probability and informed-minus-uninformed hit gap; $\Delta$ is also the increment symbol in $\Delta p, \Delta_j^s$ and the option value $\Delta_{k+1}$ <sup>†</sup> | Prop. 10          |
| $L_E, R_N, m(E)$                            | per-round uncovered deficit, cumulative deficit, expected number of occupied cells                                                                                                     | Sec. F.1, Cor. 18 |
| $F_k, p_k(F_k), \gamma_k, w_k, d_k$         | allocation to family $k$ , its coverage curve, unit price, weight, deficit scale                                                                                                       | Sec. F.2          |
| $\Lambda, B$                                | budget multiplier and budget                                                                                                                                                           | Eq. (46)          |
| $r$                                         | residual coverage $r_m(S)$ (Sec. 3.1); write rate $r_j$ ; coverage exponent in $(1 + F)^{-r}$ (Sec. F.2); prefix charge in App. J.2 <sup>†</sup>                                       | Sec. F            |

Table 11: Symbols used only in the appendices, and further overloaded letters.

| Symbol                        | Meaning                                                                                                                                             | Where    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| $T, t$                        | ex-ante branch indicator and its probability; $t$ is also the Charnes–Cooper scale in App. I.3 <sup>†</sup>                                         | App. H.4 |
| $\varrho$                     | randomization probability of a router in one state                                                                                                  | App. I.3 |
| $\kappa_g, \kappa_h, c_p$     | gate charge, auxiliary routing charge, primary price; $\kappa$ is the floor multiplier in Thm. 5 and $\kappa_i$ a task cost in Prop. 7 <sup>†</sup> | App. H.6 |
| $L, \gamma, \delta, c_m$      | prefix length, reusable fraction, cached-input multiplier, input price of model $m$ ; $\gamma_k$ and $\delta_j$ elsewhere <sup>†</sup>              | App. H.8 |
| $f(s), g(s)$                  | uninformed cost $\min\{a + (1-s)b, b\}$ and perfect-information cost $sa + (1-s)b$                                                                  | App. I.5 |
| $\hat{s}$                     | a reported, possibly miscalibrated, sufficiency score                                                                                               | App. I.5 |
| $C_{\text{dec}}$              | cost of the refined cascade in Cor. 12                                                                                                              | App. I.5 |
| $\mathcal{G}, v, w$           | selection information, calibrated probability $\mathbb{E}[Y   \mathcal{G}]$ , stake; $\mathcal{G}$ is the set of routers in App. I.3 <sup>†</sup>   | App. J.1 |
| $\ell_\kappa, \ell_p, \ell_b$ | controller, primary and reference completion times                                                                                                  | App. J.2 |

| Symbol                                                  | Meaning                                                                                                                                                                                                                      | Where              |
|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| $L_p, L_b, m_h, m_m, l_b$                               | random completion times, their branch-conditional means, mean reference latency                                                                                                                                              | App. J.2           |
| $\mu, g_k, \mathcal{K}_+$                               | budget scale, $(w_k d_k / \gamma_k)^{1/(1+r)}$ , funded set; $\mu_m$ are arm values in App. M.10 <sup>†</sup>                                                                                                                | App. K.2           |
| $\rho_k, \tau_k$                                        | variable resource charge, nonnegativity multiplier                                                                                                                                                                           | App. K.3, App. K.2 |
| $\mathcal{A}, F_k^{\min}$                               | activation set and activation threshold of family $k$                                                                                                                                                                        | App. K.3           |
| $q, p_1, p_2, h(\rho), W_0,$                            | answer-type law, proposal laws, residual coverage, region masses, normalizer, standalone qualities; $W_\lambda, Z$ and $a$                                                                                                   | App. L.3           |
| $W_1, Z(\rho), a_1, a_2$                                | elsewhere <sup>†</sup>                                                                                                                                                                                                       |                    |
| $M_{j,n}$                                               | record state of cell $j$ at round $n$ ; $M$ is the FIFO length in Sec. F.1 <sup>†</sup>                                                                                                                                      | App. M             |
| $s_j^U, s_j^I, s_j^W, s_j^O; \text{Reg}_N$              | utility means by record state and oracle mean; regret                                                                                                                                                                        | App. M.7           |
| $\mu_m, e_m, \hat{\mu}_m, \Delta_m, \beta_{\text{fa}},$ | arm values, logging probabilities, sample means, gaps, verifier false-accept and false-reject rates                                                                                                                          | App. M.10          |
| $\beta_{\text{fr}}$                                     | Dinkelbach function; also the Poissonized occupancy function in App. M.8 <sup>†</sup>                                                                                                                                        | App. I.3           |
| $\Phi$                                                  | number of attempts (Prop. 7); also the number of families in App. K <sup>†</sup>                                                                                                                                             | Prop. 7            |
| $K$                                                     | staleness hazard and retention factor $\alpha_j(1 - \vartheta_j)$ ; $\beta$ is total write probability and $\beta_1, \beta_0$ reference success probabilities <sup>†</sup>                                                   | App. M.4           |
| $\vartheta_j, \beta_j$                                  | constants and the tolerance in the occupancy asymptotics, $\alpha = 1/z$ , and a slowly varying function in App. M.8; distinct from $\kappa, \kappa_i, \kappa_z, \tau_k, \alpha_j$ , and the latencies $\ell$ . <sup>†</sup> | App. M.8           |
| $\kappa_j, \tau, \alpha, \ell$                          | coefficients of the two-state record recurrence in Remark 16; elsewhere $A$ is the primary resource and $b_j$ a fallback cost <sup>†</sup>                                                                                   | Rem. 16            |
| $A_j, b_j$ (matrix, vector)                             |                                                                                                                                                                                                                              |                    |

## H ROUTER COMPARISON AND HOMOGENEOUS CASCADES

### H.1 PROOF OF LEMMA 1

*Proof.* For any router, the definitions  $Q = \mathbb{E}[V - Y_B]$  and  $\mathcal{S} = \mathbb{E}[C_B - D]$  give  $\mathbb{E}[V] = \bar{v} + Q$  and  $\mathbb{E}[D] = \bar{c} - \mathcal{S}$ . Under verified execution with escalation, Equation (12) further gives the pointwise expansions

$$V - Y_B = H(Y_P - Y_B) + (1 - H)(Y_F - Y_B),$$

and

$$C_B - D = HC_B - A - (1 - H)(C_F - C_B).$$

Taking expectations yields Equation (13). In either case,  $\bar{c} - \mathcal{S} > 0$  and

$$\eta - \eta_B = \frac{\bar{v} + Q}{\bar{c} - \mathcal{S}} - \frac{\bar{v}}{\bar{c}} = \frac{Q + (\bar{v}/\bar{c})\mathcal{S}}{\bar{c} - \mathcal{S}},$$

which is (10). The denominator is positive, giving the efficiency comparison. For (11), substitute  $\bar{v} = \eta_B \bar{c}$ :

$$Q + \eta_B \mathcal{S} = (\mathbb{E}[V] - \bar{v}) + \eta_B (\bar{c} - \mathbb{E}[D]) = \mathbb{E}[V - \eta_B D],$$

and, for any  $\lambda$ ,  $\mathbb{E}[V - \lambda D] = \mathbb{E}[D] (\mathbb{E}[V]/\mathbb{E}[D] - \lambda) = \mathbb{E}[D](\eta - \lambda)$ . The quality condition follows separately from  $\mathbb{E}[V] = \bar{v} + Q$ . These equalities require neither independence nor stationarity. For the long-run statement, the ergodic theorem applied to integrable  $V_n$  and  $D_n$  gives convergence of their sample means to their expectations. Their ratio converges because the limiting mean cost is positive.  $\square$

The comparison uses a reference policy on the same task population. If routing changes subsequent within-task states, taking the entire task as the round includes all consequences in  $(V, D)$ , including both the primary and fallback paths. The null action is an available choice; we do not derive an optimal stopping rule. Evaluation on intermediate states visited by the router instead gives a comparison on that particular state distribution.

For example, if fallback restores the reference on every miss, values lie in  $[0, 1]$ , and  $e = \mathbb{P}(H = 1, Y_P < Y_B)$ , then  $Q \geq -e$ . Savings satisfying  $\eta_B \mathcal{S} \geq e$  suffice for an efficiency guarantee, while

$e \leq \epsilon$  suffices for the quality constraint. Neither conclusion requires the verifier to preserve the reference distribution. Appendix H.5 expresses  $e$  through the verifier’s false-accept rate.

## H.2 HOW THE RESULTS SPECIALIZE THE GAIN IDENTITY

Table 12 applies the gain identity against each result’s reference and price. The memory row is per cell and record; the aggregate criterion is  $\sum_j \nu_j G_j \geq 0$ . The latency row measures resource in time, and the information row preserves value. In every case the quality floor  $Q \geq -\epsilon$  remains a separate condition.

Table 12: Specializations of the gain identity (11): reference policy and price, the terms  $Q$  and  $\mathcal{S}$ , and the criterion  $Q + \lambda \mathcal{S} \geq 0$ .

| Result                           | Reference (price $\lambda$ )                             | $Q$                   | $\mathcal{S}$                     | Criterion                                               |
|----------------------------------|----------------------------------------------------------|-----------------------|-----------------------------------|---------------------------------------------------------|
| Eq. (15)                         | direct path with $\bar{v}$ , $\bar{c} = b$               | $p(v_h - \bar{v})$    | $pb - a$                          | $pv_h b \geq \bar{v}a$                                  |
| Prop. 2                          | $(\eta_B)$<br>current portfolio $S$                      | $r_m(S)(v_h - v_m)$   | $br_m(S) - d_m(S)$                | Eq. (18)                                                |
| Prop. 4,<br>refinement<br>clause | $(\eta(S))$<br>unrefined cascade ( $\lambda$ )           | $\Delta p(v_h - v_b)$ | $\Delta p b - k$                  | $\Delta p W_\lambda \geq \lambda k$                     |
| Eq. (25)                         | uninformed store, state<br>$U$ ( $\eta_U$ )              | $v_j^s - v_j^U$       | $c_j^U - c_j^s$                   | $g_j^s > 0$ ; stationary<br>$\sum_j \nu_j G_j \geq 0$   |
| Eq. (43)                         | reference path, latency<br>$\ell_b$ ( $\bar{v}/\ell_b$ ) | $p(v_h - \bar{v})$    | $p\ell_b - \ell_\kappa - p\ell_p$ | $p(\ell_b v_h - \bar{v}\ell_p) \geq \bar{v}\ell_\kappa$ |
| Prop. 6                          | cheap-first cascade (any<br>price)                       | 0                     | $a(1 - s) - k$                    | $s < 1 - k/a$                                           |

The main clause of Proposition 4 is the first row at an exogenous price: with the direct path as reference,  $Q = p(v_h - v_b)$ ,  $\mathcal{S} = pb - a$  and  $Q + \lambda \mathcal{S} = pW_\lambda - \lambda a$ . Against the direct path, Proposition 6 has  $\mathcal{S} = s(b - a) - k$ , and the band (21) is the intersection of the two savings conditions.

## H.3 WHY THE QUALITY FLOOR AND THE VALUE ZERO MATTER

The “if and only if” statements of the paper compare efficiencies; the floor  $Q \geq -\epsilon$  is a separate condition, and values must be measured on a scale whose zero is meaningful. Two examples show that neither qualification is idle.

**Blind acceptance passes the ratio test.** Let  $a = 1/10$  and  $b = \bar{c} = \bar{v} = 1$ , and let the primary be correct with probability  $q = 3/20$ . Accepting every primary output gives  $\eta = q/a = 3/2 > \eta_B = 1$ , yet  $Q = q - 1 = -17/20$ . It therefore passes the efficiency comparison while violating every quality floor with  $\epsilon < 17/20$ . A perfect free verifier instead accepts only correct outputs and restores the reference otherwise, giving  $\eta = 20/19$  and  $Q = 0$ . Theorem 5(iii) enforces this separate quality constraint.

**The zero of the value scale.** Adding a constant to every value leaves  $Q$  unchanged but changes  $\eta_B$ , hence  $Q + \eta_B \mathcal{S}$  whenever  $\mathcal{S} \neq 0$ . In the homogeneous cascade with  $a = 1/10$ ,  $b = \bar{c} = 1$  and  $p = 1/2$ , the values  $(v_h, \bar{v}) = (7/10, 1)$  give  $Q = -3/20$ ,  $\mathcal{S} = 2/5$  and net gain  $1/4 > 0$ . Subtracting  $7/10$  from every value preserves  $Q$  and  $\mathcal{S}$  but changes  $\eta_B$  to  $3/10$  and the net gain to  $-3/100 < 0$ . The record threshold (29) has the same dependence through the cost channel of (25); with equal record-state costs that channel vanishes and the threshold is shift-invariant.

## H.4 HOMOGENEOUS THRESHOLD AND ENDPOINT BOUNDS

Total expectation gives

$$\eta(p) = \frac{pv_h + (1 - p)v_m}{a + (1 - p)b}.$$

Suppose the fallback restores the reference on every miss and the reference's conditional means do not depend on  $H$ , so that  $v_m = \bar{v}_m = \bar{v}$  and  $b = \bar{c}_m = \bar{c}$ . This is the reference-restoring fallback of (15); we call it the homogeneous case below. With  $v_h, \bar{v}, a, b > 0$ ,

$$\begin{aligned} \eta(p) \geq \eta_B &\iff b[pv_h + (1-p)\bar{v}] \geq \bar{v}[a + (1-p)b] \\ &\iff pbv_h \geq \bar{v}a. \end{aligned}$$

Strict comparison follows by replacing both inequalities with strict inequalities. A floor above one cannot be attained by an admissible acceptance probability. If the reference means depend on  $H$ , the condition is (36) in Appendix H.6.

For  $G = v_h/\bar{v}$  and  $P = b/a$ , write

$$R(p) = \frac{pG + 1 - p}{1/P + 1 - p}, \quad R'(p) = \frac{G(1 + P) - 1}{P(1/P + 1 - p)^2}.$$

Since the derivative has constant sign and the endpoint values are  $R(0) = P/(1+P)$  and  $R(1) = GP$ ,

$$\min \left\{ GP, \frac{P}{1+P} \right\} \leq R(p) \leq \max \left\{ GP, \frac{P}{1+P} \right\}, \quad 0 \leq p \leq 1.$$

Also  $R(p) \geq [P/(1+P)](pG + 1 - p)$  because its denominator is at most  $(1+P)/P$ . When  $G(1+P) < 1$ , the ratio decreases and remains below one. The crossing  $R(p) = 1$  is  $p = 1/(GP)$ , feasible if and only if  $GP \geq 1$ .

More generally, holding conditional values and positive resources fixed,

$$\frac{d}{dp} \frac{pv_h + (1-p)v_m}{pd_h + (1-p)d_m} = \frac{v_h d_m - v_m d_h}{[pd_h + (1-p)d_m]^2}.$$

Thus higher acceptance improves efficiency if and only if a hit has greater value per resource than a miss. Changing the accepted task population need not preserve these conditional means.

For an ex-ante router, let  $T$  indicate choosing the primary branch before execution and  $t = \mathbb{P}(T)$ . With no routing overhead, primary cost  $a$ , reference cost  $b$ , and conditional values  $v_h, \bar{v}$ , normalized efficiency is

$$\frac{tG + 1 - t}{ta/b + 1 - t}.$$

The always-paid wasted-primary term is absent. Branch selection  $T$  is not the verified acceptance event  $H$ .

## H.5 THRESHOLDED VERIFIERS AND THE OPERATING POINT

Let the verifier accept when a score clears a threshold,  $H = \mathbf{1}\{\Sigma \geq \theta\}$ , let  $Y \in \{0, 1\}$  indicate that the primary output is correct, with  $\pi_0 = \mathbb{P}(Y = 1)$ , and write  $\text{TPR}(\theta) = \mathbb{P}(\Sigma \geq \theta \mid Y = 1)$  and  $\text{FPR}(\theta) = \mathbb{P}(\Sigma \geq \theta \mid Y = 0)$ ; both are nonincreasing in  $\theta$ . Give a correct output value one and an incorrect output value zero, and let the fallback restore the reference on a miss. Then

$$p(\theta) = \pi_0 \text{TPR}(\theta) + (1 - \pi_0) \text{FPR}(\theta), \quad p(\theta) v_h(\theta) = \mathbb{E}[HY_P] = \pi_0 \text{TPR}(\theta),$$

so the product  $pv_h$  that enters the floor (15) counts true accepts only. Moving  $\theta$  changes  $p$ ,  $v_h$  and the erroneous-acceptance probability  $e$  together; the refinement clause of Proposition 4 holds the other quantities fixed and therefore does not describe a change of threshold, which is instead one more action in the menu  $\mathcal{U}(x)$  of Theorem 5. Choosing  $\theta$  is the error-reject trade-off of Chow (1970), priced here by the cascade.

**Efficiency.** Let  $\beta_1 = \mathbb{P}(Y_B = 1 \mid Y = 1)$  and  $\beta_0 = \mathbb{P}(Y_B = 1 \mid Y = 0)$  be the reference's success probabilities given a correct and an incorrect primary, so that  $\bar{v} = \pi_0 \beta_1 + (1 - \pi_0) \beta_0$ ; suppose the reference cost is a constant  $b$  and the score  $\Sigma$  is conditionally independent of  $Y_B$  given  $Y$ . Then  $\mathbb{P}(H = 1, Y_B = 1) = \pi_0 \beta_1 \text{TPR} + (1 - \pi_0) \beta_0 \text{FPR}$ , and with  $Q = \mathbb{E}[H(Y_P - Y_B)]$  and  $S = \mathbb{E}[HC_B] - \mathbb{E}[A] = pb - a$ ,

$$Q = \pi_0(1 - \beta_1) \text{TPR} - (1 - \pi_0) \beta_0 \text{FPR}, \quad e = (1 - \pi_0) \beta_0 \text{FPR}, \quad (33)$$

$$Q + \eta_B \mathcal{S} = \pi_0(1 - \beta_1 + \bar{v})\text{TPR} + (1 - \pi_0)(\bar{v} - \beta_0)\text{FPR} - \frac{\bar{v}a}{b}. \quad (34)$$

Efficiency improves if and only if the right side of (34) is positive. In the homogeneous case,  $\beta_1 = \beta_0 = \bar{v}$ , reference success is independent of the primary's correctness. The FPR term then vanishes and the condition is  $\pi_0 \text{TPR}(\theta) b \geq \bar{v}a$ , which is (15) with  $p v_h = \pi_0 \text{TPR}$ . False accepts do not enter the efficiency comparison: an accepted incorrect output costs  $a$  instead of  $a + b$ , and the value it forgoes is value that, at  $\beta_0 = \bar{v}$ , the reference delivers with the same probability as on any other task. When  $\beta_1 > \beta_0$  and  $\text{TPR}(\theta) > \text{FPR}(\theta)$ , acceptance selects tasks the reference also solves,  $\bar{v}_h = \mathbb{P}(Y_B = 1 \mid H = 1) > \bar{v}$ , and condition (36) is stricter than (15): it reads  $\pi_0 \text{TPR}(\theta) b - p b(\bar{v}_h - \bar{v}) \geq \bar{v}a$ , with the selection term  $p b(\bar{v}_h - \bar{v})$  on the left.

**Accepting everything.** In the homogeneous case,  $\eta(\theta) = [\pi_0 \text{TPR} + (1 - p)\bar{v}]/[a + (1 - p)b]$  with  $p = p(\theta)$ , and accepting every output gives  $\eta = \pi_0/a$ . Cross-multiplying,

$$\eta(\theta) \leq \frac{\pi_0}{a} \iff a(1 - p)\bar{v} \leq \pi_0 a(1 - \text{TPR}) + \pi_0(1 - p)b.$$

Thus  $a\bar{v} \leq \pi_0 b$  suffices for accepting every output to maximize efficiency over all thresholds. More generally,  $\pi_0 \mathbb{E}[D] - a \mathbb{E}[V]$  is affine on the operating-point square  $(\text{TPR}, \text{FPR}) \in [0, 1]^2$ ; checking its four vertices shows that accepting everything dominates this whole square if and only if  $a\beta_0 \leq \pi_0 b$ . For a fixed score whose operating points cover only part of the square, this condition is sufficient but need not be necessary. The quality floor remains separate: by (33),  $Q \geq -\epsilon$  is

$$\text{FPR}(\theta) \leq \frac{\epsilon + \pi_0(1 - \beta_1)\text{TPR}(\theta)}{(1 - \pi_0)\beta_0}, \quad (35)$$

an upper bound on the false-accept rate; because the right side contains  $\text{TPR}(\theta)$ , the admissible thresholds need not form an up-set in general, but for a score with a monotone likelihood ratio they form an interval  $[\theta_{\min}, \infty)$ , so the floor bounds the threshold from below. The sufficient condition  $e \leq \epsilon$  of Appendix H.1 is (35) without the TPR term; the two coincide when  $\beta_1 = 1$ .

**An instance.** Let  $\pi_0 = 3/5$ ,  $\beta_1 = 19/20$ ,  $\beta_0 = 4/5$ ,  $a = 1/10$  and  $b = 1$ , so  $\bar{v} = \eta_B = 89/100$ . The score has values 1, 2, 3, 4 with probabilities  $(1, 2, 3, 4)/10$  for correct outputs and  $(4, 3, 2, 1)/10$  for incorrect ones. Table 13 shows that accepting everything maximizes efficiency but loses 29/100 in quality. At  $\epsilon = 1/20$ , the smallest admissible threshold is  $\theta = 4$ . Allowing randomization over thresholds and the direct path gives the constrained optimum: weights 6/11 and 5/11 on  $\theta = 3, 4$  bind the floor at  $\mathbb{E}[V] = 21/25$  and yield  $\eta = 462/373$ . At  $\theta = 4$ , acceptance selects easier reference tasks ( $\bar{v}_h = 13/14 > \bar{v}$ ), so the homogeneous formula predicts  $\eta \approx 1.074$  instead of  $87/82 \approx 1.061$ .

Table 13: Operating points of the instance in Appendix H.5:  $\pi_0 = 3/5$ ,  $\beta_1 = 19/20$ ,  $\beta_0 = 4/5$ ,  $a = 1/10$ ,  $b = 1$ ,  $\eta_B = 89/100$ .

| $\theta$ | TPR  | FPR  | $p$   | $Q$     | $e$    | $\eta$  | $Q + \eta_B \mathcal{S}$ |
|----------|------|------|-------|---------|--------|---------|--------------------------|
| 1        | 1    | 1    | 1     | -29/100 | 8/25   | 6       | 511/1000                 |
| 2        | 9/10 | 3/5  | 39/50 | -33/200 | 24/125 | 145/64  | 2201/5000                |
| 3        | 7/10 | 3/10 | 27/50 | -3/40   | 12/125 | 163/112 | 1583/5000                |
| 4        | 2/5  | 1/10 | 7/25  | -1/50   | 4/125  | 87/82   | 701/5000                 |
| 5        | 0    | 0    | 0     | 0       | 0      | 89/110  | -89/1000                 |

## H.6 STATE-MATCHED REFERENCE VALUES AND COSTS

Suppose fallback restores the reference on misses:  $Y_F = Y_B$  and  $C_F = C_B$  there. Let  $\bar{v}_h, \bar{v}_m$  and  $\bar{c}_h, \bar{c}_m$  denote the reference value and cost means conditional on  $H = 1, 0$ , respectively. Let  $v_h = \mathbb{E}[Y_P \mid H = 1]$  and  $a = \mathbb{E}[A]$ . For  $0 < p < 1$ ,

$$Q = p(v_h - \bar{v}_h), \quad \mathcal{S} = p\bar{c}_h - a,$$

while  $\bar{v} = p\bar{v}_h + (1 - p)\bar{v}_m$  and  $\bar{c} = p\bar{c}_h + (1 - p)\bar{c}_m$ . The efficiency condition is

$$p(v_h - \bar{v}_h) + \eta_B(p\bar{c}_h - a) \geq 0. \quad (36)$$

In particular,  $v_h \geq \bar{v}_h$  and  $a \leq p\bar{c}_h$  suffice. If  $\eta_B > 0$ , either strict inequality yields strict improvement. The event  $H$  may be random conditional on the observable state.

When the reference means do not depend on  $H$ , so that  $\bar{v}_h = \bar{v}$  and  $\bar{c}_h = \bar{c}$ , condition (36) reduces to (15) with  $b = \bar{c}$ :  $p(v_h - \bar{v}) + \eta_B(p\bar{c} - a) = pv_h - \bar{v}a/\bar{c}$ . If accepted outputs preserve value,  $v_h = \bar{v}_h$ , the condition is  $p \geq a/\bar{c}_h$  whatever  $\bar{v}_h$  is: the floor is  $a/\bar{c}_h$ , and it exceeds  $a/\bar{c}$  if and only if acceptance selects tasks that the reference solves cheaply,  $\bar{c}_h < \bar{c}$ . With  $\bar{c}_h = \bar{c}/4$ , a tenfold price advantage  $a = \bar{c}/10$  requires  $p \geq 2/5$  rather than  $p \geq 1/10$ .

**Thresholds and the information band with a correlated reference.** The same conditional means give the correlated form of Propositions 4 and 6. With the reference as the direct path, its objective is  $J_{\text{direct}} = p\bar{v}_h + (1-p)\bar{v}_m - \lambda[p\bar{c}_h + (1-p)\bar{c}_m]$ , and attempting the primary first gives  $J_{\text{cascade}} = pv_h + (1-p)\bar{v}_m - \lambda[a + (1-p)\bar{c}_m]$ . Subtracting,

$$J_{\text{cascade}} - J_{\text{direct}} = p(\lambda\bar{c}_h + v_h - \bar{v}_h) - \lambda a, \quad \text{so that} \quad W_\lambda = \lambda\bar{c}_h + v_h - \bar{v}_h.$$

The miss-conditional quantities  $b = \bar{c}_m$  and  $v_b = \bar{v}_m$  cancel: the margin is priced by what the reference would have spent and delivered on the accepted tasks. When the means are  $H$ -independent this is (19). For the routing-information band, let  $U$  be the sufficiency event with  $\mathbb{P}(U = 1) = s$ , let the reference cost  $\bar{c}_h$  on sufficient tasks and  $\bar{c}_m$  on the others, and let values be preserved. Then  $C_{\text{direct}} = s\bar{c}_h + (1-s)\bar{c}_m$ ,  $C_{\text{cascade}} = a + (1-s)\bar{c}_m$ ,  $C_{\text{info}} = k + sa + (1-s)\bar{c}_m$ , and

$$C_{\text{direct}} - C_{\text{info}} = s(\bar{c}_h - a) - k, \quad C_{\text{cascade}} - C_{\text{info}} = a(1-s) - k,$$

and  $C_{\text{direct}} - C_{\text{cascade}} = s\bar{c}_h - a$ , so (21) holds with  $b$  replaced by  $\bar{c}_h$  (for  $\bar{c}_h > a$ ), the uninformed crossover is  $s = a/\bar{c}_h$ , and the band is nonempty if and only if  $k < a(\bar{c}_h - a)/\bar{c}_h$ .

The same reference-restoring fallback gives

$$\frac{\mathbb{E}[HC_B]}{\mathbb{E}[C_B]} = 1 - \frac{\mathbb{E}[D]}{\mathbb{E}[C_B]} + \frac{\mathbb{E}[A]}{\mathbb{E}[C_B]}.$$

Therefore cost savings lower-bound a fraction weighted by reference cost. They lower-bound ordinary acceptance frequency only under an additional condition such as constant  $C_B$ .

An auxiliary routing call need not have an invocation event independent of  $H$ . Suppose primary plus gate cost  $a = c_p + \kappa_g$  is paid every round, while auxiliary routing cost  $\kappa_h$  is paid only on misses. If the fallback backend restores reference value at reference cost  $b$ , its total additional resource is  $C_F = b + \kappa_h$ , and the comparison gives

$$\eta \geq \eta_B \iff p \geq \frac{\bar{v}(c_p + \kappa_g + \kappa_h)}{bv_h + \bar{v}\kappa_h}.$$

This follows by the same cross-multiplication and incorporates the dependence of average overhead on the miss rate.

## H.7 REJECTED ATTEMPTS THAT LEAVE REUSABLE WORK

The acceptance floor (15) charges a rejected attempt in full: the fallback then pays the whole reference cost  $b$  on top of the primary cost  $a$ . In a router that continues from the state a rejected attempt leaves behind, part of that work is reusable. A failed patch may leave a configured environment, a partial edit, or informative test output; the model that continues does not start from nothing. Let  $\sigma \in [0, 1]$  be the fraction of the reference's work that remains usable after a rejected attempt, and let  $\kappa_s \geq 0$  be the cost of switching to the fallback path (a new context, a cold cache, a hand-over). The fallback then restores the reference value  $\bar{v}$  at additional cost  $(1-\sigma)b + \kappa_s$  instead of  $b$ .

**Proposition 11** (Salvageable attempts). *Let the reference cost  $b = \bar{c}$  and value  $\bar{v}$  be independent of acceptance, let the primary cost  $a$  include its verification, let an accepted attempt deliver conditional value  $v_h$  with probability  $p$ , and let a rejected attempt be followed by a fallback that restores  $\bar{v}$  at additional cost  $(1-\sigma)b + \kappa_s$ . The router is at least as efficient as the reference if and only if*

$$pbv_h + (1-p)\sigma b\bar{v} \geq \bar{v}[a + (1-p)\kappa_s]. \quad (37)$$

*If  $bv_h + \kappa_s\bar{v} > \sigma b\bar{v}$ , which holds whenever  $v_h \geq \bar{v}$  unless  $\sigma = 1$  and  $\kappa_s = 0$ , this is*

$$p \geq p_\sigma := \frac{\bar{v}(a + \kappa_s - \sigma b)}{bv_h - \sigma b\bar{v} + \kappa_s\bar{v}}, \quad (38)$$

which reduces to (15) at  $\sigma = \kappa_s = 0$ . If, in addition to  $bv_h + \kappa_s \bar{v} > \sigma b \bar{v}$ , also  $\sigma b \geq a + \kappa_s$ , then  $p_\sigma \leq 0$ : the router is at least as efficient at every acceptance probability, because a rejected attempt leaves behind more reusable reference work than it costs.

*Proof.* Substitute  $\mathbb{E}[V] = pv_h + (1-p)\bar{v}$  and  $\mathbb{E}[D] = a + (1-p)[(1-\sigma)b + \kappa_s]$  into  $b\mathbb{E}[V] \geq \bar{v}\mathbb{E}[D]$ . Canceling the common miss terms gives (37), or equivalently

$$p(bv_h - \sigma b \bar{v} + \kappa_s \bar{v}) \geq \bar{v}(a + \kappa_s - \sigma b).$$

Dividing by the positive coefficient of  $p$  gives (38). Its numerator is nonpositive exactly when  $\sigma b \geq a + \kappa_s$ , proving the final assertion.  $\square$

With  $a = 1/10$  and  $b = v_h = \bar{v} = 1$  the floor is 10% without salvage and disappears at  $\sigma = 3/10$ ,  $\kappa_s = 1/20$  ( $p_\sigma = -1/5$ ). The salvage fraction is a property of the harness, not of the model: a router that re-routes inside a trajectory keeps the environment and the observations of the failed attempt, whereas a cascade that restarts the task from its prompt has  $\sigma = 0$ , and a per-task router such as ACRouter (Zhou et al., 2026a), which commits its single attempt, has no rejected attempt to salvage and lies outside the proposition. Proposition 11 is the one-attempt case; with several attempts the option value of later attempts lowers the bar further (Proposition 7).

A test suite or sandbox check is a verifier with a fixed operating point, covered by Appendix H.5; its execution cost and latency belong in the attempt cost.

## H.8 GATING, AUXILIARY ROUTING, AND MODEL SWITCHING

For additive charges, write the always-incurred gate cost as  $\kappa_g$  and an auxiliary routing call’s cost as  $\kappa_h$ . If  $J_h$  indicates invoking that call, expected routing overhead is

$$\mathbb{E}[\kappa_g + J_h \kappa_h].$$

For constant charges this becomes  $\kappa_g + \mathbb{P}(J_h = 1)\kappa_h$ . It needs no independence between the call and primary acceptance. Put each charge in  $A$  if paid before the primary verdict, or in  $C_F$  if incurred only on fallback. The miss-only threshold above is the homogeneous instance.

A change of model can also alter input-cache cost. Suppose a fixed prefix has  $L$  tokens, fraction  $\gamma$  is reusable, and cached input has price multiplier  $\delta \in [0, 1]$ . Let  $c_m, c_{m'}$  be full input prices per token. Staying on model  $m$ , whose cache holds the prefix, costs  $c_m[(1-\gamma) + \gamma\delta]L$ ; switching to a destination  $m'$  whose cache does not hold it costs  $c_{m'}L$ . The immediate input-cost difference is

$$c_{m'}L - c_m[(1-\gamma) + \gamma\delta]L.$$

The destination’s lost discount relative to its own cached execution is instead  $c_{m'}(1-\delta)\gamma L$ . These quantities coincide when prices agree. Switching to a sufficiently cheaper model can yield a negative net difference despite losing its cache discount. Apply the gain identity of Lemma 1 to the costs actually incurred on the same task; do not add a cache penalty again if it is already in the billed input cost. This is a one-step accounting calculation, not an optimal dynamic switching policy.

## I THRESHOLDS AND THE VALUE OF ROUTING INFORMATION

### I.1 PROOF OF PROPOSITION 4

Here  $b$  and  $v_b$  are the fallback’s miss-conditional cost and value, and they are also the cost and value of the direct path: the reference means do not depend on acceptance. Appendix H.6 treats the correlated case, in which  $W_\lambda = \lambda \bar{c}_h + v_h - \bar{v}_h$ .

*Proof.* The direct path has objective  $J_{\text{direct}} = v_b - \lambda b$ . Attempting the primary first has

$$J_{\text{cascade}} = pv_h + (1-p)v_b - \lambda[a + (1-p)b].$$

Subtracting,

$$J_{\text{cascade}} - J_{\text{direct}} = p(v_h - v_b + \lambda b) - \lambda a = pW_\lambda - \lambda a.$$

This proves the comparison and, for  $W_\lambda > 0$ , the threshold. If  $W_\lambda \leq 0$ , then the difference is strictly negative because  $\lambda, a > 0$ .

A routing refinement whose stipulated effects are to change  $p$  to  $p + \Delta p$  and add cost  $k$  changes the attempt-then-escalate objective by

$$\Delta p(v_h - v_b + \lambda b) - \lambda k = \Delta p W_\lambda - \lambda k.$$

Its sign proves the second assertion. The clause compares the refined cascade with the unrefined one, both attempting first; it is worthwhile overall only if the refined cascade also clears the direct-path comparison,  $(p + \Delta p)W_\lambda \geq \lambda(a + k)$ . A refinement can satisfy  $\Delta p W_\lambda \geq \lambda k$  while the cascade remains inferior to dispatching directly, and then the right decision is direct. The refinement changes the success probability with the other quantities held fixed; it does not model a signal allowing contingent choice between fixed actions, nor a change of the verifier's threshold, which moves  $p, v_h$  and  $e$  together (Appendix H.5).  $\square$

## I.2 PROOF OF PROPOSITION 7

A task is handled by at most  $K$  attempts and, if all of them fail, by a terminal path of cost  $b$  and value  $v_b$ . Attempt  $k$  costs  $a_k$ , including its verification, and is accepted with probability  $p_k$  conditional on attempts  $1, \dots, k-1$  having failed. An accepted attempt delivers  $v_h$  and ends the task. A policy decides, after each failure, whether to make the next attempt or to stop and use the terminal path. Its decision may depend on the history, which at that point is the number of failed attempts, and may randomize. Write  $J^{\text{direct}} = v_b - \lambda b$ , let  $\Pi_k$  be the supremum of  $J_\lambda = \mathbb{E}[V - \lambda D]$  over policies from the state in which  $k-1$  attempts have failed, and set  $\Pi_{K+1} = J^{\text{direct}}$ , since after  $K$  failures only the terminal path remains. All costs and values are expectations conditional on the state, as in Proposition 4.

*Proof. Recursion.* Fix  $k \leq K$  and the state with  $k-1$  failures. Stopping yields  $J^{\text{direct}}$ . Attempting pays  $a_k$ ; with probability  $p_k$  the attempt is accepted and the task ends with value  $v_h$ , and with probability  $1 - p_k$  it fails and the process reaches the state with  $k$  failures. From there the best continuation is worth  $\Pi_{k+1}$ : the probabilities  $p_{k+1}, \dots, p_K$  are by hypothesis already conditional on this failure, so the continuation problem is the one that defines  $\Pi_{k+1}$ . Since  $J_\lambda$  is linear in value and cost, and  $a_k$  is paid whatever follows, attempting and then continuing optimally is worth  $p_k v_h - \lambda a_k + (1 - p_k)\Pi_{k+1}$ ; no continuation is worth more. A randomized decision yields a convex combination of the two numbers and cannot exceed the larger one. Hence

$$\Pi_k = \max\{J^{\text{direct}}, p_k v_h - \lambda a_k + (1 - p_k)\Pi_{k+1}\}, \quad k = K, K-1, \dots, 1,$$

which is backward induction from  $\Pi_{K+1} = J^{\text{direct}}$ . The finite-horizon dynamic program has an optimal policy that is deterministic and depends only on  $k$ : in each state it takes an action attaining the maximum.

*The decision rule.* Write  $\Delta_{k+1} = \Pi_{k+1} - J^{\text{direct}}$  and  $J^{\text{direct}} = p_k J^{\text{direct}} + (1 - p_k)J^{\text{direct}}$ . Subtracting the two branches,

$$\begin{aligned} [p_k v_h - \lambda a_k + (1 - p_k)\Pi_{k+1}] - J^{\text{direct}} &= p_k (v_h - v_b + \lambda b) - \lambda a_k + (1 - p_k)\Delta_{k+1} \\ &= p_k W_\lambda - \lambda a_k + (1 - p_k)\Delta_{k+1}, \end{aligned}$$

with  $W_\lambda = \lambda b + v_h - v_b$ . Its sign gives the decision rule. Since stopping is always available,  $\Delta_{k+1} \geq 0$ , and  $\Delta_{K+1} = 0$  recovers Proposition 4 at the last attempt. Earlier attempts receive the nonnegative option value  $(1 - p_k)\Delta_{k+1}$ . For example, with  $v_h = v_b = b = \lambda = 1$ ,  $(p_1, a_1) = (1/10, 1/5)$  and  $(p_2, a_2) = (9/10, 1/4)$ , the first attempt fails its own threshold but is optimal because  $1/10 + (9/10)(13/20) > 1/5$ . Retry probabilities are conditional on the preceding failures; a model switch may change both  $p_k$  and  $a_k$ .

*Regret.* Zhou et al. (2026a) score task  $i$ , handled by model  $a_i$ , with the reward  $r_i = \epsilon_1 s_i + \epsilon_2 \kappa_i$ , where  $s_i$  is the verifier's score for the output,  $\kappa_i$  its cost,  $\epsilon_1 > 0$  and  $\epsilon_2 < 0$ , and they define the cumulative regret against a per-task oracle as  $\sum_i [\max_j R_{ij} - R_{ia_i}]$ , where  $R_{ij}$  is the reward model  $j$  earns on task  $i$ . With  $\lambda = -\epsilon_2/\epsilon_1 > 0$ ,

$$r_i = \epsilon_1 (s_i - \lambda \kappa_i) = \epsilon_1 J_\lambda(x_i, a_i),$$

where  $J_\lambda(x_i, u) = s_{iu} - \lambda \kappa_{iu}$  is the score of model  $u$  on task  $i$  net of its priced cost. The reward is thus  $J_\lambda$  up to the positive factor  $\epsilon_1$ . This factor changes neither the optimal action nor the sign of any gap, and

$$\sum_i [\max_j R_{ij} - R_{ia_i}] = \epsilon_1 \sum_i [\max_u J_\lambda(x_i, u) - J_\lambda(x_i, u_i)],$$

the accumulated per-task gap of the proposition, up to the factor  $\epsilon_1$ . The identity concerns recorded outcomes, as in an outcome matrix. When outcomes are random, the oracle is a hindsight oracle, and since  $\mathbb{E} \max_u \geq \max_u \mathbb{E}$  the expected regret is at least the gap computed from expected values.  $\square$

In Zhou et al. (2026a),  $K = 1$  with no within-task fallback: one model's output is committed and verification writes records for later tasks. Step-level agentic routing permits  $K > 1$  (Liu et al., 2026); a single verify-then-escalate cascade is  $K = 1$  with a terminal fallback.

### I.3 PROOF OF THEOREM 5

The proof applies the fractional-programming characterization of Dinkelbach (1967) to finitely many observable states, including the quality floor.

Throughout,  $x$  ranges over a finite set  $\mathcal{X}$  of observable states with router-independent probabilities  $\pi(x) > 0$ ; each state offers a finite menu  $\mathcal{U}(x)$  of actions with expected value  $v(x, u)$  and expected cost  $c(x, u) > 0$ ; a router  $g$  chooses  $u$  with probability  $g(u | x)$ . Let  $\mathcal{G}$  be the set of routers, a product of simplices and hence a compact polytope, and put  $c_{\min} = \min_{x,u} c(x, u) > 0$ . For  $g \in \mathcal{G}$ ,

$$\mathbb{E}_g[V] = \sum_x \pi(x) \sum_{u \in \mathcal{U}(x)} g(u | x) v(x, u), \quad \mathbb{E}_g[D] = \sum_x \pi(x) \sum_{u \in \mathcal{U}(x)} g(u | x) c(x, u) \geq c_{\min},$$

$\eta(g) = \mathbb{E}_g[V] / \mathbb{E}_g[D]$ , and for every real  $\lambda$

$$\mathbb{E}_g[V - \lambda D] = \mathbb{E}_g[D] (\eta(g) - \lambda). \quad (39)$$

A router is deterministic if every  $g(\cdot | x)$  is a point mass; there are  $\prod_x |\mathcal{U}(x)|$  of them. The reference policy is admissible if it is an element  $g_B \in \mathcal{G}$ ; then  $\eta(g_B) = \eta_B$  and  $\mathbb{E}_{g_B}[V] = \bar{v}$ , so the reference satisfies the floor  $\mathbb{E}[V] \geq \bar{v} - \epsilon$  for every  $\epsilon \geq 0$ .

*Proof. Two preliminary facts.* (F1) State-wise separation: for any real numbers  $\phi(x, u)$ , the linear functional  $g \mapsto \sum_x \pi(x) \sum_u g(u | x) \phi(x, u)$  on  $\mathcal{G}$  is maximized exactly by the routers that, in every state  $x$ , put all their mass on  $\arg \max_u \phi(x, u)$ , and its maximum is  $\sum_x \pi(x) \max_u \phi(x, u)$ . Indeed, for each  $x$  the inner sum is a convex combination of the numbers  $\phi(x, u)$ , hence at most  $\max_u \phi(x, u)$ , with equality if and only if  $g(\cdot | x)$  is supported on the maximizers; and  $\pi(x) > 0$ . In particular a deterministic maximizer exists. (F2) Sign of the Dinkelbach function: applying (F1) with  $\phi = v - \lambda c$ ,

$$\Phi(\lambda) = \sum_x \pi(x) \max_{u \in \mathcal{U}(x)} [v(x, u) - \lambda c(x, u)] = \max_{g \in \mathcal{G}} \mathbb{E}_g[V - \lambda D], \quad (40)$$

so by (39) and  $\mathbb{E}_g[D] > 0$ ,  $\Phi(\lambda) \geq 0$  if and only if some router has  $\eta(g) \geq \lambda$ , and  $\Phi(\lambda) > 0$  if and only if some router has  $\eta(g) > \lambda$ .

*Part (i).* Each  $\phi_x(\lambda) = \max_u [v(x, u) - \lambda c(x, u)]$  is the maximum of finitely many affine functions of  $\lambda$  with slopes  $-c(x, u) < 0$ . It is therefore convex, piecewise linear and continuous, and strictly decreasing: if  $\lambda < \lambda'$  and  $u'$  attains  $\phi_x(\lambda')$ , then  $\phi_x(\lambda) \geq v(x, u') - \lambda c(x, u') > v(x, u') - \lambda' c(x, u') = \phi_x(\lambda')$ . Moreover  $\phi_x(\lambda) \rightarrow +\infty$  as  $\lambda \rightarrow -\infty$ , and  $\phi_x(\lambda) \leq \max_u v(x, u) - \lambda c_{\min} \rightarrow -\infty$  as  $\lambda \rightarrow +\infty$ . The positive combination  $\Phi = \sum_x \pi(x) \phi_x$  inherits all these properties and has a unique root  $\eta^*$ . By (F2),  $\Phi(\eta^*) = 0$  yields a router with  $\eta(g) \geq \eta^*$  and forbids  $\eta(g) > \eta^*$ ; hence  $\eta^* = \max_g \eta(g)$ , and the maximum is attained. A router  $g$  is efficiency-optimal if and only if  $\eta(g) = \eta^*$ , if and only if  $\mathbb{E}_g[V - \eta^* D] = 0 = \Phi(\eta^*)$  by (39), if and only if  $g$  maximizes the  $\eta^*$ -objective over  $\mathcal{G}$ , if and only if, by (F1), in every state it uses only actions maximizing  $v(x, u) - \eta^* c(x, u)$ . For the two-action menu of Proposition 4, direct has  $(v, c) = (v_b, b)$  and attempt-then-escalate has  $(pv_h + (1-p)v_b, a + (1-p)b)$ , all quantities depending on  $x$ ; the difference of their  $\lambda$ -objectives is  $p(v_h - v_b) - \lambda a + \lambda p b = pW_\lambda - \lambda a$  (Appendix I.1). At  $\lambda = \eta^*$  this is the rule  $p(x)W_{\eta^*}(x) \geq \eta^* a(x)$ , either action being allowed at equality.

*Part (ii).* Set  $\lambda_0 = \eta_B$  and, given  $\lambda_t$ , let  $g_t$  be a deterministic maximizer of the  $\lambda_t$ -objective, which exists by (F1), and  $\lambda_{t+1} = \eta(g_t)$ . By (40) and (39),  $\Phi(\lambda_t) = \mathbb{E}_{g_t}[D](\lambda_{t+1} - \lambda_t)$ . Admissibility gives  $\Phi(\lambda_0) \geq \mathbb{E}_{g_B}[V - \eta_B D] = 0$ , and for  $t \geq 1$ ,  $\Phi(\lambda_t) \geq \mathbb{E}_{g_{t-1}}[V - \lambda_t D] = 0$  because  $\lambda_t = \eta(g_{t-1})$ . Hence  $\lambda_{t+1} \geq \lambda_t$  at every step, with equality if and only if  $\Phi(\lambda_t) = 0$ , that is  $\lambda_t = \eta^*$ , where the iteration stops. As long as it has not stopped,  $\lambda_1 < \lambda_2 < \dots$  are efficiencies of distinct deterministic routers, of which there are finitely many; so the iteration stops after at most  $\prod_x |\mathcal{U}(x)|$  steps with  $\lambda_T = \eta^*$ , and any maximizer of the  $\eta^*$ -objective is efficiency-optimal by part (i). For the first step, write  $Q_g = \mathbb{E}_g[V] - \bar{v}$  and  $S_g = \bar{c} - \mathbb{E}_g[D]$ ; since  $\bar{v} = \eta_B \bar{c}$ ,  $Q_g + \eta_B S_g = \mathbb{E}_g[V - \eta_B D]$ , which is (11), and (40) at  $\lambda = \eta_B$  reads  $\Phi(\eta_B) = \max_g [Q_g + \eta_B S_g]$ . The first Dinkelbach step thus maximizes the net gain of Lemma 1 over routers, and  $\Phi(\eta_B) > 0$  if and only if some router has positive net gain. Without admissibility the iteration still terminates, since  $\Phi(\lambda_t) \geq 0$  for all  $t \geq 1$ ; only the monotone start from  $\eta_B$  uses it.

*Part (iii): existence and a router that randomizes in at most one state.* Let  $\mathcal{G}_\epsilon = \{g \in \mathcal{G} : \mathbb{E}_g[V] \geq \bar{v} - \epsilon\}$ , a compact polytope containing  $g_B$ . Since  $\eta$  is continuous on  $\mathcal{G}$  (its denominator is at least  $c_{\min}$ ), a constrained maximizer exists; let  $\eta_\epsilon^* = \max_{\mathcal{G}_\epsilon} \eta$ . To see that a maximizer can be chosen that randomizes in at most one state, use the Charnes–Cooper transformation. For  $g \in \mathcal{G}_\epsilon$  put  $t = 1/\mathbb{E}_g[D] > 0$ ,  $y(x, u) = t \pi(x) g(u | x)$  and  $\sigma = t(\mathbb{E}_g[V] - \bar{v} + \epsilon) \geq 0$ . Then  $(y, t, \sigma)$  is feasible for the linear program

$$\begin{aligned} & \max \quad \sum_{x,u} y(x, u) v(x, u) \\ & \text{subject to} \quad \sum_{x,u} y(x, u) c(x, u) = 1, \quad \sum_{u \in \mathcal{U}(x)} y(x, u) = \pi(x) t \quad (x \in \mathcal{X}), \\ & \quad \sum_{x,u} y(x, u) v(x, u) - (\bar{v} - \epsilon) t - \sigma = 0, \quad y \geq 0, t \geq 0, \sigma \geq 0, \end{aligned} \quad (41)$$

with objective value  $\eta(g)$ . Conversely, a feasible  $(y, t, \sigma)$  has  $t > 0$ , since  $t = 0$  forces  $y = 0$  and contradicts the first constraint, and  $g(u | x) = y(x, u)/(\pi(x)t)$  is a router with  $\mathbb{E}_g[D] = 1/t$ ,  $\mathbb{E}_g[V] = t^{-1} \sum yv = \bar{v} - \epsilon + \sigma/t \geq \bar{v} - \epsilon$  and  $\eta(g) = \sum yv$ . Thus (41) is feasible, its value is  $\eta_\epsilon^*$ , and it is bounded because  $\sum y \leq 1/c_{\min}$ . A feasible and bounded linear program in standard form has an optimal basic solution, and a basic solution has at most as many positive variables as there are equality constraints, here  $|\mathcal{X}| + 2$ . Since  $t > 0$ , at most  $|\mathcal{X}| + 1$  of the variables  $y(x, u)$  and  $\sigma$  are positive, and each state has at least one positive  $y(x, \cdot)$  because  $\sum_u y(x, u) = \pi(x)t > 0$ . Therefore either every state has exactly one positive  $y(x, \cdot)$ , a deterministic router, or exactly one state has two positive entries and  $\sigma = 0$ . The corresponding optimal router randomizes in at most one state, between two actions, and when it does the floor binds.

*Part (iii): the multiplier.* Define  $\Phi_\epsilon(\lambda) = \max_{g \in \mathcal{G}_\epsilon} \mathbb{E}_g[V - \lambda D]$ . As a maximum over a compact set of affine functions of  $\lambda$  with slopes  $-\mathbb{E}_g[D] \leq -c_{\min}$ , it is convex, continuous and strictly decreasing, tends to  $\pm\infty$  as  $\lambda \rightarrow \mp\infty$ , and has a unique root. By (39),  $\Phi_\epsilon(\lambda) \geq 0$  if and only if some  $g \in \mathcal{G}_\epsilon$  has  $\eta(g) \geq \lambda$ , so the root is  $\eta_\epsilon^*$ . Consequently a router is constrained-optimal if and only if it solves

$$\max_{g \in \mathcal{G}} f(g) := \mathbb{E}_g[V - \eta_\epsilon^* D] \quad \text{subject to} \quad h(g) := \mathbb{E}_g[V] - (\bar{v} - \epsilon) \geq 0, \quad (42)$$

whose optimal value is  $\Phi_\epsilon(\eta_\epsilon^*) = 0$ . Both  $f$  and  $h$  are affine in  $g$  and  $\mathcal{G}$  is a polytope, so (42) is a linear program; it is feasible ( $g_B$ ) and bounded. By linear-programming strong duality applied to the single constraint  $h \geq 0$ , there is a multiplier  $\kappa \geq 0$  such that

$$\max_{g \in \mathcal{G}} [f(g) + \kappa h(g)] = 0.$$

If  $g$  is optimal for (42), then  $h(g) \geq 0$  and  $f(g) = 0$ , so  $0 \leq f(g) + \kappa h(g) \leq 0$ : hence  $\kappa h(g) = 0$  and  $g$  maximizes  $f + \kappa h$  over  $\mathcal{G}$ . (Conversely, a feasible maximizer of  $f + \kappa h$  with  $\kappa h(g) = 0$  has  $f(g) = 0$  and is optimal.) Now  $f(g) + \kappa h(g) = \mathbb{E}_g[(1 + \kappa)V - \eta_\epsilon^* D] + \kappa(\epsilon - \bar{v})$ , so by (F1) every constrained-optimal router uses, in every state, only actions maximizing  $(1 + \kappa)v(x, u) - \eta_\epsilon^* c(x, u)$ , and  $\kappa(\mathbb{E}_g[V] - \bar{v} + \epsilon) = 0$ : if  $\kappa > 0$  the floor binds at every optimal router. The value  $\kappa = 0$  is a valid multiplier if and only if some efficiency-optimal router of part (i) satisfies the floor, and then  $\eta_\epsilon^* = \eta^*$ . For the two-action menu the difference of the modified objectives is

$$(1 + \kappa)p(v_h - v_b) - \eta_\epsilon^*(a - pb) = p[\eta_\epsilon^* b + (1 + \kappa)(v_h - v_b)] - \eta_\epsilon^* a,$$

and dividing by  $1 + \kappa > 0$  gives  $pW_{\lambda'} - \lambda'a$  with  $\lambda' = \eta_\epsilon^*/(1 + \kappa)$ : this is Proposition 4 at price  $\eta_\epsilon^*/(1 + \kappa)$ . The floor thus enters only through  $\kappa$ , which raises the value of a hit relative to its cost; a state in which the modified rule ties is the one state in which an optimal router may randomize, with the randomization probability set by the floor.  $\square$

**A two-state example.** Let two states be equally likely, with  $p = 9/10$ ,  $a = 1/10$ ,  $b = v_b = 1$ , and  $v_h = 1/2$  in state 1 and  $v_h = 1$  in state 2. The reference dispatches directly in both states. The unconstrained optimum attempts in both, giving  $\eta^* = 31/8$  and quality loss  $9/40$ . Under a tighter floor, state 2 still attempts; attempting in state 1 with probability  $\varrho$  gives  $\mathbb{E}[V] = 1 - 9\varrho/40$  and  $\mathbb{E}[D] = 3/5 - 2\varrho/5$ . Their ratio increases in  $\varrho$ , so the constrained optimum binds the floor at  $\varrho = 40\epsilon/9$ :

$$\eta_\epsilon^* = \frac{45(1 - \epsilon)}{27 - 80\epsilon}, \quad 0 \leq \epsilon \leq \frac{9}{40},$$

At  $\epsilon = 9/80$ , this gives  $\varrho = 1/2$ ,  $\eta_\epsilon^* = 71/32$  and multiplier  $\kappa = 53/18$ : exactly one state randomizes, as part (iii) asserts.

#### I.4 PROOF OF PROPOSITION 6

*Proof.* Subtract informed cost from each uninformed alternative:

$$C_{\text{direct}}(s) - C_{\text{info}}(s) = s(b - a) - k, \quad C_{\text{cascade}}(s) - C_{\text{info}}(s) = a(1 - s) - k.$$

Both differences are strictly positive if and only if  $k/(b - a) < s < 1 - k/a$ , which is (21). The interval is nonempty if and only if

$$\frac{k}{b - a} < 1 - \frac{k}{a} \iff k < \frac{a(b - a)}{b}.$$

This same condition places  $a/b$  strictly between the two endpoints. Since  $k \geq 0$ , the band lies within  $0 < s < 1$ . Equality at an endpoint means that information ties one uninformed alternative.  $\square$

#### I.5 IMPERFECT INFORMATION AND THE PERFECT-INFORMATION BOUND

Let  $U$  denote the binary sufficiency event, with  $\mathbb{P}(U = 1) = s$ . An experiment  $Z$  produces posterior  $s_Z = \mathbb{P}(U = 1 \mid Z)$ . Put

$$f(s) = \min\{a + (1 - s)b, b\}, \quad g(s) = sa + (1 - s)b.$$

After paying for  $Z$ , choosing the cheaper available action has expected cost  $k + \mathbb{E}[f(s_Z)]$ . Thus its net saving is  $f(s) - \mathbb{E}[f(s_Z)] - k$ .

The minimum of two affine functions is concave. Since  $\mathbb{E}[s_Z] = s$ , Jensen's inequality gives  $\mathbb{E}[f(s_Z)] \leq f(s)$ . Also  $f(t) \geq g(t)$  for every  $t \in [0, 1]$ : cascade minus  $g(t)$  is  $a(1 - t)$ , and direct minus  $g(t)$  is  $t(b - a)$ . Since  $g$  is affine,

$$g(s) = \mathbb{E}[g(s_Z)] \leq \mathbb{E}[f(s_Z)] \leq f(s).$$

Gross information value is therefore at most  $f(s) - g(s)$ . Exact revelation attains this bound because  $f(0) = g(0) = b$  and  $f(1) = g(1) = a$ . This proves directly the relevant perfect-information comparison (Blackwell, 1953). In particular,

$$f(s) - g(s) = \min\{s(b - a), a(1 - s)\},$$

maximized at  $s = a/b$  with value  $a(b - a)/b$ . At the same experiment cost  $k$ , no imperfect experiment strictly reduces cost outside the perfect-information band. Experiments with different costs require separate comparisons. In the language of decision theory,  $f(s) - g(s)$  is the expected value of perfect information and  $f(s) - \mathbb{E}[f(s_Z)]$  the expected value of the experiment  $Z$  (Raiffa & Schlaifer, 1961; Howard, 1966), here for two actions and a binary state.

**Information can pay only if  $k < a$ .** Exact information saves  $a(1 - s) - k$  against cheap-first execution, so it can pay only if  $k < a(1 - s) < a$ : it avoids at most one unsuccessful primary attempt per task.

**Corollary 12** (Routing information versus a better proposer). *Let a refinement raise sufficiency from  $s$  to  $s + \Delta p \leq 1$  at cost  $k$ , so that the refined cascade costs  $C_{\text{dec}} = k + a + (1 - s - \Delta p)b$ , and let exact routing information cost the same  $k$ . Then*

$$C_{\text{info}} - C_{\text{dec}} = b\Delta p - a(1 - s).$$

*At equal cost, paying for the routing signal beats improving the proposer if and only if  $b\Delta p < a(1 - s)$ .*

*Proof.*  $C_{\text{info}} - C_{\text{dec}} = [k + sa + (1 - s)b] - [k + a + (1 - s)b - \Delta p b] = -a(1 - s) + b\Delta p$ .  $\square$

Information avoids wasted attempts, worth  $a$  on each of the  $1 - s$  failing tasks; a better proposer avoids fallbacks, worth  $b$  on each of the  $\Delta p$  newly solved tasks. Both comparisons hold the reference fixed; with a correlated reference,  $b$  is replaced by  $\bar{c}_h$  as in Appendix H.6.

**A miscalibrated signal can be worse than no signal.** Equation (22) values the rule that thresholds the true posterior  $s_Z$  at  $a/b$ . A learned score  $\hat{s}$ , a function of  $Z$ , used through the rule “attempt if and only if  $\hat{s} \geq a/b$ ”, has expected cost  $k + \mathbb{E}[C_{\alpha(\hat{s})}(s_Z)]$ , where  $C_{\alpha}(t)$  is the cost of action  $\alpha$  when the posterior is  $t$ . Since  $C_{\alpha}(t) \geq \min\{a + (1 - t)b, b\} = f(t)$  for both actions,

$$k + \mathbb{E}[C_{\alpha(\hat{s})}(s_Z)] \geq k + \mathbb{E}[f(s_Z)],$$

with equality if and only if the rule attempts on  $\{s_Z > a/b\}$ , up to ties. Calibration is sufficient; a monotone transform with the correspondingly transformed threshold also attains the bound. For a miscalibration example, let  $a = 1/10$ ,  $b = 1$ ,  $s = 1/2$  and  $k = 0$ . The uninformed cascade costs  $3/5$ . Equiprobable posteriors  $s_Z \in \{1/20, 19/20\}$  reduce Bayes cost to  $23/40$ , but reporting  $\hat{s} = s_Z/10$  and retaining threshold  $a/b$  rejects both, costing 1. A free informative signal can therefore worsen decisions when its score is used at the wrong threshold.

## J VERIFICATION AND PARALLEL RESOURCE ACCOUNTING

### J.1 WHAT A CALIBRATED ACCEPTANCE EVENT GUARANTEES

Let  $Y$  indicate primary correctness and let  $\mathcal{G}$  contain all information used for commitment, including controller randomization. Suppose  $v = \mathbb{E}[Y \mid \mathcal{G}]$  and the measurable event  $\{H = 1\}$  is contained in  $\{v \geq \theta\}$ . For a positive-probability acceptance event,

$$\mathbb{P}(Y = 1 \mid H = 1) = \mathbb{E}[v \mid H = 1] \geq \theta.$$

For a nonnegative integrable  $\mathcal{G}$ -measurable stake  $w$ ,

$$\mathbb{E}[wY \mid H = 1] = \mathbb{E}[wv \mid H = 1] \geq \theta \mathbb{E}[w \mid H = 1].$$

These conclusions require calibration relative to all selection information. A score calibrated only marginally need not retain calibration after selection using additional predictive information. Lemma 1 does not assume calibration.

### J.2 CONCURRENT (HEDGED) EXECUTION: LATENCY AND CANCELLATION CHARGES

Concurrent (hedged) execution launches the primary and the reference at the same time, after controller overhead  $\ell_\kappa$ , and cancels the reference when the primary is accepted; below we say concurrent execution. Suppose deterministic completion times satisfy  $0 < \ell_p \leq \ell_b$ , with primary verification included in  $\ell_p$ , and the controller waits for the primary verdict. It terminates at  $\ell_\kappa + \ell_p$  on a hit and  $\ell_\kappa + \ell_b$  on a miss. With hit value  $v_h$  and a reference-restoring fallback, so that the miss value is  $\bar{v} > 0$ ,

$$\begin{aligned} \eta_{\text{hedge}}^\ell &= \frac{pv_h + (1 - p)\bar{v}}{\ell_\kappa + p\ell_p + (1 - p)\ell_b}, \\ \eta_{\text{hedge}}^\ell &\geq \frac{\bar{v}}{\ell_b} \iff p(\ell_b v_h - \bar{v}\ell_p) \geq \bar{v}\ell_\kappa. \end{aligned} \tag{43}$$

This follows by multiplying the two positive denominators and canceling the common miss term. If  $\ell_\kappa = 0$ ,  $v_h \geq \bar{v}$ , and  $\ell_p \leq \ell_b$ , the comparison holds, strictly when  $p > 0$  and either component comparison is strict. Sequential escalation instead has mean latency  $\ell_\kappa + \ell_p + (1-p)\ell_b$  and condition  $p\ell_b v_h \geq \bar{v}(\ell_\kappa + \ell_p)$ .

For dollars, let  $a > 0$  be the primary and controller cost,  $b > 0$  the full reference cost, and  $r$  the conditional expected charge for a prefix consumed before cancellation on a hit. Then

$$\eta_{\text{hedge}}^c = \frac{pv_h + (1-p)\bar{v}}{a + (1-p)b + pr}, \quad \eta_{\text{hedge}}^c \geq \frac{\bar{v}}{b} \iff p(bv_h - \bar{v}r) \geq \bar{v}a. \quad (44)$$

This is the hedged-request mechanism of Dean & Barroso (2013), with verification and billing for the consumed fallback prefix. Relative to sequential escalation with the same value, it shortens mean latency by  $(1-p)\ell_p$  and raises expenditure by  $pr$ .

For random completion times the two resources behave differently. Dollars add across calls, so the constants  $a$ ,  $b$  and  $r$  in (44) may be read as conditional expected charges without further change. Latency cannot be handled this way, because the controller waits for the primary verdict and, on a miss, for the fallback as well: the round ends at  $\ell_\kappa + L_p$  on a hit and at  $\ell_\kappa + \max(L_p, L_b)$  on a miss, and in general  $\mathbb{E}[\max(L_p, L_b)] \neq \max(\mathbb{E}L_p, \mathbb{E}L_b)$ . With  $m_h = \mathbb{E}[L_p \mid H = 1]$ ,  $m_m = \mathbb{E}[\max(L_p, L_b) \mid H = 0]$  and reference latency  $l_b = \mathbb{E}[L_b]$ , the mean latency of concurrent execution is  $\ell_\kappa + pm_h + (1-p)m_m$  and

$$\eta_{\text{hedge}}^\ell \geq \frac{\bar{v}}{l_b} \iff p(l_b v_h - \bar{v}m_h) \geq \bar{v}\ell_\kappa + (1-p)\bar{v}(m_m - l_b), \quad (45)$$

which is (43) plus the miss-branch term  $(1-p)\bar{v}(m_m - l_b)$ ; the two agree when  $m_m = l_b$ , as in the deterministic case  $\ell_p \leq \ell_b$ . Substituting the mean  $\mathbb{E}[L_p]$  for  $\ell_p$  is not valid in general. Take  $L_p \in \{1/2, 3/2\}$  equally likely and independent of  $H$ ,  $L_b = 1$ ,  $\ell_\kappa = 0$ ,  $v_h = \bar{v} = 1$  and  $p = 1/2$ . The mean-substituted criterion reads  $0 \geq 0$ , whereas  $m_h = 1$ ,  $m_m = 5/4$ , the mean latency is  $9/8 > 1$  and the efficiency is  $8/9 < 1$ ; criterion (45) correctly fails, since  $0 \geq 1/8$  is false. Lemma 1 applies verbatim to the latency resource with  $A = \ell_\kappa + L_p$ ,  $C_F = (L_b - L_p)^+$  and  $C_B = L_b$ ; what fails is only the substitution of unconditional means for the branch-conditional ones.

## K COVERAGE ALLOCATION ACROSS INDEPENDENT POSITIONS

### K.1 SETTING AND STATEMENT

Suppose  $K$  independent positions or task families have increasing, strictly concave differentiable coverage curves  $p_k(F_k)$ , where  $F_k \geq 0$  is a continuous allocation and  $\gamma_k > 0$  its unit price. Let  $w_k > 0$  be the value of coverage at position  $k$ . For  $J_\lambda$ , this weight can include the family frequency multiplied by the margin  $W_\lambda$  in (19); families with nonpositive margin are not attempted first and are excluded. A hard-budget subproblem is

$$\max_{F \geq 0} \sum_{k=1}^K w_k p_k(F_k) \quad \text{subject to} \quad \sum_k \gamma_k F_k \leq B. \quad (46)$$

This is separable allocation across positions, not a model of mutually redundant proposers within one position.

**Proposition 13** (Coverage allocation). *For  $B > 0$ , (46) has a unique optimizer. There is a multiplier  $\Lambda > 0$  such that funded positions satisfy  $w_k p'_k(F_k) = \Lambda \gamma_k$ , and unfunded positions satisfy  $w_k p'_k(0) \leq \Lambda \gamma_k$ . If  $1 - p_k(F) = d_k(1 + F)^{-r}$  with  $d_k \in (0, 1]$  and common  $r > 0$ , then*

$$F_k^* = \left[ \mu \left( \frac{w_k d_k}{\gamma_k} \right)^{1/(1+r)} - 1 \right]_+, \quad (47)$$

where  $\mu$  makes the budget bind.

The exponent depends on the capacity unit. For clean Zipf families, Proposition 10 gives  $r = 1 - 1/z$  for a retention horizon or event FIFO length, yielding large-allocation proportions  $(w_k d_k / \gamma_k)^{z/(2z-1)}$ . For distinct stored records, Corollary 18 gives  $r = z - 1$  and proportions  $(w_k d_k / \gamma_k)^{1/z}$ . These proportions are asymptotic.

## K.2 PROOF OF PROPOSITION 13

*Proof.* The feasible region  $\{F \geq 0: \sum_k \gamma_k F_k \leq B\}$  is nonempty, convex, and compact. A maximizer exists by continuity. Positive weights and strict concavity of each separate  $p_k$  make the sum strictly concave, so the maximizer is unique.

The budget binds: otherwise some coordinate could be increased feasibly and improve the objective. Since  $B > 0$ , a strictly feasible point with every coordinate positive exists. The concave problem therefore satisfies the Karush–Kuhn–Tucker conditions, which are also sufficient. With budget multiplier  $\Lambda \geq 0$  and nonnegativity multipliers  $\tau_k \geq 0$ , stationarity and complementary slackness give

$$w_k p'_k(F_k) - \Lambda \gamma_k + \tau_k = 0, \quad \tau_k F_k = 0.$$

A funded coordinate thus satisfies  $w_k p'_k(F_k) = \Lambda \gamma_k$ , and an unfunded one satisfies  $w_k p'_k(0) \leq \Lambda \gamma_k$ . A differentiable strictly increasing concave function on  $[0, \infty)$  has positive derivative at every finite argument: a zero derivative would preclude any subsequent increase by concavity. Since some coordinate is funded,  $\Lambda > 0$ .

For the shifted power law,  $p'_k(F) = r d_k (1 + F)^{-r-1}$ . Set  $g_k = (w_k d_k / \gamma_k)^{1/(1+r)}$  and  $\mu = (r/\Lambda)^{1/(1+r)}$ . Funded stationarity gives  $1 + F_k = \mu g_k$ , whereas exclusion gives  $\mu g_k \leq 1$ . Combining them proves

$$F_k^* = [\mu g_k - 1]_+.$$

The budget function  $\sum_k \gamma_k [\mu g_k - 1]_+$  is continuous, zero for sufficiently small  $\mu$ , strictly increasing once positive, and unbounded above. Exactly one  $\mu$  therefore gives  $B > 0$ .  $\square$

For the funded set  $\mathcal{K}_+ = \{k: \mu g_k > 1\}$ ,

$$\mu = \frac{B + \sum_{k \in \mathcal{K}_+} \gamma_k}{\sum_{k \in \mathcal{K}_+} \gamma_k g_k}. \quad (48)$$

The funded set is an upper level set of  $g_k$ , found by sorting and checking the corresponding sets. Equivalently, start with all coordinates, solve the displayed budget equation, and drop a smallest- $g$  coordinate when  $\mu g_k < 1$ . Such a removal decreases  $\mu$ : if  $T = B + \sum \gamma_k$  and  $U = \sum \gamma_k g_k$ , then  $(T - \gamma_j)/(U - \gamma_j g_j) < T/U$  if and only if  $(T/U)g_j < 1$ . A coordinate with equality has zero allocation and can be excluded without changing the solution.

## K.3 CONNECTING A HARD-BUDGET SUBPROBLEM TO EFFICIENCY

Proposition 13 maximizes coverage value under a capacity constraint. It does not imply that an efficiency-optimal policy exhausts a computation budget. To connect the weights to  $J_\lambda$ , let family  $k$  have frequency  $\nu_k$ , fixed hit/miss values, and fallback cost. Its benefit from increasing the hit probability has weight

$$w_k = \nu_k (v_{h,k} - v_{m,k} + \lambda b_k).$$

Positive-margin families enter the concave coverage problem. If effort  $F_k$  also adds  $\rho_k F_k$  to expected per-round resource, the appropriate objective, up to an additive constant, is

$$\sum_k w_k p_k(F_k) - \lambda \sum_k \rho_k F_k, \quad \sum_k \gamma_k F_k \leq B.$$

The funded KKT condition becomes

$$w_k p'_k(F_k) = \lambda \rho_k + \Lambda \gamma_k,$$

with the corresponding weak inequality at zero. The budget need not bind in this penalized problem. The hard-budget result is recovered when allocation resource is separately constrained and has no additional variable charge in this objective, or when that charge is fixed across compared allocations.

**Activation thresholds.** The weight  $w_k = \nu_k W_{\lambda,k}$  prices coverage only for a family that is routed cheap-first. By Proposition 4, family  $k$  is routed cheap-first at price  $\lambda$  if and only if  $p_k(F_k)W_{\lambda,k} \geq \lambda a_k$ ; otherwise it dispatches directly and gains nothing from coverage. Relative to always dispatching directly, its benefit is therefore

$$\nu_k \max\{0, p_k(F_k)W_{\lambda,k} - \lambda a_k\}, \quad (49)$$

which is zero up to the activation threshold  $F_k^{\min}$  defined by  $p_k(F_k^{\min}) = \lambda a_k / W_{\lambda,k}$  (with  $F_k^{\min} = \infty$  if this ratio exceeds one) and concave increasing beyond it. The kink at  $F_k^{\min}$  makes (49) non-concave, so the Karush–Kuhn–Tucker conditions of Proposition 13 are not sufficient for the  $J_\lambda$  problem. Problem (46) is the inner problem for a fixed activation set  $\mathcal{A}$ : maximize  $\sum_{k \in \mathcal{A}} w_k p_k(F_k)$  subject to  $\sum_k \gamma_k F_k \leq B$ ,  $F_k \geq F_k^{\min}$  for  $k \in \mathcal{A}$  and  $F_k = 0$  otherwise. For the shifted power law its solution is  $F_k^* = \max\{F_k^{\min}, \mu g_k - 1\}$  on  $\mathcal{A}$ , with  $\mu$  set by the budget. The outer problem chooses  $\mathcal{A}$  among the sets with  $\sum_{k \in \mathcal{A}} \gamma_k F_k^{\min} \leq B$ ; at the resulting allocation the per-family routing decision is the state-wise rule of Theorem 5 with the family as the observable state, and only the coupling through the budget makes the outer choice combinatorial.

For a counterexample to allocating first and routing afterwards, take two equally frequent families with  $\lambda = b_k = \gamma_k = 1$ ,  $v_h = v_b$ ,  $B = 2$ ,  $p_k(F) = F/(1+F)$  and  $a_1 = a_2 = 3/5$ . The equal-weight solution of (46) is  $F = (1, 1)$ , but  $p_k(1) = 1/2 < 3/5$  activates neither family, giving zero gain. Allocating  $F = (2, 0)$  instead activates family 1 and gains  $\frac{1}{2}(\frac{2}{3} - \frac{3}{5}) = 1/30 > 0$ . The activation choice must therefore precede marginal equalization; pruning inactive families after the fact does not recover the optimum.

#### K.4 STATIC COVERAGE AND ADAPTIVE SELECTION

For  $S \subseteq T$  and  $m \notin T$ ,

$$\begin{aligned} p(S \cup \{m\}) - p(S) &= \mathbb{P} \left( E_m \cap \bigcap_{i \in S} E_i^c \right) \\ &\geq \mathbb{P} \left( E_m \cap \bigcap_{i \in T} E_i^c \right) = p(T \cup \{m\}) - p(T). \end{aligned}$$

This proves diminishing returns under every fixed joint law. Nonnegative marginals also prove monotonicity. The classical static cardinality result consequently applies (Nemhauser et al., 1978); a price budget instead requires an appropriate submodular-knapsack method (Sviridenko, 2004).

Static submodularity does not establish adaptive submodularity. Let exactly one of two sources succeed, each with probability  $1/2$ . Before observations, the second source’s marginal success probability is  $1/2$ . After the first source fails, the conditional marginal becomes one. Information has increased its marginal benefit, violating adaptive diminishing returns. An adaptive guarantee requires the additional property studied by Golovin & Krause (2011). Likewise, subtracting modular call cost preserves submodularity but need not preserve monotonicity, so the monotone-coverage approximation guarantees do not apply unchanged to the penalized objective.

## L PORTFOLIO COMPLEMENTARITY

### L.1 PROOF OF PROPOSITION 2

Fix a portfolio  $S$  and let

$$p = p(S), \quad \mathcal{V} = v_m + p(v_h - v_m), \quad D = a(S) + (1 - p)b > 0.$$

For a candidate  $m \notin S$ , write

$$r = r_m(S) = \mathbb{P} \left( E_m \cap \bigcap_{i \in S} E_i^c \right), \quad d = d_m(S) = a(S \cup \{m\}) - a(S).$$

The enlarged portfolio has expected value  $\mathcal{V} + r(v_h - v_m)$  and positive expected cost  $D + d - rb$ . Subtracting efficiencies gives

$$\eta(S \cup \{m\}) - \eta(S) = \frac{r\{v_h - v_m + \eta(S)b\} - \eta(S)d}{D + d - rb}. \quad (50)$$

The numerator gives the criterion;  $d$  includes changes in aggregation and verification cost. The exact selecting verifier realizes the portfolio's virtual best solver, and  $r_m(S)$  is its component marginal contribution (Xu et al., 2012; Fréchette et al., 2016).

This calculation requires no independence among success events. It does require the same conditional values  $v_h, v_m$  and fallback cost  $b$  for the two portfolios. If a new portfolio changes which kinds of tasks reach the fallback, those conditional quantities may change; (50) must then be replaced by the corresponding comparison of unconditional expected values and costs.

Static coverage is monotone submodular under any fixed joint law of the events (Appendix K.4); conditional marginals after observing outcomes need not satisfy the analogous adaptive property.

## L.2 PROOF OF PROPOSITION 3

Let  $\mathbb{P}(E_0) = 1/2$ . Set  $(E_1, E_2) = (E_0, E_0^c)$  in one world and  $(E_0^c, E_0)$  in the other. Both worlds have marginal successes  $(1/2, 1/2, 1/2)$  and candidate costs  $c$ . A marginal-only selector therefore chooses  $m_1$  with the same probability  $q$  in both worlds. Its costs are  $2c + qb/2$  and  $2c + (1 - q)b/2$ , so

$$\max \left\{ 2c + \frac{qb}{2}, 2c + \frac{(1-q)b}{2} \right\} \geq 2c + \frac{b}{4}.$$

An informed selector chooses the complement and pays  $2c$ . The perfect fallback gives every policy unit value. The result covers randomized selection before the primary's output; it does not cover learning dependence through additional observations.

*Remark 14* (Identifiable range of residual coverage). For  $S = \{m_0\}$ , the Fréchet bounds are  $\max\{0, p_m - p_0\} \leq r_m(S) \leq \min\{p_m, 1 - p_0\}$ . Both endpoints are attainable by varying the overlap of  $E_m$  and  $E_0$  while fixing their marginals. Thus marginals identify an interval, not the residual coverage itself.

## L.3 RESIDUAL STEERING: SETTING AND STATEMENT

A role or context view can change where a candidate allocates its effort. To isolate that effect, let a task's accepted answer type  $T^*$  have law  $q$  on a finite set  $\mathcal{T}$ , and let the two proposals  $X_1 \sim p_1$  and  $X_2 \sim p_2$  be independent of each other and of  $T^*$  conditional on the observable state. Fix a region  $T_0 \subseteq \mathcal{T}$  to de-emphasize, chosen before the proposals are realized, and write  $T_1 = T_0^c$ . For  $\rho \in [0, 1]$ , define

$$p_2^\rho(t) = \frac{p_2(t)}{Z(\rho)} \begin{cases} \rho, & t \in T_0, \\ 1, & t \in T_1, \end{cases} \quad Z(\rho) = \rho W_0 + W_1, \quad (51)$$

where  $W_k = \sum_{t \in T_k} p_2(t)$ . Lower  $\rho$  moves effort out of  $T_0$ . If  $a_1 = \sum_t q(t)p_1(t) < 1$ , the target law after primary failure is

$$q^-(t) = \frac{q(t)(1 - p_1(t))}{1 - a_1}, \quad h(\rho) = \sum_t q^-(t)p_2^\rho(t).$$

**Proposition 15** (Residual steering). *Assume that  $\mathcal{T}$  is finite; that, conditional on the observable state,  $T^* \sim q$ ,  $X_1 \sim p_1$  and  $X_2 \sim p_2$  are mutually independent; that  $a_1 < 1$ ; that  $T_0$  is fixed before the proposals are realized; and that  $W_0, W_1 > 0$ . For a real function  $f$  on  $\mathcal{T}$  let  $\bar{f}_k = W_k^{-1} \sum_{t \in T_k} f(t)p_2(t)$  be its  $p_2$ -weighted mean over  $T_k$  and  $M_f(\rho) = \sum_t f(t)p_2^\rho(t)$ . Then*

$$M'_f(\rho) = \frac{W_0 W_1}{Z(\rho)^2} (\bar{f}_0 - \bar{f}_1), \quad \text{in particular} \quad h'(\rho) = \frac{W_0 W_1}{Z(\rho)^2} (\bar{q}_0^- - \bar{q}_1^-).$$

*Hence reducing  $\rho$  strictly increases the conditional coverage  $h = M_{q^-}$  if and only if  $\bar{q}_1^- > \bar{q}_0^-$ , and strictly reduces candidate 2's standalone quality  $a_2(\rho) = M_q(\rho)$  if and only if  $\bar{q}_0 > \bar{q}_1$ ; both can hold at once. Under an exact selecting verifier the pair succeeds with probability  $a_1 + (1 - a_1)h(\rho)$ .*

This is a finite answer-type model, not an assumption about arbitrary language-model distributions. It isolates a distinction that marginal scoring misses. Applying Proposition 2 or differentiating (14) converts the coverage change into an efficiency comparison only when the compared values and costs satisfy their stated conditions; in particular, changing the failure set may change the fallback's conditional quality (Appendix L.5).

#### L.4 PROOF OF PROPOSITION 15

Bayes' rule gives  $q^-(t) = \mathbb{P}(T^* = t \mid X_1 \neq T^*)$ . For any real  $f$ ,

$$M_f(\rho) = \frac{\rho W_0 \bar{f}_0 + W_1 \bar{f}_1}{Z(\rho)},$$

$$M'_f(\rho) = \frac{W_0 W_1}{Z(\rho)^2} (\bar{f}_0 - \bar{f}_1).$$

Conditional independence identifies  $M_{q^-} = h$  and  $M_q = a_2$ , proving both monotonicity statements. Exact selection gives  $p_{\text{pair}} = a_1 + (1 - a_1)h$ . With costs and values fixed,

$$\frac{d}{dp} \frac{v_m + p(v_h - v_m)}{a + (1 - p)b} = \frac{v_h(a + b) - v_m a}{[a + (1 - p)b]^2},$$

so the numerator determines the efficiency effect. This is a static reweighting chosen before  $X_1$ ; sequential adaptation to the realized proposal lies outside (51). Proposition 2 itself does not require proposal independence.

#### L.5 A FOUR-TYPE CONSTRUCTION

The following finite construction separates individual quality from complementarity without estimation error. Take

$$\begin{aligned} q &= (0.30, 0.20, 0.40, 0.10), & p_1 &= (0.45, 0.45, 0.05, 0.05), \\ p_2 &= (0.40, 0.40, 0.10, 0.10), & T_0 &= \{1, 2\}. \end{aligned}$$

Both members retain standalone quality  $1/4$ , while

$$\bar{q}^-_0 = \frac{11}{60} < \frac{19}{60} = \bar{q}^-_1, \quad h(\rho) = \frac{0.110\rho + 0.0475}{0.60\rho + 0.15}.$$

Thus steering away from  $T_0$  increases pair coverage. With each member costing one, fallback costing ten, and fixed  $v_h = 1$ ,  $v_m = 4/5$ ,

$$\eta(\rho) = \frac{\frac{4}{5} + \frac{1}{5}p_{\text{pair}}(\rho)}{12 - 10p_{\text{pair}}(\rho)}.$$

Table 14 gives the resulting trade-off. The candidate  $p_3 = (0.60, 0.40, 0, 0)$  has higher standalone

Table 14: The four-type construction: steered proposal law, standalone quality, conditional coverage, pair success, and efficiency at three steering levels.

| $\rho$ | $p_2^\rho$           | $a_2$ | $h(\rho)$ | $p_{\text{pair}}$ | $\eta$     |
|--------|----------------------|-------|-----------|-------------------|------------|
| 1      | (.4, .4, .1, .1)     | 1/4   | 21/100    | 163/400           | 1763/15850 |
| 1/4    | (.25, .25, .25, .25) | 1/4   | 1/4       | 7/16              | 71/610     |
| 0      | (0, 0, .5, .5)       | 1/4   | 19/60     | 39/80             | 359/2850   |

quality  $13/50$ , but lower pair success  $393/1000$  than even the unsteered pair's  $163/400$ . This establishes a limitation of marginal ranking within this pool, not a general advantage of steering over adding models.

The fixed-fallback assumption matters. For Figure 3, instead let fallback succeed exactly on types 3 and 4. Final value becomes

$$v(\rho) = 0.725 + \frac{0.11\rho}{Z(\rho)}.$$

Efficiency  $v(\rho)/[12 - 10p_{\text{pair}}(\rho)]$  falls from  $167/1585$  to  $29/285$  as  $\rho$  decreases from one to zero, despite increasing coverage: the fallback's conditional value changes with the residual task distribution.

## M INFORMATION-STATE DYNAMICS

### M.1 PROOF OF PROPOSITION 8

Index the first decision by  $n = 0$ . Arrivals are independent of the pre-decision store; expiration precedes a replacing write. A record of either type is freshly written or survives expiration without replacement, giving

$$i_{j,n+1} = \nu_j \omega_j + \alpha_j i_{j,n}, \quad w_{j,n+1} = \nu_j \psi_j + \alpha_j w_{j,n}. \quad (52)$$

Their fixed points and empty-start solutions are

$$\begin{aligned} i_j^* &= \frac{\nu_j \omega_j}{1 - \alpha_j}, & w_j^* &= \frac{\nu_j \psi_j}{1 - \alpha_j}, \\ i_{j,n} &= i_j^* (1 - \alpha_j^n), & w_{j,n} &= w_j^* (1 - \alpha_j^n). \end{aligned}$$

These are (27). Using the record gains of (25) and conditioning on the arriving cell,

$$\begin{aligned} \mathbb{E}[V_n - \eta_U D_n] &= \sum_j \nu_j [v_j^U - \eta_U c_j^U + i_{j,n} g_j^I + w_{j,n} g_j^W] \\ &= \sum_j \nu_j G_j (1 - \alpha_j^n), \end{aligned} \quad (53)$$

where, as in (26),

$$G_j = i_j^* g_j^I + w_j^* g_j^W = \frac{\nu_j (\omega_j g_j^I + \psi_j g_j^W)}{1 - \alpha_j}.$$

The uninformed-policy terms cancel after averaging over cells.

Let  $\bar{c}_\infty > 0$  be the stationary expected cost. Passing to the limit in (53) gives the stronger comparison

$$\eta_\infty - \eta_U = \frac{\sum_j \nu_j G_j}{\bar{c}_\infty}. \quad (54)$$

Summing (53) as a geometric series gives (28), an identity for expected aggregates.

If  $g_j^I > 0 > g_j^W$ , the sign of  $G_j$  is the sign of

$$\omega_j g_j^I + \psi_j g_j^W = r_j \left[ g_j^I - \frac{\psi_j}{r_j} (g_j^I - g_j^W) \right].$$

It is positive if and only if  $\psi_j / r_j < g_j^I / (g_j^I - g_j^W)$ .

For finitely many cells, convergence is geometric with factor at most  $\max_j \alpha_j < 1$ . Independent arrivals do not make the resulting hit sequence independent or Markov.

### M.2 RECORDS INSIDE A VERIFIED CASCADE: PROOF OF PROPOSITION 9

The record changes only  $p_j^s$ ;  $a_j, b_j, v_h, v_b$  are fixed. Thus

$$v_j^s = p_j^s v_h + (1 - p_j^s) v_b, \quad c_j^s = a_j + (1 - p_j^s) b_j.$$

Subtracting the uninformed state, with  $\Delta_j^s = p_j^s - p_j^U$ , gives

$$g_j^s = \Delta_j^s (v_h - v_b) + \eta_U \Delta_j^s b_j = \Delta_j^s W_{\eta_U, j}, \quad W_{\eta_U, j} = \eta_U b_j + v_h - v_b,$$

If  $W_{\eta_U, j} > 0$  and  $\Delta_j^I > 0 > \Delta_j^W$ , the common factor cancels:

$$\frac{g_j^I}{g_j^I - g_j^W} = \frac{\Delta_j^I}{\Delta_j^I - \Delta_j^W},$$

This proves the noise threshold. When  $W_{\eta_U, j} < 0$ , increasing acceptance instead lowers gain.

**Serving a record without verification.** Serving an erroneous record of value zero directly at lookup cost  $c_\ell$  gives

$$g_j^W = \eta_U(c_j^U - c_\ell) - v_j^U,$$

which is nonnegative exactly when  $v_j^U/c_j^U \leq \eta_U(1 - c_\ell/c_j^U)$ . The saving can outweigh the lost value, so  $g_j^W < 0$  must not be assumed for an unverified served record.

**Relation to verification.** The write probabilities of Section F are parameters. They acquire a verification meaning under three hypotheses: (i) a record is written for the arriving cell if and only if the primary is accepted, so that  $r_j = \mathbb{P}(H = 1 \mid \text{cell } j)$ ; (ii) the record is erroneous if and only if the accepted output is worse than the reference,  $Y_P < Y_B$ ; (iii) as in the standing assumption of Section F, acceptance and the accepted output’s quality relative to the reference do not depend on the record state, as when the records are written by attempts on which the router ignores the store (a logging or exploration stream) while the remaining rounds use the store without writing. Then

$$\frac{\psi_j}{r_j} = \mathbb{P}(Y_P < Y_B \mid H = 1, \text{ cell } j).$$

Calibration at level  $\theta$  for  $\{Y_P \geq Y_B\}$  relative to all selection information, including cell identity, gives  $\psi_j/r_j \leq 1 - \theta$  (Appendix J.1). Thus  $\theta > -g_j^W/(g_j^I - g_j^W)$  suffices under the signed-gain assumptions. If the record changes acceptance and every accepted attempt writes, hypothesis (iii) fails and Remark 16 applies instead.

### M.3 WRITES THAT DEPEND ON THE RECORD

*Remark 16* (State-dependent writes). Suppose the write type depends on the record the decision used: in record state  $s \in \{U, I, W\}$  the arriving cell  $j$  receives an informative write with probability  $\omega_j^s$  and an erroneous one with probability  $\psi_j^s$ , with  $r_j^s = \omega_j^s + \psi_j^s$ , while expiration keeps its probability  $\delta_j$ . Conditioning on the record state at the start of the round, the record probabilities  $x_{j,n} = (i_{j,n}, w_{j,n})^\top$  obey the affine recurrence  $x_{j,n+1} = A_j x_{j,n} + b_j$  with

$$A_j = \begin{pmatrix} \nu_j(\omega_j^I - \omega_j^U) + (1 - \delta_j)(1 - \nu_j r_j^I) & \nu_j(\omega_j^W - \omega_j^U) \\ \nu_j(\psi_j^I - \psi_j^U) & \nu_j(\psi_j^W - \psi_j^U) + (1 - \delta_j)(1 - \nu_j r_j^W) \end{pmatrix}$$

and  $b_j = \nu_j(\omega_j^U, \psi_j^U)^\top$ . This follows by conditioning on the old record and substituting  $\mathbb{P}(U) = 1 - i_{j,n} - w_{j,n}$ . If  $\delta_j > 0$  and  $\nu_j r_j^s < 1$  for every  $s$ , every state can reach U, which has a self-loop. The chain therefore converges to a unique stationary law,  $x_j^* = (I - A_j)^{-1} b_j$ . From an empty store,  $x_{j,n} = (I - A_j^n) x_j^*$ .

The gain calculation is unchanged:  $\eta_\infty \geq \eta_U$  exactly when  $\sum_j \nu_j(i_j^* g_j^I + w_j^* g_j^W) \geq 0$ . Under  $g_j^I > 0 > g_j^W$ , the cell threshold applies to the stationary erroneous share  $w_j^*/(i_j^* + w_j^*)$ , not a state-independent write fraction.

For ten equally likely cells,  $r_j^s = 1/2$ ,  $\delta_j = 1/100$ , unit costs and values (0.6, 0.9, 0.1), the gain threshold is  $3/8$ . State-independent erroneous fractions  $e^U = e^I = e^W = 0.3$  give gain +0.050. Keeping  $e^U = e^I = 0.3$  but raising  $e^W$  to 0.6 leaves coverage unchanged while raising the stationary erroneous share to  $0.401 > 3/8$  and changing gain to  $-0.018$ . Thus the base write error of 0.3 is insufficient to determine the sign once erroneous records reinforce erroneous writes.

*Remark 17* (Accumulating stores). Keeping every record without expiration preserves coverage  $1 - (1 - \nu_j r_j)^n$ , but changes readout. With iid write types, the stored erroneous fraction converges to  $\psi_j/r_j$ . An idealized majority readout consequently becomes informative below  $1/2$  and erroneous above  $1/2$ , whereas newest-record readout has threshold (29). For  $g^I = 0.3$ ,  $g^W = -0.5$ , these disagree for error fractions in  $(3/8, 1/2)$ . Equations (27) and (28) concern newest-record readout.

### M.4 RECORDS THAT BECOME STALE

Section F lets records expire but never lets an informative record become erroneous, although a change in the capability pool or in the task distribution can invalidate a stored route. Suppose that in each round, after the decision and the expiration step, an informative record that has survived expiration

turns erroneous with probability  $\vartheta_j \in [0, 1)$ , independently of everything else (a memoryless hazard), before the arrival's write; a record written in the round is fresh. With  $\alpha_j = (1 - \delta_j)(1 - \nu_j r_j)$  as in (23) and  $\beta_j = \alpha_j(1 - \vartheta_j)$ , the record probabilities obey

$$i_{j,n+1} = \nu_j \omega_j + \beta_j i_{j,n}, \quad w_{j,n+1} = \nu_j \psi_j + \alpha_j w_{j,n} + \alpha_j \vartheta_j i_{j,n},$$

The fixed points are

$$i_j^* = \frac{\nu_j \omega_j}{1 - \beta_j}, \quad w_j^* = \frac{\nu_j \psi_j + \alpha_j \vartheta_j i_j^*}{1 - \alpha_j}, \quad i_j^* + w_j^* = \frac{\nu_j r_j}{1 - \alpha_j} :$$

staleness changes record type, not coverage. Under  $g_j^I > 0 > g_j^W$ , a cell contributes positively exactly when its stationary erroneous share

$$e_j^{\text{eff}} = \frac{w_j^*}{i_j^* + w_j^*} = \frac{\alpha_j \vartheta_j + (\psi_j/r_j)(1 - \alpha_j)}{1 - \alpha_j + \alpha_j \vartheta_j}$$

is below  $g_j^I/(g_j^I - g_j^W)$ .

At  $\nu_j = 0.1$ ,  $r_j = 0.5$ ,  $\delta_j = 0.01$ ,  $g^I = 0.3$ ,  $g^W = -0.5$  and write error  $\psi_j/r_j = 0.2$ , no staleness gives gain +0.118. A hazard  $\vartheta_j = 0.02$  raises the stationary erroneous share to  $0.392 > 3/8$  and gives gain  $-0.012$ ; the sign changes at  $\vartheta^* \approx 0.0177$ . This stationary memoryless hazard does not describe a deterministic pool change at a fixed time.

#### M.5 AN ARBITRARY REFERENCE AND MEMORY-MAINTENANCE OVERHEAD

The uninformed state of a router with memory can already incur a maintenance or logging cost. To compare with a reference that does not pay that cost, fix any  $\lambda \in \mathbb{R}$  and define

$$\begin{aligned} J_U(\lambda) &= \sum_j \nu_j (v_j^U - \lambda c_j^U), \\ g_j^s(\lambda) &= (v_j^s - v_j^U) - \lambda(c_j^s - c_j^U), \quad s \in \{I, W\}, \\ G_j(\lambda) &= \frac{\nu_j [\omega_j g_j^I(\lambda) + \psi_j g_j^W(\lambda)]}{1 - \alpha_j}. \end{aligned}$$

The empty-start finite-horizon identity is

$$\sum_{n=0}^{N-1} \mathbb{E}[V_n - \lambda D_n] = N J_U(\lambda) + \sum_j \nu_j G_j(\lambda) \left[ N - \frac{1 - \alpha_j^N}{1 - \alpha_j} \right]. \quad (55)$$

At stationarity the expected gain is  $J_U(\lambda) + \sum_j \nu_j G_j(\lambda)$ ; with positive stationary expected cost, its sign is the sign of  $\eta_\infty - \lambda$ .

In particular, let a no-memory reference have the same mean value as the uninformed state, mean cost  $c_0 > 0$ , and efficiency  $\eta_0$ . If the uninformed state of the router with memory costs  $c_0 + h$  with  $h > 0$ , then  $J_U(\eta_0) = -\eta_0 h$ . Memory is therefore at least as efficient as that reference if and only if

$$\sum_j \nu_j G_j(\eta_0) \geq \eta_0 h. \quad (56)$$

Here  $h$  is the difference in expected cost per round; any further record-dependent costs remain inside  $g_j^s(\eta_0)$ .

The proof is (53) with the uninformed term retained, followed by the same geometric sum.

#### M.6 COVERAGE UNDER TIME-BASED AND EVENT-BASED RETENTION

Consider first a record that remains usable for exactly  $E$  global rounds after it is written, with no other expiration. Assume  $E$  is a positive integer. Once the initial transient has passed, cell  $j$  is covered if and only if at least one of the preceding  $E$  rounds wrote a record for it. The iid write probability per round is  $\nu_j r_j$ , and hence

$$\chi_j^{\text{TTL}} = 1 - (1 - \nu_j r_j)^E, \quad i_j^{\text{TTL}} = \frac{\omega_j}{r_j} \chi_j^{\text{TTL}}, \quad w_j^{\text{TTL}} = \frac{\psi_j}{r_j} \chi_j^{\text{TTL}}. \quad (57)$$

Conditional on a write to  $j$ , its type is informative with probability  $\omega_j/r_j$ . This remains true for the most recent such write, which proves the last two identities.

Now consider an ordinary FIFO that retains the most recent  $M$  write events globally, including repeated entries for the same cell, and answers a query using the newest retained matching record. Put

$$\beta = \sum_j \nu_j r_j > 0, \quad q_j = \frac{\nu_j r_j}{\beta}.$$

Discarding rounds with no write produces an iid sequence of cell marks with law  $q$ . Thus, after the FIFO has filled,

$$\chi_j^{\text{FIFO}} = 1 - (1 - q_j)^M, \quad i_j^{\text{FIFO}} = \frac{\omega_j}{r_j} \chi_j^{\text{FIFO}}, \quad w_j^{\text{FIFO}} = \frac{\psi_j}{r_j} \chi_j^{\text{FIFO}}. \quad (58)$$

These are independent-reference caching formulas (Fagin, 1977). Replacing  $E$  by the mean span  $M/\beta$  underestimates FIFO coverage: Bernoulli's inequality gives  $(1 - \beta q_j)^{M/\beta} \geq (1 - q_j)^M$ . TTL fixes elapsed rounds while FIFO fixes write events. Capacity in distinct cells is different again (Corollary 18).

## M.7 ORACLE REGRET AND ITS SIGNED TRANSIENT

The next identity concerns scalar utility, not necessarily raw accuracy. Let  $s_j^U, s_j^I, s_j^W$  be fixed conditional means, and let  $s_j^O$  denote a per-cell oracle mean. For example,  $s = v - \lambda c$  gives the additive objective. Assume no expiration, empty initial memory, and finitely many cells. Put

$$s_j^\infty = \frac{\omega_j s_j^I + \psi_j s_j^W}{r_j}.$$

Let  $\text{Reg}_N$  denote cumulative utility regret against these cell-oracle means. Before round  $n$ , cell  $j$  is unwritten with probability  $(1 - \nu_j r_j)^n$ ; otherwise its mean utility is  $s_j^\infty$ . Conditioning on that event and summing the geometric series gives

$$\begin{aligned} \mathbb{E}[\text{Reg}_N] &= N \sum_j \nu_j (s_j^O - s_j^\infty) \\ &\quad + \sum_j \frac{1 - (1 - \nu_j r_j)^N}{r_j} (s_j^\infty - s_j^U). \end{aligned} \quad (59)$$

The first term includes both imperfect informative records and write errors. The transient is signed and satisfies

$$\left| \sum_j \frac{1 - (1 - \nu_j r_j)^N}{r_j} (s_j^\infty - s_j^U) \right| \leq \sum_j \frac{|s_j^\infty - s_j^U|}{r_j}.$$

It is a nonnegative learning cost only when every  $s_j^\infty \geq s_j^U$ ; the limiting record policy can otherwise be worse than the uninformed policy.

## M.8 PROOF OF PROPOSITION 10

Let  $\kappa_z = 1/\zeta(z)$  and  $\nu_j = \kappa_z j^{-z}$  for  $j \geq 1$ , where  $z > 1$ . Assume clean writes with a common probability  $\omega > 0$  and a common positive informed-minus-uninformed hit gap  $\Delta$ . By (57) with  $r_j = \omega$ , a cell is uncovered under a TTL of  $E$  rounds with probability  $(1 - \omega \nu_j)^E$ , so the TTL coverage deficit is

$$L_E = \Delta \sum_{j \geq 1} \nu_j (1 - \omega \nu_j)^E,$$

and from an empty store without expiration the uncovered probability at round  $n$  is  $(1 - \omega \nu_j)^n$  by (27) with  $\delta_j = \psi_j = 0$ , so the per-round deficit is  $L_n$  (with  $L_0 = \Delta$ ) and, summing the geometric series in  $n$ , the cumulative deficit is

$$R_N := \sum_{n=0}^{N-1} L_n = \frac{\Delta}{\omega} \sum_{j \geq 1} [1 - (1 - \omega \nu_j)^N]. \quad (60)$$

**Occupancy form.** Round  $n$  writes cell  $j$  with probability  $\omega\nu_j$ . If  $K_N$  counts distinct written cells, then (60) gives  $R_N = (\Delta/\omega)\mathbb{E}[K_N]$ , while  $\omega L_N/\Delta$  is the expected unseen-cell write mass. The singleton identity  $\mathbb{E}[K_{N,1}]/(\omega N) = L_{N-1}/\Delta$  also connects the deficit to Good–Turing missing mass (Good, 1953).

**Rate and constants.** The write-frequency counting function is  $\bar{\nu}(x) = \lfloor (\omega\kappa_z/x)^{1/z} \rfloor \sim (\omega\kappa_z)^{1/z}x^{-1/z}$ . The occupancy and missing-mass asymptotics of Karlin (1967) and Gneden et al. (2007, Prop. 17 and Lemma 1), at index  $1/z$ , give

$$L_E \sim \frac{\Delta\kappa_z^{1/z}}{z}\Gamma\left(1 - \frac{1}{z}\right)(\omega E)^{1/z-1}, \quad R_N \sim \Delta\kappa_z^{1/z}\Gamma\left(1 - \frac{1}{z}\right)\omega^{1/z-1}N^{1/z}, \quad (61)$$

and hence the claimed rates.

**A self-contained derivation of the constants.** For a direct derivation, squeeze  $(1-x)^E$  between  $e^{-Ex}$  and  $e^{-(1+\varepsilon)Ex}$  for sufficiently small  $x$ ; the remaining finite head contributes exponentially little. For  $f_c(x) = \kappa_z x^{-z} e^{-c\kappa_z x^{-z}}$ ,

$$\int_1^\infty f_c(x) dx \sim \frac{\kappa_z^{1/z}}{z}\Gamma(1 - 1/z)c^{1/z-1}.$$

Unimodality bounds the sum–integral error by  $O(c^{-1}) = o(c^{1/z-1})$ , proving the first equivalent at  $c = \omega E$ . Applying the same squeeze to  $1 - (1 - \omega\nu_j)^N$  gives  $\mathbb{E}[K_N] \sim \Gamma(1 - 1/z)(\omega\kappa_z N)^{1/z}$ ; the monotone sum–integral error is at most one. This proves the second equivalent.

**Relation to Proposition 8.** With  $\delta_j = \psi_j = 0$ , constant  $\omega$ , hit gap  $\Delta$  and margin  $W_{\eta_U}$ , one has  $1 - \alpha_j = \omega\nu_j$  and  $G_j = \Delta W_{\eta_U}$ . Substitution in (28) gives shortfalls  $W_{\eta_U} L_n$  per round and  $W_{\eta_U} R_N$  cumulatively. Any persistent informed-versus-oracle gap contributes a separate linear term. Zipf arrivals alone do not determine the rate if  $\Delta_j$  or  $\omega_j$  varies with the cell.

## M.9 CAPACITY MEASURED IN RECORDS

**Corollary 18** (Capacity measured in records). *Under the assumptions of Proposition 10, the expected number of cells holding a record under a TTL of  $E$  rounds is  $m(E) = \sum_j [1 - (1 - \omega\nu_j)^E] \sim \kappa_z^{1/z}\Gamma(1 - 1/z)(\omega E)^{1/z}$ , and the stationary deficit expressed through occupied capacity is*

$$L_E \sim \frac{\Delta\Gamma(1 - 1/z)^z}{z\zeta(z)} m(E)^{-(z-1)}.$$

*A store holding at most  $M$  distinct cells, updated by the write events with least-recently-used replacement, obeys the same law with  $M$  in place of  $m(E)$  for every  $\omega \in (0, 1]$  (Jelenković, 1999; Che et al., 2002), whereas the event FIFO of (32) with  $M$  slots has deficit  $\Theta(M^{-(z-1)/z})$ .*

*Proof.* By (57) with  $r_j = \omega$ , cell  $j$  holds a record under the TTL if and only if it was written during the last  $E$  rounds, so the expected number of cells holding a record is

$$m(E) = \sum_j [1 - (1 - \omega\nu_j)^E] = \mathbb{E}[K_E] = \frac{\omega}{\Delta} R_E$$

by (60). Eliminating  $\omega E$  between this asymptotic and (61) gives the TTL law.

Clean constant-probability writes have iid cell marks with law  $\nu$ , regardless of  $\omega$ . The LRU store therefore follows the independent reference model. Its cited fault asymptotic is  $(1 - 1/z)\Gamma(1 - 1/z)^z$  times  $\sum_{j>M} \nu_j \sim \kappa_z M^{1-z}/(z-1)$  (Jelenković, 1999), giving the same constant as TTL.

For event FIFO, (58) instead gives deficit  $\Delta \sum_j \nu_j (1 - \nu_j)^M$ , equal to  $L_M$  at  $\omega = 1$ , hence exponent  $(z-1)/z$ . The  $M$  slots hold only  $\mathbb{E}[K_M] = \Theta(M^{1/z})$  distinct cells; the remaining slots are duplicates.  $\square$

**Remark 19** (Break-even horizon). With  $C_z = \Delta\kappa_z^{1/z}\Gamma(1 - 1/z)\omega^{1/z-1}/z$ , an empty-start deficit  $L_N \sim C_z N^{-(z-1)/z}$  falls below a tolerance  $\tau$  after  $N^* \sim (C_z/\tau)^{z/(z-1)}$  rounds. The exponent makes this horizon large when  $z$  is close to one; the rate is asymptotic, not a finite-capacity allocation guarantee.

## M.10 WHAT ARM COVERAGE DOES, AND DOES NOT, ESTABLISH

Cell coverage is not equivalent to identifying the best allocation within a cell. The distinction can be made explicit through an idealized record model.

**Lemma 20** (Exact-reveal arm coverage). *Fix a cell and a finite set of arms with values  $\mu_m$ . On each visit, an independent logger evaluates arm  $m$  with probability  $e_m$ . A logged evaluation reveals  $\mu_m$  exactly. The deployed recommendation chooses a highest-valued recorded arm, using an arbitrary initial action if no arm has been recorded. If a best arm  $m^*$  satisfies  $e_{m^*} > 0$ , then after  $v$  visits the probability of a suboptimal recommendation is at most  $(1 - e_{m^*})^v$ . If no best arm is in the logger’s support, the recommendation converges to the best value within that support.*

*Proof.* The probability that  $m^*$  has never been recorded is  $(1 - e_{m^*})^v$ . Once it is recorded, its exact value is available and no recorded arm has greater value. This proves the first assertion. For the second, each arm with positive logging probability is eventually recorded almost surely; finiteness of the arm set implies that eventually all such arms have been recorded. The recommendation then maximizes over precisely that set.  $\square$

The lemma assumes exact values and exogenous logging. With noisy samples, identifying the best arm additionally depends on quality gaps and estimation error (Bubeck et al., 2011). Logging costs and exploration losses remain part of the router’s resource and value accounting; accurate recommendations alone do not bound cumulative regret.

Finally, binary verifier noise can alter the estimand. With arm-independent false-accept and false-reject probabilities  $\beta_{fa}$  and  $\beta_{fr}$ , an arm’s recorded success mean is

$$\tilde{\mu}_m = \beta_{fa} + (1 - \beta_{fr} - \beta_{fa})\mu_m.$$

Rankings are preserved when  $\beta_{fr} + \beta_{fa} < 1$ . Arm-dependent errors need not preserve rankings, regardless of sample size. Thus adequate coverage, accurate mean estimation, and faithful verification are distinct requirements.

## N ANALYTICAL ILLUSTRATIONS

Figures 3 and 4 evaluate the stated models with illustrative parameters; they contain no model calls or benchmark measurements. The source README documents the parameters, numerical recipes, library versions, and the reproduction entry point, `scripts/reproduce.py`.

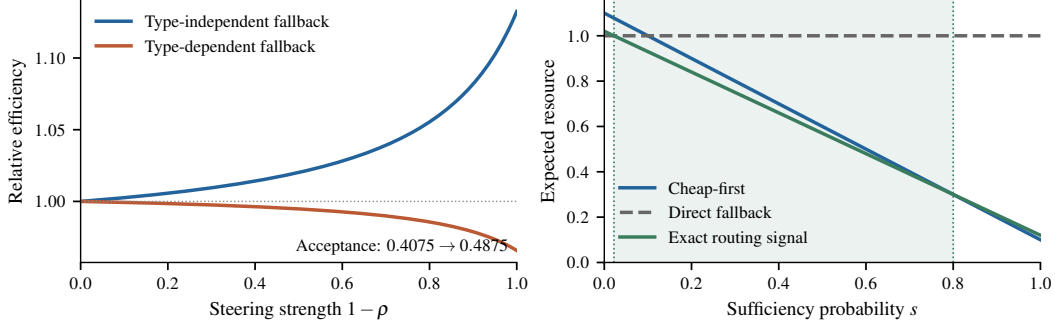


Figure 3: **Left:** the four-type construction of Appendix L.5, with unit proposer cost and fallback cost ten. Steering raises acceptance from 0.4075 to 0.4875. Efficiency rises by 13.2% with a type-independent fallback of value 0.8, but falls by 3.4% when the fallback succeeds exactly on types 3 and 4. Each curve is normalized by its unsteered efficiency. **Right:** the costs in (20) for  $a = 0.1$ ,  $b = 1$ , and  $k = 0.02$ . Exact routing information is cheapest inside  $(1/45, 4/5)$ , containing the uninformed crossover  $s = 0.1$ .

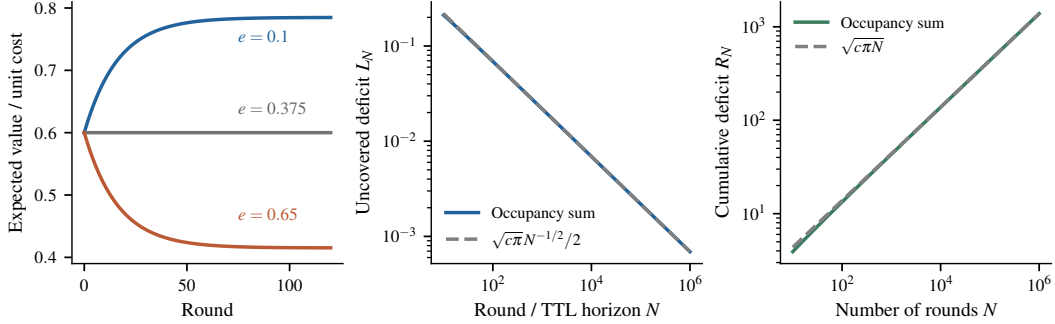


Figure 4: **Left:** the record recurrence for ten equally likely cells, write rate 0.5, expiration 0.01, unit costs, and values  $(v^U, v^I, v^W) = (0.6, 0.9, 0.1)$ . Gains are  $g^I = 0.3$ ,  $g^W = -0.5$ ; records improve efficiency exactly when the erroneous write fraction  $e < 3/8$ . **Center/right:** Zipf arrivals with  $z = 2$ , clean writes on every visit, no expiration, and unit hit gap. The uncovered deficit decays as  $N^{-1/2}$  and its cumulative sum grows as  $N^{1/2}$  (Proposition 10). Solid curves are occupancy sums; dashed curves are their leading asymptotic equivalents.

## O EXTENDED RELATED WORK

**Model selection, cascades, and portfolios.** Cascades and routers formalize parts of the problem (Chen et al., 2024b; Ong et al., 2025; Dekoninck et al., 2025); the acceptance step is a reject option (Chow, 1970), studied as confidence-based or budgeted deferral (Jitkrittum et al., 2023; Gupta et al., 2024; Aggarwal et al., 2024; Chen et al., 2020; Trapeznikov & Saligrama, 2013; Verma et al., 2023) and linked to speculative decoding (Narasimhan et al., 2025). Algorithm portfolios select complementary solvers per instance (Rice, 1976; Huberman et al., 1997; Gomes & Selman, 2001; Kotthoff, 2014): the exact selecting verifier of Section 3.1 is their virtual best solver, residual coverage is the marginal contribution of a component (Xu et al., 2012; Fr  chette et al., 2016), and steering relates to the Hydra objective (Xu et al., 2010). Ensemble diversity and multi-proposer aggregation motivate combining outputs (Kuncheva & Whitaker, 2003; Brown et al., 2024; Jiang et al., 2023;

Wang et al., 2025); boosting adapts component weights (Schapire, 1990; Freund & Schapire, 1997). Our portfolio comparison asks when additional coverage justifies the complete calling cost.

**Information, verification, and memory.** Paying for a routing signal is a value-of-information and metareasoning question (Raiffa & Schlaifer, 1961; Howard, 1966; Russell & Wefald, 1991; Hay et al., 2012). Router memory relates to caches under the independent reference model (Fagin, 1977; Breslau et al., 1999; Jelenković, 1999; Che et al., 2002), LLM response caches and model multiplexing (Bang, 2023; Zhu et al., 2023; Gaspar Schroeder et al., 2026), and erroneous agent memory (Chen et al., 2024c). Imperfect verifiers constrain resampling gains (Stroebl et al., 2026; Chen et al., 2024a; Snell et al., 2025), while hedged execution trades expenditure for tail latency (Dean & Barroso, 2013). These connections motivate measuring the decision improvement supplied by information together with its acquisition and retention costs.

**Agent orchestration and routing granularity.** Routing decisions recur throughout inference: which frontier model an agent should call using verified outcome records (Zhou et al., 2026a) or OpenSquilla’s harness context and data flywheel (Liu et al., 2026); which vision–language model to select under a budget (Yu et al., 2026); which token-pruning strategy to use (Liang et al., 2026); which skills to select from a typed graph (Bai et al., 2026); which submodel to use per video frame (Tang et al., 2026); which acceleration to apply per denoising step (Zhao et al., 2026); and which model emits each token (Fu et al., 2025). Conductor and Trinity learn coordination over model pools (Nielsen et al., 2026; Xu et al., 2026); Fugu and Fusion further motivate allocating computation across heterogeneous models (Sakana AI, 2026; Zhou et al., 2026b). These systems determine the state and action menu in Eq. (4). The model-routing experiments make one such menu measurable on common tasks; applying the framework at another granularity requires the corresponding outcomes and costs.

**Cross-model state transfer.** Cache-to-Cache projects and fuses KV caches to avoid intermediate text generation (Fu et al., 2026); cross-model KV transfer and CacheBridge reduce repeated prefill (Heo et al., 2026; Qu et al., 2026); LatentMAS communicates through a shared latent working memory (Zou et al., 2026). Mostik’s technical note describes a learned bridge from a large model’s hidden states to a small model’s decoder (Mostik AI, 2026). These mechanisms change both the resource cost and the value retained across a hand-off, so their benefit enters the general comparison through measured  $V$  and  $D$ . Proposition 11 specializes this comparison when reusable work lowers fallback cost and the fallback restores reference value. Appendix P.2 summarizes the correspondences.

## P ILLUSTRATIVE APPLICATIONS OF THE FRAMEWORK

### P.1 STRUCTURED DECISION MODELS

Jev returns distributions for typed decision questions (TypeSafe AI, 2026c;d). A pre-execution query can supply the signal  $Z$  at cost  $k$  in (22); a post-execution check can define acceptance  $H$ , with its cost included in  $A$ , as in TypeSafe’s extraction cascade (TypeSafe AI, 2026a). Its generic confidence field is not automatically the posterior probability of primary sufficiency (TypeSafe AI, 2026b). Applying the criteria requires measuring this probability and the resulting value and cost on matched tasks. Our direct-choice experiment evaluates ex-ante dispatch on recorded backend outcomes (Appendix B.5); it does not evaluate Jev as a post-output verifier or establish a quality floor.

### P.2 APPLICATIONS ACROSS GRANULARITIES

Table 15 connects recent mechanisms to the framework’s actions, states, and value–cost comparison. Applying a specialized portfolio, cascade, or memory result additionally requires its stated assumptions. In particular, a full-computation reference need not be an executed fallback after rejection.

Table 15: Illustrative correspondences. A reference supplies the comparison outcome; a fallback is additional computation actually executed after rejection.

| System                                                                                                             | Decision and accounting                                                                                        | Assumption requiring separate measurement                                                                                              |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| ACRouter (Zhou et al., 2026a)                                                                                      | One model per task; verified outcomes enter a bounded store.                                                   | No within-task fallback. Embedding-based retrieval is not the fixed, independent cell process of Proposition 8.                        |
| OpenSquilla (Liu et al., 2026)                                                                                     | Model choices at agent steps use harness state; execution and verification costs enter $D$ .                   | The retry recursion uses success probabilities conditional on preceding failures.                                                      |
| SCOPE-Router (Yu et al., 2026)                                                                                     | Ex-ante VLM dispatch; compare chosen and reference outcomes.                                                   | Rank Score is not the net value of information in (22).                                                                                |
| VIP-Router; RDS; RAPID <sup>3</sup> (Liang et al., 2026; Tang et al., 2026; Zhao et al., 2026)                     | Token pruning, video submodels, or denoising accelerations; full computation can be the reference.             | Training rewards do not supply an online acceptance event or a rejection-triggered fallback.                                           |
| R2R (Fu et al., 2025)                                                                                              | Small/large model selection at each generated token.                                                           | Earlier decisions change later token states; Theorem 5 assumes a policy-independent state distribution.                                |
| SkillDAG (Bai et al., 2026)                                                                                        | Skill selection and execution-validated graph updates.                                                         | Learning graph structure need not follow the independent-write record model.                                                           |
| Conductor; Trinity; Fugu; Fusion (Nielsen et al., 2026; Xu et al., 2026; Sakana AI, 2026; Zhou et al., 2026b)      | Role assignments, portfolios, and further evidence acquisition.                                                | A learned verifier or synthesizer is not the exact selecting verifier of Proposition 2; sequential stopping needs continuation values. |
| Cache and latent hand-over (Fu et al., 2026; Heo et al., 2026; Qu et al., 2026; Zou et al., 2026; Mostik AI, 2026) | Reused state changes switching cost and retained value; Proposition 11 gives a quality-restoring special case. | The value retained and the transfer cost must be measured for the particular model pair.                                               |

**Published aggregates are application illustrations.** The supplementary cross-granularity comparison combines fixed cheap-path quantities with adaptive-router efficiencies. These describe different policies and illustrate applications; testing the cascade criteria requires matched conditional outcomes at each system’s decision granularity. Our empirical studies focus on model routing.

## Q WHEN DOES ROUTER MEMORY HELP?

**Question.** A record of a previously successful model can avoid another routing decision, but a mistaken write can also persist. Does a store improve the policy it replaces, and how do verification and retention affect that gain?

**Design.** The memory study combines real CodeRouterBench outputs and verifier judgments with simulated record arrivals and retention. The main execution population contains 2,044 probing and 867 test tasks in 12 dimension $\times$ source cells; tasks scored zero by every recorded model in the affected execution sources are excluded. Variants use 21 cells or 50 TF-IDF  $k$ -means clusters fitted on probing. On each round, the fixed writer explores a uniformly sampled model with probability  $\varepsilon = 0.25$  and writes its identity to the arriving cell if verification accepts. The newest retained model serves subsequent tasks in that cell; uncovered cells use always-Opus or a DimensionBest prior fitted on probing. Arrivals, exploration, and expiration are constructed because the release contains no memory-write log.

Writers use ACRouter’s doctest/syntax tools, locally executed hidden tests, or a DeepSeek-v4-pro judge on a subsample. The 401 empty responses among 6,936 main-population task-model pairs are assigned their released acceptance truth and excluded from verifier error-rate estimates. Fifty streams of 60,000 task draws use the final 10,000 rounds for stationary measurements. We compare geometric expiration, TTL, event FIFO, and LRU retention. Probing estimates predict the gain before the test streams are evaluated; intervals resample streams and, in a separate analysis, task clusters. The record store remains semi-synthetic even when its writer is a live judge.

**Findings.** **The same memory policy helps relative to Opus and hurts relative to an already efficient prior.** In the 12-cell hidden-test setting without expiration, where all cells become covered, stationary efficiency is 135.8 score per dollar. The record criterion correctly predicts opposite gains against the two priors (Table 16). Across the correlated setting grid, it agrees with all 279 interval-decided signs; agreement at the cell level is 858/909 (94.4%). These counts share tasks and simulated streams.

Table 16: The same hidden-test-written store compared with two priors. Gain is the expected per-round surplus  $\mathbb{E}[V - \eta_B D]$  at each prior’s own efficiency. Brackets are paired 95% intervals.

| Prior                        | Probing-predicted gain | Realized gain [CI]      |
|------------------------------|------------------------|-------------------------|
| Always-Opus                  | +0.304                 | +0.275 [+0.272, +0.279] |
| Probing-fitted DimensionBest | −0.148                 | −0.120 [−0.126, −0.113] |

Writer errors are substantial. Against released-score truth, doctest/syntax TPR/FPR is 0.827/0.539 and hidden-test TPR/FPR is 0.960/0.205 on 6,535 non-empty response pairs; the judge gives 0.776/0.341 on 2,320 subsample pairs. For event FIFO, using realized write rates gives mean absolute coverage error 0.0064 for the exact retention expression and 0.0085 for replacing the random span by its mean. This is an in-sample mechanism check. Using probing-estimated write rates instead gives held-out errors 0.0495 and 0.0507; uncertainty in write rates is therefore much larger than the correction to the retention approximation.

**Oracle-write control.** The idealized control uses free oracle writes on 50 i.i.d. streams of 20,000 test tasks over 21 cells, measuring the final 5,000 rounds. It raises dollar efficiency from 42.9 to 72–83 score per dollar against Opus, but lowers efficiency against DimensionBest and lowers seconds-based efficiency against both priors. Thus even perfect writes do not make memory uniformly beneficial; the verifier-written study supplies the comparison with actual verifier judgments.

**Takeaway.** **Memory is useful when retained choices improve on the current prior enough to pay for exploration and mistaken writes.** More coverage alone is not the objective. Verification quality, the prior policy, and the retention rule jointly determine whether reuse helps.